The Political Legacy of Fiction*

Alex Armand Paul Atwell Joseph F. Gomes
Giuseppe Musillo Yannik Schenk

Abstract

We examine whether childhood exposure to fictional stories can shape long-run political preferences. We study an unprecedented experiment in post–WWII United States: in 1946, amid entrenched racial divisions, the popular children's radio series *The Adventures of Superman* suddenly replaced its fantasy plots with stories confronting intolerance in American society, explicitly designed to shape moral values among young listeners. This within-program shift generated variation in potential exposure across cohorts and locations, allowing us to isolate the causal effect of changing narrative content during childhood on later outcomes. We find that childhood exposure increased support for civil rights decades later, in the 1960s and 1970s, when racial equality was a contested dimension of American politics. These effects extend to support for civil rights leaders, political engagement, and lower institutional trust. At the county level, greater exposure increased political mobilization. Fictional narratives can therefore be a lasting channel of political socialization. (*JEL* D7, D83, I24, J15, L82, N32)

Keywords: political preference formation; fictional narratives; political socialization; mass media; racial attitudes; civil rights.

*__Armand__: Nova School of Business and Economics – Universidade Nova de Lisboa, CEPR, IZA, and Institute for Fiscal Studies (e-mail: alex.armand@novasbe.pt); **Atwell**: Ohio State University (e-mail: atwell.54@osu.edu); **Gomes**: UCLouvain, LIDAM, IRES, Belgium; and CEPR (e-mail: joseph.gomes@uclouvain.be); **Musillo**: Tilburg University (e-mail: g.musillo@tilburguniversity.edu); **Schenk**: Nova School of Business and Economics – Universidade Nova de Lisboa, NOVAFRICA (e-mail: yannik.schenk@novasbe.pt). We would also like to thank Leonardo Bursztyn, Davide Cantoni, Irma Clots-Figueras, Victoire Girard, Andreas Madestam, Mounu Prem, Imran Rasul, Christopher Rauh, David Strömberg, Mathias Thoenig, Noam Yuchtman, and seminar participants at CUNEF, LISER, CAGE, Virginia Tech, Paris Dauphine Workshop on Social Norms, University of Cambridge, Royal Holloway, King's College, UCL, UAB, SAEe, ULB, LMU, University of Essex, Tilburg, Maastricht, Media Bias Workshop in Cologne, PSE, and Workshop in Economic History at ENS de Lyon for helpful comments. Armand acknowledges the funding from Fundação para a Ciência e a Tecnologia (UID/00124/2025, UID/PRR/124/2025, Nova School of Business and Economics DOI: https://doi.org/10.54499/UID/00124/2025) and LISBOA2030 (DataLab2030 - LISBOA2030-FEDER-01314200). Schenk acknowledges funding from the Fonds National de la Recherche Scientifique (FNRS–T025320F). A version of this paper was previously circulated with the title "*Media Narratives and the Rise of Civil Rights*".

1 Introduction

Childhood and adolescent socialization shape beliefs and political preferences that can persist into adulthood. Recent work focuses largely on individuals' lived experiences during these formative years, such as the neighborhoods in which they grow up or the schools they attend (Giuliano and Spilimbergo, 2025). Yet, formative influences can also operate indirectly. Fictional stories expose children to people, places, and experiences beyond their immediate environment and have long been used to transmit norms and cultural values across generations (Michalopoulos and Xue, 2021).[1]

During childhood and adolescence, fictional stories can transmit moral values, foster empathy, and influence attitudes, all of which are core components of political preferences (Enke, 2020; Bonomi et al., 2021).[2] These formative years are central to *political* socialization, the process through which individuals develop political attitudes, values, and identities (Dahl et al., 2018). Fictional stories encountered during these years may therefore have lasting consequences. Yet evidence of such persistence remains scarce. The broader literature on the political effects of narratives focuses largely on adult audiences with already-formed political attitudes (Strömberg, 2015; Campante et al., 2022). In contrast, we focus on whether the narrative content encountered during formative years can shape the values and identities that underpin political preferences.

We provide causal evidence that fictional stories encountered during childhood can leave persistent effects on political preferences in adulthood. Our focus is the rise of civil rights in the United States, one of the defining political transformations of the twentieth century, as demands for racial equality moved to the center of national politics and culminated in landmark legislative reforms such as the Civil Rights Act of 1964 and the Voting Rights Act of 1965. We examine the years following these major reforms, when support for racial equality was far from universal and civil rights remained a contested dimension of American politics.

We ask whether the stories encountered during childhood amid the cultural changes that followed World War II (WWII) helped shape support for civil rights when these issues moved to the center of American politics decades later, in the 1960s and 1970s. We exploit an unusual natural experiment: the 1946 transformation of *The Adventures of Superman*, a popular radio series for children and adolescents. In a groundbreaking experiment known as *Operation Intolerance (OI)*, the producers of the series deliber-

[1] We use (fictional) *story* to refer to a narrative involving fictional events or characters, and *storytelling* to refer to its communication. A *narrative* is a representation of a sequence of events, real or fictional.

[2] The psychological mechanisms include narrative transportation, identification with characters, and vicarious intergroup contact (see, e.g., Gasser et al., 2022). In vicarious intergroup contact, individuals are indirectly exposed to out-group members through fictional narratives. In contrast, the direct-contact literature emphasizes actual social interaction (see, e.g., Lowe, 2025).

ately replaced its usual fantasy plots with stories that confronted racism and prejudice in an attempt to reshape values among children while leaving the medium, format, and target audience largely unchanged. Superman was no longer battling mythical villains but fighting intolerance within everyday American life. Unlike conventional educational programming, *OI* did not seek to transmit knowledge or cognitive skills. Its messages were conveyed entirely through fictional plots. Millions of children tuned in daily to stories promoting equality in a society deeply divided along racial lines. Later accounts suggest that these stories had a deep cultural influence on this generation (Wall, 2009).

We investigate whether exposure to *OI* during childhood and adolescence influenced support for civil rights later in life. Importantly, our goal is not to estimate the effect of exposure to an influential program such as *The Adventures of Superman*, as is more common in the media literature, but rather to isolate the effect of the change in narratives within the same program. Several features of the setting make long-run effects particularly plausible. First, the intervention targeted children and adolescents, whose social and moral values are particularly malleable. Second, the new narratives were designed to evoke identification, empathy, and social learning—psychological processes that contribute to value formation. Third, the values of equality, tolerance, and opposition to prejudice promoted by *OI* became highly salient later in life, as racial equality moved to the center of American political conflict. This later political environment provided repeated opportunities for underlying differences in attitudes to be expressed and translated into political behavior, consistent with evidence that the social and political context can shape the expression of underlying attitudes (e.g., Bursztyn et al., 2020).

Our empirical strategy combines geographic variation in potential exposure to *OI* in 1946 with differences across cohorts in their ages when the narrative changed. Using historical records, we identify the radio stations that broadcast the program in 1946 and use an engineering-based propagation model to predict signal reception across the United States. We validate these predictions using historical listenership data. We combine this variation with repeated cross-sectional surveys collected between 1964 and 1980 and implement a cohort-based difference-in-differences (DiD) design. The design estimates how the long-run relationship between potential exposure to *OI* and later political preferences varies across cohorts of different ages in 1946.

The identifying variation comes from the fact that the new stories were primarily directed at school-age children and adolescents. Younger cohorts were too young to meaningfully engage with the stories or were not yet born, whereas older cohorts were less likely to listen and had moved beyond the ages generally considered most formative. We do not assume that the effects of *OI* are confined to children and adolescents. Instead, we test whether potential exposure in 1946 predicts later outcomes more

strongly for cohorts at the program's target ages. We refer to the relationship between potential exposure and later outcomes within a cohort as the *exposure gradient*. Identification requires that, absent *OI*, this gradient would not have differed systematically across cohorts in a way that aligns with the program's target ages. We present extensive evidence, including stable exposure gradients across older cohorts, flexible cohort-, survey-, and geography-specific controls, alternative measures of radio exposure, and placebo checks. Together, these exercises support our identifying assumption.

We find that exposure to *OI* during childhood and adolescence significantly shifts political preferences in favor of civil rights. Among those aged 7–18 in 1946, higher potential exposure to *OI* later translates into greater support for civil rights policies and more favorable views toward African Americans. The exposure gradient in this cohort significantly differs from that of those aged 19–24 in 1946, who had been part of the intended audience of *The Adventures of Superman* in earlier years but were no longer targeted when the storylines changed. No systematic difference is observed among cohorts who were too old or too young in 1946 to have followed the new storylines, and no significant effect is found for the target cohorts in the placebo sample of internal migrants. The results are also not driven by urban cultural trends, station placement, exposure to radio in general, other radio content, individual survey waves or states, or any changes in educational levels.

Increased support for civil rights is also reflected in social relationships. Greater potential exposure to *OI* during childhood increases the likelihood of having an interracial friendship in adulthood but does not extend to interracial marriage, a more consequential form of social integration that remained subject to strong institutional barriers during much of the period we study (Fryer Jr, 2007).

Individuals exposed to *OI* during childhood later display a broader shift in political preferences and behavior. They express greater support for leaders championing civil rights and become more politically engaged, expressing greater approval of protests and participating more in political campaigning. They also report lower trust in institutions, including the police and military. These patterns are consistent with political conflicts that became increasingly intertwined with the Civil Rights Movement during the 1960s and 1970s, as policing became central to struggles over racial equality and civil rights activism increasingly intersected with opposition to the Vietnam War. Consistent with this broader shift, exposed individuals are also less likely to serve in the Vietnam War.

We examine whether these cohort-level effects translated into broader social and political changes. Using local historical records from 1930 to 2020, we implement an event-study DiD design comparing counties with varying broadcast exposure in 1946 over time. More exposed counties later show significantly lower vote shares for segrega-

tionist presidential candidates, consistent with the political impact of the target cohorts once they reached voting age. The effects extend to organized political mobilization around racial issues. More exposed counties experience an increase in the presence of civil rights organizations and a decline in the presence of Ku Klux Klan (KKK) chapters, in line with qualitative evidence on the role of *OI* in undermining the organization (Bowers, 2012; Levitt and Dubner, 2014). Finally, civil rights themes become more salient in local newspapers, indicating that the effects extend to public discourse.

Our findings broaden our understanding of how political preferences form over the long run. Existing work identifies several formative influences during childhood and adolescence. Lived experiences and socialization through family environments, neighborhoods, peers, schools, and racial environments can leave persistent effects on political participation, ideology, partisanship, and racial attitudes (Chetty and Hendren, 2018; Rao, 2019; Billings et al., 2021; Chyn and Haggag, 2023; Brown et al., 2023; Kaplan et al., 2025). Related work highlights the role of prolonged exposure to political events and institutions (Sears and Valentino, 1997; Alesina and Fuchs-Schündeln, 2007; Ghitza et al., 2023). A second strand shows that deliberately designed educational content can also have lasting effects. School curricula and access to educational and edutainment content can durably alter ideology, social attitudes, and voting (Cantoni et al., 2017; Voigtländer and Voth, 2015; Fouka, 2020; Adukia et al., 2023; Zeng and Duquennois, 2026).

We identify fictional narratives as a distinct formative influence on political preferences. Unlike direct social experience, stories expose children indirectly to social groups, conflicts, and moral choices they may never encounter in their immediate environment. Unlike educational content, they engage children through characters, plots, identification, narrative transportation, and imagination rather than instruction. We provide one of the first causal studies showing that fictional stories encountered during childhood can shape political preferences in the long run.

Our findings also uncover new evidence on the historical origins of support for the Civil Rights Movement. Existing work emphasizes how political mobilization was shaped by changes in local racial composition (Bazzi et al., 2023; Calderon et al., 2023), military service (Ang and Chinoy, 2024), the leadership of social movements (Dippel and Heblich, 2021), and the political mobilization induced by civil rights reforms (Bernini et al., 2025). Historians have long proposed an additional channel based on cultural change (see Section 2 for details). According to these accounts, WWII catalyzed a broader shift in American discourse around race and equality, giving rise to progressive narratives that challenged prevailing racial hierarchies and initially circulated within progressive circles. These emerging ideas, which also motivated *OI*,

helped lay the cultural foundations for the Civil Rights Movement. We provide causal evidence for one such channel, showing how earlier cultural change contributed to the formation of the constituency that later supported mass mobilization around civil rights. This result also provides novel evidence on the long-run roots of political mobilization, an area for which causal evidence remains limited (Cantoni et al., 2024).

More broadly, our findings contribute to the political economy of mass media. A large literature shows that media can shape political and social attitudes, whether through news and propaganda (DellaVigna et al., 2014; Adena et al., 2015; Yanagizawa-Drott, 2014; Wang, 2021; Ang, 2023; Esposito et al., 2023) or entertainment (Olken, 2009; Chong and La Ferrara, 2009; Jensen and Oster, 2009; DellaVigna and La Ferrara, 2015; Durante et al., 2019). This literature largely studies how media influence audiences with already-formed preferences. In contrast, we study whether media content encountered during formative years can shape values and attitudes in the long run.

This distinction may be particularly important when content challenges prevailing norms. Among adults, content is often more persuasive when it resonates with existing cultural values (Michalopoulos and Rauh, 2024), while changing entrenched prejudice requires internalizing alternative social or moral values. This process involves both cognitive and emotional dimensions (e.g., Aboud, 2012). Existing attempts to use media to counter prejudice, typically studied among adults over short horizons, illustrate the difficulty. Paluck (2009) evaluates a radio drama in Rwanda and finds shifts in perceived social norms but little change in listeners' personal beliefs. Siddique et al. (2026) show that a documentary designed to humanize an ethnic minority increased prosocial behavior but did not shift stereotypes. Blouin and Mukand (2019) show that radio can reduce ethnic salience and increase interethnic trust in post-genocide Rwanda, albeit through government propaganda rather than fiction.

We show that fictional entertainment designed to challenge prejudice can have lasting effects when encountered during formative years. Compared with previous studies, our setting also provides unusually clean variation in narrative content: *OI* abruptly changed the stories told within an existing program while leaving its characters, format, broadcasting network, and existing audience largely unchanged. To our knowledge, this is the first study to exploit such a within-program narrative shift to identify the long-run effects of the stories themselves.

2 Historical background

The Civil Rights Movement marked one of the defining political transformations of the twentieth century. Although demands for racial equality had much deeper roots, racial

issues became central to national political divisions by the late 1940s when Southern Democrats opposed President Truman's civil rights agenda and broke away to form the segregationist States' Rights Democratic Party (Dixiecrats) under Strom Thurmond. At the time, racial prejudice, both diffuse and institutionalized, persisted throughout the country but took its most severe form through Jim Crow laws in the South.

Mass mobilization intensified in the 1950s and early 1960s. Civil rights leaders, such as Martin Luther King Jr., became prominent national advocates of racial integration. Their demands increasingly found support within the national Democratic Party, although civil rights legislation continued to receive support from Republicans and faced strong opposition from Southern Democrats. At the same time, explicit opposition to integration became nationally prominent, contributing to a broader realignment of partisan politics around racial issues (Schickler, 2016). Major federal legislative reforms followed in the 1960s, including the Civil Rights Act of 1964 and the Voting Rights Act of 1965. These did not, however, settle the political conflict over racial equality. Throughout the 1960s and 1970s, contestation took multiple forms, from segregationist mobilization and resistance to desegregation to civil rights protests, boycotts, and opposition to the Vietnam War, which became increasingly intertwined with demands for racial and social justice (Hall, 2005). The implementation, enforcement, and scope of these reforms remained contested well into the late twentieth century (Anderson, 2004).

A large literature has examined the political consequences of the Civil Rights Movement, including how mass mobilization shaped electoral politics and how racial attitudes contributed to the transformation of partisan alignments (e.g., Kuziemko and Washington, 2018). One potential antecedent of these changes, however, has received considerably less attention in the empirical literature: the cultural transformation in attitudes toward race that began during WWII. In the early twentieth century, popular culture largely reflected and reinforced the prevailing racial hierarchy. WWII helped disrupt this equilibrium. The contrast between American democratic ideals abroad and racial discrimination at home heightened attention to what contemporaries described as the "American Dilemma". By the end of the war, explicit defenses of white supremacy had begun to lose public legitimacy, particularly outside the South. A counter-narrative grounded in the 'American Creed"—civil liberties, equality of opportunity, and opposition to racial discrimination—spread within progressive intellectual, political, and cultural circles (Kellogg, 1979). Historians have argued that these postwar shifts helped provide part of the cultural foundation on which the Civil Rights Movement subsequently expanded (Hall, 2005).

2.1 Superman and *Operation Intolerance*

Popular culture became one channel through which the counter-narrative that emerged from WWII reached a broader audience. The postwar years saw the emergence of radio and film productions that explicitly confronted racial discrimination (Benshoff and Griffin, 2021). *OI*, introduced in 1946, was an unusually early and deliberate example of this transformation, bringing the emerging ideas about equality and tolerance into one of the country's most popular forms of children's entertainment.

In the 1940s, radio was the dominant mass medium in the United States. Hundreds of local stations were affiliated with four major national networks: National Broadcasting Company (NBC), Columbia Broadcasting System (CBS), American Broadcasting Company (ABC), and Mutual Broadcasting System (MBS). Affiliations were fluid, with stations frequently switching networks under agreements lasting 1–5 years and specifying program time slots and remuneration. Appendix A.1 documents listenership across networks in 1945–1947.

One of MBS's most successful productions was *The Adventures of Superman*, an adventure drama series featuring the DC Comics character. The show began airing regionally in 1940, was acquired by MBS in 1942, and distributed nationally. It regularly drew millions of listeners until 1949, when it first moved to the competing ABC network and was discontinued in 1951 (De Haven, 2010).

The program primarily targeted children and adolescents. It aired for 15 minutes five times a week during the after-school slot (5–6pm), when children's and adolescents' listenership was highest. During this slot, MBS productions were entirely targeted at this group, while adult listeners, mostly non-working women, tuned primarily into other networks, making direct adult exposure unlikely. Story arcs ranged from 15 minutes to over three hours across multiple episodes, with scripts written specifically for radio rather than being adapted from the comics. Before 1946, Superman's adventures combined wartime patriotism with the triumph of good over evil as he battled mad scientists, atomic weapons, and supernatural threats (Daniels, 1998).

In 1946, the series underwent a sudden and unprecedented narrative transformation. Conceived in late 1945, the producers of *The Adventures of Superman* reoriented the narrative of the show to promote a vision of a united American society embracing individuals regardless of race, religion, or background. Superman now confronted bigotry within American society in stories designed to make these themes accessible to children (Bowers, 2012). The initiative, known as *OI*, was described by its producers and sponsor as an experiment of interest to the radio industry.[3] The decision was

[3] In an article published on Broadcasting (1946, p. 75), W. B. Lewis, then radio director at Kenyon &

not pre-announced and was not widely advertised afterward, making the change largely unanticipated by the audience. We provide evidence in favor of such feature later in this section.

The producers began developing scripts in October 1945, and on April 15th, 1946, *OI* began airing on the MBS-affiliated stations that carried *The Adventures of Superman*. The first story arc, *The Hate Mongers Organization*, followed a white supremacist group's attempt to prevent the creation of *Unity House*, a community center "where children of every race, ethnic background, and spiritual belief can play and interact to learn that all people are the same" (*The Adventures of Superman*, episodes 1254–1278). After its positive reception, the creators produced *The Clan of the Fiery Cross*, in which Superman battled an implicit representation of the KKK, mocking its rituals, code words, and bigotry. Other notable story arcs include *George Latimer, Crooked Political Boss* (September 1946), in which Superman thwarts efforts to impose discriminatory hiring rules favoring white Protestants, and *Knights of the White Carnation* (February–March 1947), where he exposes a secret group plotting to expel foreign students from a school basketball team. Appendix A.3 provides detailed summaries.

The narratives of *OI* were explicitly designed to shape children's values and contained several features that psychological research associates with attitude and value formation. First, the stories presented clear moral contrasts, portraying advocates of justice and equality as heroes and members of hate groups as villains, consistent with social learning through role models (Bandura and Walters, 1977). Second, they humanized minority characters by depicting them as sympathetic and hardworking, consistent with mediated intergroup exposure and the parasocial contact hypothesis (Schiappa et al., 2005). Third, emotionally charged plots encouraged narrative transportation and identification with fictional characters, making their experiences more psychologically salient and potentially strengthening the narratives' influence (Green and Brock, 2000).

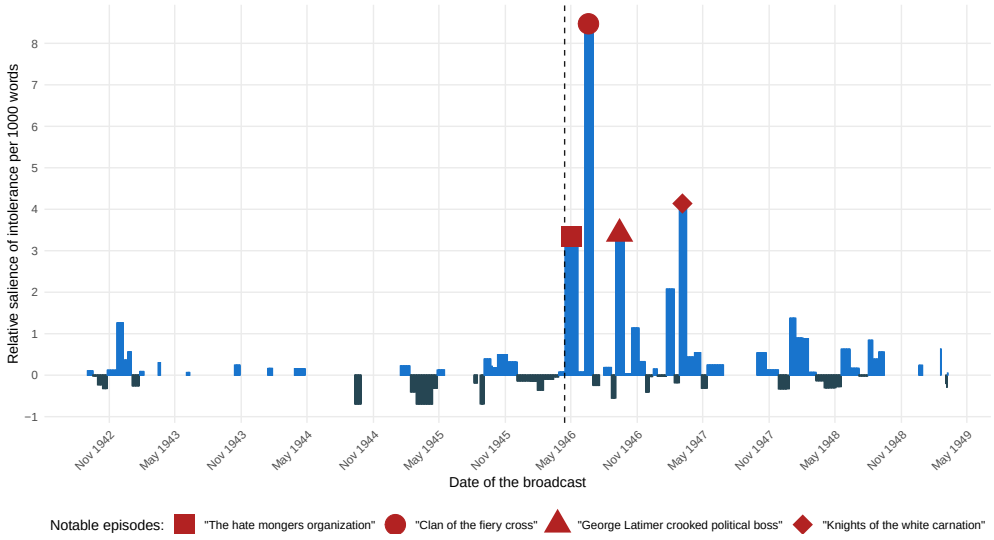
The content analysis, summarized in Figure 1, illustrates the scale of the narrative change after April 1946. The proportion of words related to intolerance rose more than threefold in the 12 months after *OI* began and nearly eightfold during the four main story arcs.[4] These patterns confirm a deliberate reorientation of the show's narrative. Section 5.2 further shows that the shift was unique to *The Adventures of Superman* rather than part of a broader change in youth radio programming. Most MBS program-

Eckhardt (the advertising agency of the show), explained that "Operation Intolerance had its beginning at a board meeting last October [1945] ... We brought the matter to the attention of the Kellogg Co. [the exclusive sponsor of the series throughout the MBS period] and recommended that the experiment—for experiment it is—be tried. W. H. Vanderploeg, president of Kellogg, concurred heartily."

[4]The analysis uses a bag-of-words approach with 50 keywords related to intolerance and tolerance, applied to 1,019 episode transcripts broadcast on MBS between 1942 and 1949. See Appendix A.3.

ming in 1946 was neutral or conservative, with *The Adventures of Superman* being the only production explicitly advancing progressive views on equality (Appendix A.1).

Figure 1: The change in narrative in *The Adventures of Superman*



Note. The figure shows, for each episode transcript, the share of keywords per 1,000 words related to tolerance or intolerance, relative to the pre-*OI* average. The sample includes all available MBS episodes from August 1942 to June 1949. Each bin represents a story arc and its width the arc's duration. Appendix A.3 provides methodological details.

The launch of *OI* coincided with the increased popularity of the program. In the following year, *The Adventures of Superman* reached an estimated 4.5 million listeners in its target demographic, becoming the most popular youth program on the air (Whiteside, 1947). Over this period, the series broadcast 65.25 hours of content, including 18 hours devoted to the four main story arcs. The large, loyal audience and sustained exposure made *OI* exceptional in scale relative to other efforts to reduce prejudice (see, e.g., Bertrand and Duflo, 2017). Section 5.1 provides further evidence on the listenership patterns of the show around the launch of *OI*.

Evidence indicates that the adult population had limited direct exposure to *OI*. The program's time slot and its focus on younger audiences made adults less likely to listen directly, although *OI* could have reached them indirectly through discussions or broader public discourse. Indeed, the initiative drew reactions from both supporters and opponents, underscoring the political salience of introducing highly progressive narratives into a popular children's program. President Harry S. Truman and former Vice President Henry A. Wallace publicly endorsed the initiative (Appendix A.3), while opponents mounted localized resistance. Historical accounts document an unsuccessful attempt by the Georgia KKK to organize a boycott of Kellogg's, the show's sponsor, as well as reported threats against the show's production and local radio stations (Wall, 2009). However, we find little evidence that these reactions diffused broadly through

contemporaneous media. In newspapers, an important source of information for adults at the time, coverage of *Superman* and themes related to *OI* did not increase discontinuously following the April 1946 narrative shift, nor did it increase differentially in areas where adults had greater exposure to the show (Appendix B.7). It is unlikely that television played a role, as only about 8,000 U.S. households owned television sets in 1946. These patterns are consistent with limited broader promotion of *OI* or public backlash. We cannot rule out exposure through interpersonal channels. Whether *OI* influenced adults in the long run remains an empirical question that we discuss in Section 5.

3 Data

We combine survey, historical, and archival data with geographic measures of radio coverage. Additional data are described in Section 5.

3.1 Exposure to *Operation Intolerance*

To measure individual exposure to *OI*, we calculate the geographic coverage of its broadcasts using a propagation model that predicts signal strength from each transmitter. We describe the main elements of this approach below, with additional methodological details in Appendix A. In 1946, broadcasters had limited knowledge of their signal reach when deploying infrastructure. The first U.S. ground conductivity map was published only in 1954, and accurate propagation models were infeasible prior to this.

Identification of radio stations. *OI* was broadcast in 1946 by the radio stations airing the show *The Adventures of Superman*—a subset of all MBS-affiliated stations (Section 2.1). Using historical records, we identify these stations in three steps. First, we digitize data from Radio Daily (1946), a broadcasting-industry reference guide listing stations, ownership, and technical specifications. This yields all 998 U.S. radio stations operating in 1946, including their transmitting antennas and broadcasting frequencies (measured in hertz), which determine the channels listeners could tune into. Second, we use programming and network-affiliation information from National Authority (1945) and Broadcasting (1947) to identify 217 stations participating in the advertising deal for *The Adventures of Superman* at the start of 1946. Because affiliations or programming may have changed between this point in time and the launch of *OI*, we cross-verify stations using schedules from local newspapers on newspapers.com, adding 20 stations and excluding 16. The final set includes 221 stations. These stations were geographically dispersed across Northern and Southern states and urban and non-urban counties, while network affiliations changed substantially over time (Appendix A.2).

Computation of exposure. We geo-locate the transmitters of the stations broadcasting *OI* and use their technical characteristics to compute radio coverage. In 1946, nearly all broadcasting relied on AM technology, used by 99.6% of stations in 1940 and 74% in 1950 (US Bureau of the Census, 1975).[5] AM transmissions propagate through two components: a ground wave, predominant during the day, and a sky wave, predominant at night.[6] Unlike FM signals, AM waves are less obstructed by terrain and can travel beyond the horizon, with their reach varying sharply between day and night (Reed and Sander, 1987). Consequently, radio coverage models focused solely on topography, such as the Longley-Rice Irregular Terrain Model, are not appropriate predictors of AM radio coverage (Gagliarducci et al., 2020).

To estimate radio coverage, we partnered with ATDI, a radio-engineering firm, which predicted signal strength for each station (i.e., the measurable quality of the received signal that determines the ability to hear a broadcast). Predictions are specific to AM transmissions and follow the United Nations' ITU modeling standards (ITU, 1995, 2007). The model incorporates transmitter characteristics (frequency, output power, antenna design), terrain and soil conductivity, and temporal variables, including the relative prevalence of ground and sky waves and solar radiation during the broadcast. The output consists of 400-meter resolution rasters of signal strength for each station across the continental U.S. We overlay each raster, retaining the strongest signal at each location, to construct the composite *OI* coverage map shown in panel A of Figure 2. We define county-level *exposure* as the share of the county population covered by the signal in 1946 (panel B). Average exposure is 13%, with a standard deviation of 0.30. In all estimations, exposure is standardized for ease of interpretation. Appendix B.6 discusses alternative measures of exposure.

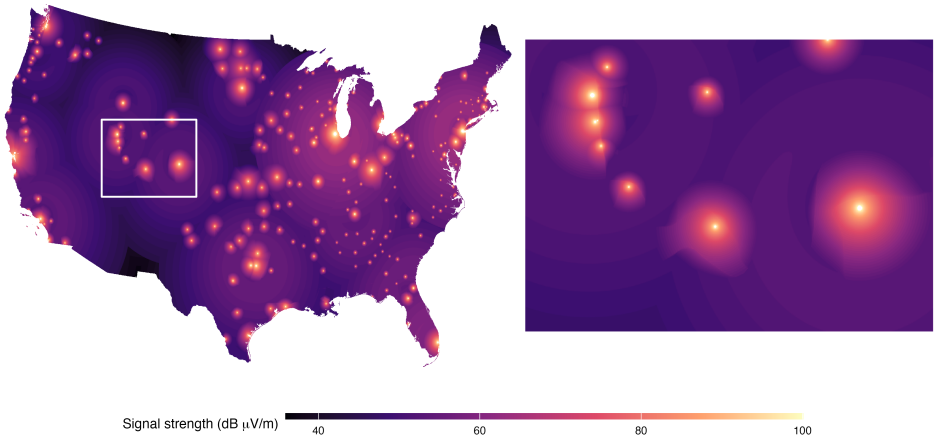
Validation of exposure. The ITU algorithm, developed from decades of empirical measurements, is an internationally validated standard routinely used for AM broadcast coordination and network planning worldwide (ITU, 2013). Nevertheless, in Appendix A.1, we conduct a context-specific empirical validation using contemporaneous listenership data. The only available county- and station-level source for the 1940s is BMB (1949), which reports the share of each county's population listening to different radio stations in 1949, three years after the launch of *OI*. We digitize this source to build the first comprehensive county-by-station dataset of mid-century radio listening. Matching

[5] AM was the dominant U.S. broadcasting technology from the 1920s to the 1970s. In 1960, 92% of all radio sets were AM-only, and FM listenership did not surpass AM until 1979.

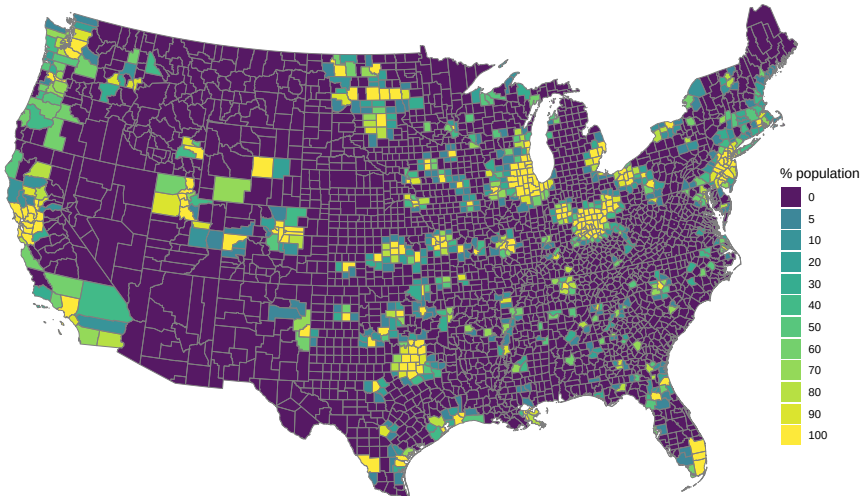
[6] Ground wave propagation depends on topography and soil conductivity (i.e., the ability of the Earth's surface to conduct radio waves) and is determined by moisture, soil composition, and mineral content. Sky wave propagation depends mainly on ionospheric refraction (i.e, the bending of radio waves in the ionosphere, which varies with solar radiation from day to day).

Figure 2: Exposure to *OI*

A. Signal strength of *OI* radio stations



B. Share of population covered by the broadcast in 1946



Note. Panel A displays *OI* signal strength at a 400-meter resolution in $dB\mu V/m$ (higher values indicate stronger signals); the right figure enlarges the area indicated by the rectangle. Panel B displays the share of each county's population covered in 1946. We define coverage as signal strength $\geq 66, dB\mu V/m$, a standard threshold for intelligible reception (ETSI, 2021). Population in each grid cell is obtained from the 1940 HYDE dataset (Klein Goldewijk et al., 2017) and interpolated to the same resolution. County exposure is the population-weighted share of covered cells. Section 3.1 describes the signal strength computation.

realized listenership to predicted signal strength reveals a tight correspondence between modeled coverage and the probability of having listeners, providing strong external validation of our exposure measure. We further validate exposure using local newspapers. Superman was more salient in areas with higher predicted exposure (Appendix B.7).

3.2 Political attitudes and behavior

We use the American National Election Studies (ANES, 2021), a biennial public-opinion survey of nationally representative samples of the voting-age population. We use all waves from 1964 to 1980, 18–34 years after the launch of *OI*. These waves consistently

include questions on civil rights, reflecting the issue's high political salience during this period. Earlier and later waves offer fewer such measures.

We focus on three sets of variables. Appendix B.1 provides definitions, descriptive statistics, and availability across waves. All attitudinal variables are standardized into z-scores in the results tables. First, we examine *support for civil rights* using eight questions across survey rounds (see Appendix B.6 for partial correlations). Six capture support for civil rights policies: desegregation, racially-integrated schools, desegregation busing, representation of Black Americans in society and politics, affirmative action, and civil rights legislation. Two measure feelings toward Black Americans and Southerners, the latter were widely associated with resistance to desegregation and civil-rights reform (Bartley and Graham, 2019). Feelings toward these groups are measured using the 1–100 Feeling Thermometer, with higher values indicating more favorable feelings. Second, we assess *interracial assimilation* using a self-reported measure of interracial friendships, complemented by census data on interracial marriage. Finally, we analyze *broader political attitudes and behavior* along three dimensions. We assess political support for leaders supporting or fighting against civil rights, political engagement, and attitudes towards institutions.

We merge survey responses with *OI* exposure using each respondent's county of residence and restrict the sample to individuals residing in the state in which they grew up, reducing measurement error from migration (see Kearney and Levine, 2019 for empirical evidence supporting this approach). We find no evidence of selective migration in our sample, and internal migrants are similar in demographic and socioeconomic characteristics.[7] Section 5.2 further discusses this restriction.

4 Empirical approach

This paper estimates the causal effect of exposure to progressive stories during childhood on long-run outcomes. For this purpose, we do not estimate the effect of *The Adventures of Superman* but rather isolate the effect of the sudden and unanticipated shift in the storylines within the same program. We exploit two sources of variation: differences across locations in access to stations broadcasting the program and differences across birth cohorts. Specifically, we implement a cohort-based DiD design using repeated cross-sections of individuals observed as adults between 1964 and 1980, 18 to 34 years after *OI*. Our design allows us to compare adult outcomes within cohorts and

[7]The samples are comparable in terms of age (46.4 in the main sample versus 44.3 among internal migrants), gender (56.3% female in both), race (88.9% white versus 84.6%), education (63.1% with secondary or higher education versus 69.4%), and marital status (67.3% married versus 68.5%).

across areas with different levels of potential exposure to *OI* in 1946 and examine how the exposure gradient—the relationship between potential exposure in 1946 and later outcomes within a given cohort—varies across age cohorts (relative to 1946).

Age cohorts vary in their likelihood of having listened to and meaningfully engaged with the new storylines at the time of the narrative shift in 1946. The cohorts most likely to have listened were school-age children and adolescents in 1946. We refer to them as the *target cohorts*. They were the program's primary audience and the group targeted by *OI* (see Section 2). In addition, political socialization is highest in this group. Children's ability to understand narrative content with moral lessons develops from the early school years (Burris and Brown, 2014), while political socialization is greatest approximately between the ages of 7 and 17, with a peak during adolescence (Raabe and Beelmann, 2011; Bartels and Jackman, 2014).

Younger cohorts were either not yet born when the show aired or were too young to understand the episodes. Any effects among these cohorts would therefore plausibly arise indirectly, for example, through older siblings, parents, or broader household and community spillovers. Older cohorts were less likely to listen to a radio program designed for younger audiences and broadcast in an after-school time slot. Some may nevertheless have been exposed incidentally through household listening or because the program also attracted listeners outside its core audience. In addition, they had moved beyond the ages generally considered most formative for political socialization.

Motivated by this cohort structure, our empirical objective is to estimate how the long-run relationship between potential exposure to *OI* and later outcomes varies across cohorts of different ages in 1946. In particular, we test whether this relationship is stronger among the target cohorts than among younger and older cohorts. We estimate the following specification:

$$Y_{ihct} = \sum_{\tau=\underline{h}}^{\bar{h}} \gamma_\tau D_\tau \cdot R_c + \mu_c + \mu_h + \mu_{a_{it}} + \mu_{t\times state} + (X_{ihct} \times D_h) + (X_{ihct} \times S_t) + \epsilon_{ihct}, \quad (1)$$

where Y_{ihct} is the outcome for individual i in cohort h (defined relative to 1946), residing in county c and interviewed in the year t (ranging from 1964 to 1980). D_τ is an indicator of whether individual i belongs to cohort $\tau = h$, with cohorts defined according to age in 1946 and grouped into 6-year intervals, while R_c denotes potential exposure to *OI*, measured as the standardized share of county c's population covered by the broadcast signal in 1946. Importantly, R_c is specific to 1946, and while it is correlated with exposure in other years, it is distinct from it (see Section 3.1). The remaining terms include a set of fixed effects (FEs) and interaction terms, which we discuss below, and

an idiosyncratic error assumed to be clustered at the county level.

The omitted category is the cohort aged 19–24 in 1946, the first cohort outside the program's primary target age. Members of this cohort were old enough not to be directly targeted by *OI*, but young enough to have been part of the target audience of *The Adventures of Superman* before the change in narrative content. Thus, persistent effects of exposure to the program or its broadcast network should also be reflected in this cohort, making it a natural comparison group for isolating the effect of the new narratives. Accordingly, each coefficient γ_τ measures the difference in the exposure gradient between cohort τ and the 19–24 reference cohort. Our approach does not impose that the effects of *OI* are confined to the target cohorts: coefficients for younger and older cohorts reveal whether the relationship between potential exposure in 1946 and later outcomes extends to individuals beyond the group targeted by *OI*. Because the specification indexes cohorts by age in 1946 rather than calendar time relative to treatment, these coefficients should not be interpreted as conventional pre-intervention event-study coefficients. Finally, the estimates have an intention-to-treat interpretation because R_c measures potential exposure rather than actual listenership; Section 5.1 examines the relationship between these two dimensions.

FEs and interaction terms in equation (1) are designed to absorb systematic differences across counties, cohorts, and survey periods, as well as observable sources of differential cohort and time patterns. The term μ_c captures county FEs, accounting for time-invariant county characteristics, such as pre-existing determinants of the radio network and its reach. Importantly, μ_c absorbs overall differences in R_c across counties but does not absorb the identifying variation in our design, since treatment varies within counties across birth cohorts through the interaction $D_h \times R_c$, nor does it restrict identification to counties observed in multiple survey waves. In Appendix B.4, we show results from dropping specific states or survey waves from the estimating sample.

The term μ_h captures cohort FEs, absorbing differences across the six-year cohorts defined by age in 1946, including factors associated with being born before, during, or after WWII.[8] We additionally include age-at-interview FEs, $\mu_{a_{it}}$, which absorb systematic differences across exact ages within these broader cohort groups, including differences associated with life-cycle stage. Because individuals from the same cohort are observed at different times, we also include state-by-survey-year FEs, $\mu_{t\times state}$, to absorb survey-specific shocks common across cohorts within a state, such as shifts in public opinion, legal or institutional changes, and election cycles.

[8]Results are robust to replacing this term with cohort-by-survey-year FEs (Appendix B.6). We do not include county-by-survey-year FEs in our baseline specification because the repeated cross-sections of the ANES contain relatively few respondents within a given county and survey wave.

The repeated cross-sectional structure provides an advantage for our design. Cohorts are defined by age in 1946 and therefore remain fixed across survey waves, while individuals from a given cohort are observed at different ages. For example, members of the target cohorts, aged 7–18 in 1946, are aged 25–36 when first observed in 1964 and 41–52 in 1980. We can therefore estimate the differential exposure gradient while accounting separately for systematic differences associated with age at interview and the survey period in which individuals are observed, and assess whether the estimated effects persist as the target cohorts age.

The terms $X_{ihct} \times D_h$ and $X_{ihct} \times S_t$ interact observable characteristics X_{ihct} with cohort (D_h) or survey-year indicators (S_t), allowing these characteristics to flexibly control for differential cohort and time patterns. X_{ihct} includes individual characteristics, such as age at interview and sex; pre-1946 county racial composition and urbanization; and technical determinants of broadcast reception associated with transmitter placement and intended market reach.[9] Estimated effects are stable in more parsimonious specifications that omit these interactions and in richer specifications that introduce additional geographic and individual-level controls (Appendix B.6).

To summarize the main effects, we complement individual cohort-based estimates with summary estimates of the pooled gradient difference in the target cohorts (relative to alternative comparison groups). We estimate the following version of equation (1):

$$Y_{ihc,t} = \gamma\, \text{Target}_h \cdot R_c + \mu_c + \mu_h + \mu_{a_{it}} + \mu_{t \times state} + (X_{ihct} \times D_h) + (X_{ihct} \times S_t) + \epsilon_{ihct}, \quad (2)$$

where Target_h is an indicator variable equal to one if the respondent is part of the target cohorts, defined by those aged 7–18 in 1946. The remaining terms are unchanged. Our baseline specification uses the full sample, thus comparing the exposure gradient of target cohorts relative to non-target cohorts. We then restrict the comparison group to older cohorts, avoiding contamination from younger cohorts that may have been indirectly exposed through reruns, cultural diffusion, or family transmission. Because coefficients are stable across older cohorts (Appendix B.3), we use all older cohorts rather than restrict the comparison to those immediately older than the target cohorts.

Identifying assumptions and validity checks. Identification requires that, absent *OI*, the relationship between potential exposure to the stations broadcasting the show in 1946 and later outcomes would not have differed systematically across cohorts in a way

[9] We refer to the latter as *propagation controls* (see Appendix B.2). In addition, due to the limited within-county variation in an individual's race (i.e., many counties contain only white respondents), combined with the large set of FEs, we control for race using the county's share of the white population in 1940. Results are robust to including an extended list of individual-level controls, including race, although this reduces the sample size (Appendix B.6).

that coincides with the program's target ages. This does not require high- and low-exposure counties to be otherwise identical, as our fixed effects and interactions absorb systematic differences across locations, cohorts, ages, and survey years. Instead, identification could be threatened if an unobserved factor correlated with R_c differentially affects the target cohorts. We assess this possibility through complementary tests.

First, Appendix B.2 assesses whether R_c is conditionally balanced with respect to a broad set of county characteristics measured before 1946. This is a standard diagnostic used in the media literature to assess whether, conditional on the geographic and technical determinants of signal reception, residual variation in exposure is systematically related to predetermined characteristics (e.g., Yanagizawa-Drott, 2014). While conditional balance is not required for identification in our cohort-based DiD, since county FEs absorb all time-invariant differences across counties (including the main effect of R_c), it helps reinforce our identifying assumption. Substantial pre-existing differences between high- and low-exposure areas could signal the possibility that those differences also produced differential patterns across cohorts independently of *OI*.

We find a high degree of conditional balance. Of the 25 pre-broadcast characteristics considered, only one—the share of the county population living in a city—remains imbalanced conditional on a basic set of controls. This is similar to what is documented by Wang (2021), due to the concentration of radio infrastructure around urban markets. However, in our setting, other measures of urbanization, such as the share of urban land, are conditionally balanced, as are correlates including agricultural prevalence and educational attainment. While the time-invariant level of the share of the county population living in a city is also absorbed by county FEs in equation (1), in our main specification, we additionally interact this variable with cohort and survey-year indicators, allowing its association with outcomes to vary flexibly across both dimensions. Adding or removing these controls has no effect on our estimates (Appendix B.6). Section 5.2 discusses additional tests to rule out the confounding effect of urbanization or general access to contemporaneous radio broadcasting.

Second, Appendix B.6 examines whether the estimates depend on particular modeling or measurement choices. We progressively vary the set of FEs and controls included in equation (1), including specifications with increasingly flexible controls for geographic characteristics, radio infrastructure, and differential cohort and time patterns. In addition, we vary the construction of R_c, using alternative measures and thresholds for signal reception. Stability across these specifications provides evidence that the estimated cohort pattern is not driven by our modeling or measurement choices.

Third, Appendix B.3 examines the behavior of the exposure gradient among older cohorts relative to the omitted cohort (aged 19–24 in 1946). Because older cohorts were

unlikely to have listened to *OI* in 1946, systematic differences would raise the concern that R_c captures broader cohort-specific differences between high- and low-reception areas. We implement two complementary diagnostics. First, an unrestricted cohort-stability test assesses whether the exposure coefficients for all older cohorts are jointly equal to zero relative to the omitted cohort. Second, using only the omitted and older cohorts, a linear cohort-gradient test replaces the cohort-specific exposure interactions with a single interaction between exposure and a linear six-year cohort score and tests whether its coefficient is zero. The first test is sensitive to arbitrary departures from stability, whereas the second has greater power against a smooth cohort pattern. Overall, we find no evidence of systematic differences among older cohorts. Given the number of outcomes examined, we do not interpret isolated rejections as evidence against the design, particularly for outcomes available only in specific survey waves and therefore based on smaller cohort samples. We discuss these cases in Section 5, while our main results are robust to dropping these variables (Appendix B.3).

5 Results

5.1 From exposure to listenership

We begin by quantifying how higher potential exposure to the radio signals of stations broadcasting *OI* translates into higher listenership. This analysis provides an empirical first stage since our exposure measure captures the probability of listening rather than actual listenership. Because the listenership data discussed in Section 3.1 do not identify specific time slots and refer to 1949, we digitized historical records produced by Crossley's Cooperative Analysis of Broadcasting (CAB; see Appendix A.1). In the 1940s, CAB published detailed listenership data for a representative sample of households living in areas with populations of 50,000 or more that had access to a telephone, a more restricted sample than BMB (1949).

We glean data from the *City Report* (CAB, 1946a), which provides bi-monthly, city-specific listenership shares by station for five time slots (9am–12noon, 12–3pm, 3–6pm, 6–8pm, and 8–10:30pm). Matching stations to those that broadcast *OI* yields a panel of listenership shares by city and time slot from July 1945 to February 1946, the latest available report before CAB ceased operations in mid-1946.

We match each observation to our exposure measure, R_c, using the county of the city and regress listenership on exposure, controlling for state and time FEs and for the same county characteristics included as controls in equation (1). This specification differs from our main cohort-based empirical specification, which we use in the following

sections. Here, our objective is simply to establish whether the residual variation in radio exposure, conditional on observable county characteristics, predicts actual listenership. This approach is standard in the media literature (see, e.g., Yanagizawa-Drott, 2014) and is particularly informative in our setting, given the conditional balance of R_c across predetermined county characteristics (Appendix B.2).

Columns (1)–(3) of Table 1 present the results. Column (1) covers all time slots, column (2) restricts the sample to the slot including *The Adventures of Superman* (3–6pm), and column (3) includes all other slots. The average listenership share was 9.6% and remained stable throughout the day. The 3–6pm period likely underestimates listenership during the show itself, since, before 5pm, most children and adolescents were not yet tuned in. The effect of exposure on listenership is consistent across sub-samples, with a one-standard-deviation increase in exposure raising the overall listenership share by 2.6 percentage points, a sizable 27% increase over the mean. Using county-level BMB (1949) data covering the full U.S. yields consistent estimates (Appendix A.1).

We next examine whether the launch of *OI* increased the listenership of *The Adventures of Superman*. Because data from the City Report are unavailable after February 1946, we digitized complementary information from the *Program Report* (CAB, 1946b), which summarizes the aggregate listenership of major national networks, rather than individual stations, during each 15-minute slot of the day. When programs were broadcast during the same slot on the same national network, such as *The Adventures of Superman*, this report allows for the recovery of the time series of listenership at a bi-weekly frequency.

Using all relevant 15-minute slots from 3pm to 6pm on weekdays (12 slots), we build a bi-weekly panel of slot-specific network listenership covering February 1945–July 1946. To estimate whether the show's audience increased after the narrative shift, we implement a DiD specification comparing the evolution of *The Adventures of Superman* to other programs before and after April 1946, controlling for time and slot FEs.

Results are presented in columns (4)–(5). Column (4) includes all 3–6pm slots, while column (5) focuses on the key 5–6pm period. Following the launch of *OI*, the series saw a 4.3–4.7 percentage point rise in listenership (an 18–19% increase over the sample mean). These results underscore the strong appeal of the new episodes among listeners. Appendix Figure A3 shows that, following the narrative shift, *The Adventures of Superman* became the most popular program for children and adolescents, without drawing listeners away from adult-oriented programs.

These results are consistent with historical accounts and reinforce the show's cultural imprint among children and adolescents. To assess the potential reach of *OI* specifically in the target cohort, we combine county-level census data on population by age and

Table 1: Potential exposure to *OI* and listenership

	Share of radio set users listening to ...				
	Radio stations of *OI*			A specific network during a 15-minute slot	
Sub-sample:	9am–10.30pm	3–6pm	9am–10.30pm (excl. 3–6pm)	3–6pm	5–6pm
	(1)	(2)	(3)	(4)	(5)
Exposure	0.026***	0.026***	0.026***		
	(0.007)	(0.006)	(0.008)		
Superman × post				0.047***	0.043***
				(0.005)	(0.007)
Dependent variable mean	0.096	0.096	0.096	0.250	0.236
R^2	0.600	0.677	0.616	0.939	0.937
Observations	1,725	345	1,380	1,148	426
Level of observation	City	City	City	Network	Network

Note. Estimates in columns (1)–(3) are based on equation (4), but with repeated observations over time, thus controlling for time FEs. The dependent variable is the city-level share of radio set users that was listening to any of the radio stations that broadcast *OI* in 1946 during a specific time slot and period of reporting. *Exposure* measures the share of the county population covered by the radio signal of stations that broadcast *OI* in 1946. Estimates in columns (4)–(5) are based on a difference-in-differences specification comparing over time the listenership of specific networks during 15-minute slots, controlling for time and slot FEs. The dependent variable is the network-level share of radio set users listening to the radio during a specific 15-minute slot and period of reporting. *Superman* is an indicator variable equal to 1 if listenership refers to *The Adventures of Superman*, and 0 if it refers to other programs. *Post* is an indicator variable set to 1 for observations capturing listenership on or after the launch of *OI* in April 1946. Standard errors are reported in parentheses and are clustered at city level in columns (1)–(3) and at network-by-slot level (adjusting for the small number of clusters) in columns (4)–(5) (* $p < 0.1$, ** $p<0.05$, *** $p < 0.01$). In columns (1)–(3), observations cover the period July 1945–February 1946, while in columns (4)–(5), they cover February 1945–July 1946.

radio ownership from Haines (2010) with our measure of radio coverage. We estimate that approximately 10.81 million individuals aged 7–18 in 1946 could have listened to the show, representing about 45% of this age group nationwide. Contemporary sources report an audience of 4.5 million, roughly 42% of the estimated population of potential listeners in this age group. Direct listenership among adults was limited (Section 2.1).

These figures are important for interpreting our subsequent estimates, which capture reduced-form effects of potential exposure rather than individual listenership. The show's substantial audience, together with the limited media alternatives available at the time, suggests that radio coverage could translate into widespread exposure. Moreover, radio listening was often a social activity (Hooper Inc., 1949), which was not directly captured by the listenership statistics used for the estimates in Table 1. In addition, the show's messages could diffuse beyond direct listeners through households, siblings, and peers. It is therefore likely that, within reception range, a much larger share than 42% of potential listeners ended up being directly or indirectly exposed to *OI* at least once in 1946. Consequently, our estimates may capture both direct exposure and broader social transmission, making them informative about the population-level consequences of *OI*.

5.2 Support for civil rights

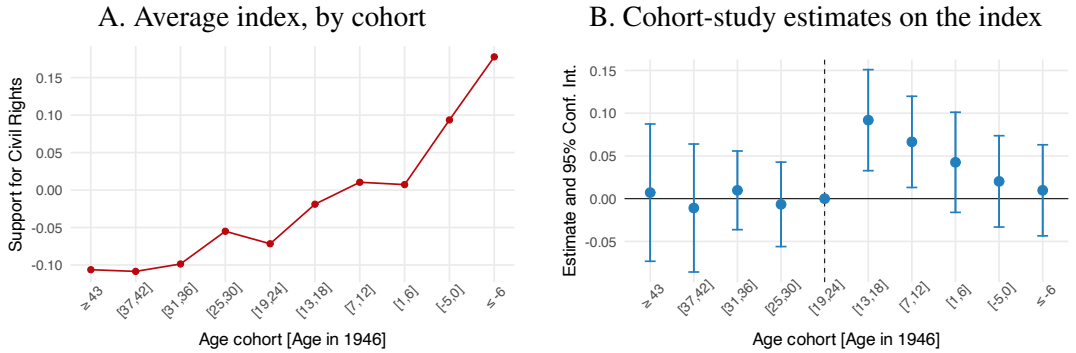
We investigate whether exposure to *OI* in 1946 impacted support for civil rights at the time of the reforms. We first focus on a single indicator, the *Support for Civil Rights Index*, which combines several survey questions related to support for policies guaranteeing civil rights and racial attitudes (see Section 3.2). Following Kling et al. (2007), the index combines responses by standardizing individual indicators and averaging the z-scores across all available variables for each respondent. This approach facilitates interpretation and addresses multiple hypothesis testing. Appendix B.6 reports an alternative aggregation method that accommodates incomplete data structures. We discuss effect magnitudes separately in Section 5.5.

On average, younger cohorts are more supportive of civil rights, with a difference of 0.28 index units between the youngest and oldest cohorts (panel A of Figure 3). Panel B presents cohort-level estimates based on equation (1) for the index. Relative to the excluded cohort (aged 19–24 in 1946), we detect a significant difference in the exposure gradient of the cohorts that constituted the program's primary audience. For the age cohorts 7–12 and 13–18, a one-standard-deviation increase in potential exposure—equivalent to a 30-percentage-point increase in the share of the county population covered by the broadcast signal—leads to gains in support of 0.07 (significant at the 5% level) and 0.09 units (significant at the 1% level), respectively. The strongest effects appear among individuals during adolescence, in line with effects being concentrated when political socialization is highest (Section 2.1).

We observe no significant differences relative to the excluded cohort for cohorts that were either too young to meaningfully engage with the program's content, not yet born, or too old at the time of the broadcast. Formal tests of the stability of the exposure gradient among older cohorts strongly support this pattern (Appendix B.3). Among these cohorts, the exposure coefficients for the index are jointly indistinguishable from zero, and there is no evidence of a linear exposure-specific cohort gradient. Because the excluded cohort could itself have been exposed to *The Adventures of Superman* before the 1946 narrative shift, this age-specific pattern weighs against persistent exposure to the show or its broadcast network as an explanation for the estimated effects and instead points to the change in narrative content experienced by the target cohorts.

Table 2 provides summary estimates of the differential effect among target cohorts using equation (2) for the full sample (panel A) and when restricting the sample to the target and older cohorts (panel B). Column (1) focuses on the index, while columns (2)–(9) provide estimates for each individual indicator comprising the index, reported as standardized z-scores for comparability. For these variables, cohort-study estimates

Figure 3: Support for civil rights



Note. Panel A plots the *Support for Civil Rights Index*, averaged by age cohort in 1946. The index aggregates z-scores from individual questions that capture racial attitudes and support for civil rights. Descriptive statistics and temporal coverage of the variables comprising the index are presented in Appendix B.1. Panel B displays the coefficients from equation (1) of the interaction term between the age cohort in 1946 and potential exposure to *OI* in the county. Each coefficient measure how the exposure gradient—the relationship between potential exposure in 1946 and the index within a given cohort—differs from that of individuals who were aged 19–24 at the time of *OI* (indicated by the vertical line). The specification includes the set of controls and FEs described in Section 4. Error bars indicate 95% confidence intervals, computed using clustered standard errors at the county level. Formal tests of the stability of coefficient for cohorts older than 19–24 are reported in Appendix B.3. Additional details about the variables are available in Appendix B.1.

using equation (1) are presented in Appendix B.3. Similar to the analysis for the index, the older-cohort coefficients are stable across nearly all index components relative to the excluded cohort. The only exception is the variable capturing favorable feelings towards Southerners, which fails both stability tests due to a single outlier cohort. Excluding this variable from the index leaves the main results unchanged (Appendix B.3).

For target cohorts, a one-standard-deviation increase in exposure raises the index by 0.07 units relative to all other cohorts and by 0.08 units relative to older cohorts. This result suggests that spillover effects on younger cohorts are minor, as the estimate is only slightly larger in panel B. The results are robust to alternative specifications, measures of exposure, and inference assumptions (Appendix B.6). In addition, the results are robust to recomputing the index by removing individual variables or by estimating the effects while excluding specific states or survey waves from the sample, indicating that the main results are not driven by individual indicators, nor by individual states or survey waves (Appendices B.3–B.4).

The increase in the index is not driven by a single indicator but by a general change in support for civil rights policies and racial equality. Compared to non-target cohorts (panel A), a one-standard-deviation increase in exposure among target cohorts significantly increases support for desegregation (0.07 standard deviations), desegregation busing (0.12), Black representation (0.09), affirmative action (0.11), and civil rights legislation (0.06). The estimate for integrated schools is also positive but not statistically significant. Exposure also increases favorable feelings toward the Black population (0.05), while reducing favorable feelings toward Southerners (0.11).

In Appendix B.8, we examine whether these estimates reflect changes in education or a broader shift in attitudes towards other groups. Among target cohorts, we do not find any detectable difference in educational attainment or college attendance, nor in attitudes toward groups representing economic inequality, feminist movements, or other whites. Attitudes toward religious minorities warrant a different interpretation. Although *OI* also condemned antisemitism and anti-Catholic discrimination, we find no effect on favorable feelings toward Jews or Catholics. Religious prejudice became less politically salient after WWII as these groups became increasingly integrated into American society (e.g., Dinnerstein, 1995), while racial equality remained central to political conflict in the 1960s and 1970s. Moreover, ANES provides only general feeling thermometers toward Jews and Catholics, rather than measures of the specific forms of discrimination targeted by the stories. The narrative shift was also less novel with respect to antisemitism, as Superman had previously confronted Nazi villains, albeit as foreign enemies rather than as expressions of domestic antisemitism (Daniels, 1998). We therefore interpret these null results as showing no detectable persistent effect on the available measures of religious attitudes, rather than no effect on religious prejudice.

Alternative mechanisms. Our approach identifies the differential effect of exposure to *OI* in 1946. Table 3 presents a variety of sample exclusions and placebo checks that address potential threats to identification and test for alternative channels. We present only estimates using all cohorts. Results are robust to excluding younger cohorts.

The first concern is that potential exposure to *OI* may capture unobservable cultural dynamics associated with urbanization or the geographical targeting of radio markets that differentially affect the target cohorts. All of our specifications control for county FEs, which account for time-invariant characteristics that may have influenced antenna placement, and the results are robust to flexible controls for both dimensions (Section 4 and Appendix B.6). Nevertheless, in columns (1)–(4), we estimate equation (2) on restricted samples excluding large urban areas, counties hosting the show's transmitters, and counties within 10 or 50 km of a transmitter. The estimates remain stable, ruling out urban-specific changes or strategic antenna placement as drivers of our results.

The second concern is that our estimates reflect broader trends in radio access or content. In columns (5)–(8), we conduct four placebo checks using equation (2) and controlling for interactions between the target cohort indicator and potential exposure to alternative contemporaneous radio content, constructed as in Section 3.1. In columns (5)–(6), we focus on exposure to the overall radio network and to any commercial stations that did not broadcast *OI*. In column (7), we control for exposure to *The Lone Ranger*, an ABC series and one of Superman's main competitors, which did not undergo a narrative shift in 1946 (Appendix A.3). In column (8), we add exposure to

stations outside the 1946 *OI* network that were affiliated with MBS by 1960, testing whether geographic network selection or later MBS programming accounts for shifts toward more progressive attitudes among the target cohorts. Across all four specifications, the estimated effects remain robust, and none of the coefficients on alternative radio exposure is statistically different from zero.

The third threat concerns whether potential exposure captures the direct experience of listening to *OI* or other unrelated experiences. In column (9), we estimate equation (2) using the sample of internal migrants (i.e., those who grew up in a different state compared to the location of the interview), which is excluded from our main analysis (see Section 3.2). For these respondents, exposure is assigned using the county of the interview rather than their childhood county. Because the survey does not identify the respondents' exact county of residence during childhood, actual childhood exposure to *OI* is likely different from the exposure matched with the county of the interview. In this sample, we find no statistically significant difference between target and non-target cohorts, reinforcing a mechanism based on direct exposure to the show.

Interracial assimilation. We analyze whether the shifts in favor of civil rights among the target cohorts affected social interactions. We begin by focusing on interracial friendship, measured by an indicator variable for whether respondents report having friends of a different race. Friendly relations across racial lines were explicitly emphasized in several *OI* scripts (Section 2.1). Panel A of Figure 4 shows the average share of interracial friendships across cohorts. In line with the broader process of racial assimilation over the 20$^{\text{th}}$ century, younger cohorts report substantially more interracial friendships: among respondents in the oldest cohort, 26% report interracial friendships, while among those in the youngest cohort, the share increases to 64%.

Panel B plots cohort-specific estimates from equation (1) for interracial friendship. Relative to the omitted cohort, the exposure gradient is significantly more positive only for individuals aged 7–12 in 1946: a one-standard-deviation increase in exposure leads to a 6-percentage-point greater probability of having a friend of a different race, significant at the 5% level. We find no significant differential gradients for the remaining cohorts, and formal tests confirm the stability of the older-cohort coefficients relative to the excluded cohort (Appendix B.3). Column (1) of Table 4 summarizes the effect for the target cohorts using equation (2). As in the previous section, Panel A uses all available cohorts, while Panel B restricts the comparison group to older cohorts. The exposure gradient for the target cohorts is 2.9 percentage points larger than for all non-target cohorts and 3.8 percentage points larger than for older cohorts only.

We complement these results with a more intensive form of assimilation: interracial marriage. For this outcome, we collect individual-level data from the 5% sample of the

Table 2: Cohort-study summary estimates on support for civil rights

	Support for Civil Rights Index	**Support for civil rights reforms**						**Favorable towards...**	
		Desegregation	Integrated schools	Desegregation busing	Black representation	Affirmative action	Civil rights legislation	Black population	Southerners
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
A. Full sample									
Target × Exposure	0.066***	0.069**	0.014	0.118**	0.086**	0.105***	0.057*	0.046*	-0.113***
	(0.015)	(0.028)	(0.030)	(0.054)	(0.037)	(0.033)	(0.030)	(0.027)	(0.032)
R^2	0.204	0.201	0.199	0.194	0.214	0.158	0.214	0.149	0.329
Observations	10,364	6,836	5,872	4,734	3,470	6,377	7,874	8,145	5,207
B. Restricted sample									
Target × Exposure	0.076***	0.082**	0.009	0.045	0.158***	0.139***	0.078**	0.056*	-0.088**
	(0.018)	(0.035)	(0.035)	(0.047)	(0.055)	(0.040)	(0.034)	(0.032)	(0.040)
R^2	0.216	0.228	0.229	0.239	0.273	0.205	0.221	0.171	0.365
Observations	6,895	4,509	4,045	2,778	2,122	3,726	5,408	5,677	3,496

Note. Estimates are based on equation (2) using the full sample (panel A) and restricting the sample to target and older cohorts, aged 7–18 and 19+ in 1946, respectively (panel B). All specifications include the set of controls and FEs described in Section 4. *Target* is an indicator variable equal to 1 for respondents who were 7–18 years old in 1946. *Exposure* denotes the share of the population in the county covered by the radio signal of *OI* in 1946. The reported coefficient on *Target* × *Exposure* measures the differential effect of a one-standard-deviation increase in exposure for the target cohorts relative to all other cohorts in the sample. Standard errors, clustered by county, are reported in parentheses (* $p < 0.1$, ** $p<0.05$, *** $p < 0.01$). The dependent variables are defined as follows: (1) index that aggregates z-scores from individual questions capturing support for civil rights, reported in columns (2)–(9); (2) preference regarding racial desegregation, ranging from 1 (strict segregation) to 3 (desegregation); (3) indicator variable equal to 1 if the respondent believes that the government should ensure racially-integrated schools; (4) beliefs over whether the racial integration of schools justifies busing children to schools out of their own neighborhoods, ranging from 1 (keep children in neighborhood schools) to 7 (busing to achieve integration); (5) beliefs about the influence of Blacks in American life and politics, ranging from 1 (too much influence) to 3 (too little influence); (6) beliefs about governmental efforts to improve the position of Blacks, ranging from 1 (Blacks should help themselves) to 7 (Government should help Blacks); (7) views on the pace of civil rights progress, ranging from 1 (too fast) to 3 (too slow); (8) feelings towards the Black population, ranging from 1 (unfavorable) to 100 (favorable); (9) feelings towards Southerners, ranging from 1 (unfavorable) to 100 (favorable). Descriptive statistics and temporal coverage of the variables are presented in Appendix B.1.

Table 3: Cohort-study summary estimates on support for civil rights — robustness to sample selection and placebo tests

	Dependent variable: Support for Civil Rights Index								
	Urbanization and local radio markets				**Exposure to other radio content**				**Unexposed**
	Exclude cities from the sample	Include in the sample only counties...			Add controls for ...				Restrict to sample of internal migrants
		Without transmitters	Further than $10km$ from a transmitter	Further than $50km$ from a transmitter	Any radio coverage	Other radio coverage	*The Lone Ranger*	MBS post-1946	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Target × Exposure	0.064***	0.071***	0.067***	0.075**	0.059***	0.064***	0.057***	0.062***	0.007
	(0.018)	(0.019)	(0.017)	(0.032)	(0.016)	(0.015)	(0.017)	(0.018)	(0.026)
Target × Exposure to radio					0.023				
					(0.026)				
Target × Exposure to other content						0.011			
						(0.028)			
Target × Exposure to *Lone Ranger*							0.015		
							(0.023)		
Target × Exposure to MBS extension								0.013	
								(0.019)	
R^2	0.21	0.21	0.20	0.23	0.21	0.21	0.21	0.21	0.29
Observations	8,837	7,717	9,315	5,466	10,364	10,364	10,364	10,364	4,647

Note. Estimates are based on equation (2). All specifications include the set of controls and FEs described in Section 4. Columns (1)–(4) restrict the baseline sample as indicated in the headers; column (1) excludes counties with more than 75% of their population residing in cities, based on the 1940 U.S. Census. Columns (5)–(8) use the full baseline sample; column (9) uses all available cohorts in the sample of internal migrants. *Target* is an indicator variable equal to 1 if the respondent was 7–18 years old in 1946, while *exposure* is the (standardized) share of the population in the county that was covered by the radio signal of *OI*. *Exposure to radio* is the (standardized) share of the population in the county covered by the radio signal of any network. *Exposure to other content* is the (standardized) share of the population in the county covered by the radio signal of any network that did not broadcast *OI*. *Exposure to Lone Ranger* is the (standardized) share of the population covered by the radio signal of the network broadcasting *The Lone Ranger* in 1946. *Exposure to MBS extension* is the (standardized) share of population covered by the radio signal of radio stations not affiliated with MBS in 1946 and affiliated with MBS in 1960. Columns (5)–(8) adds to equation (2) the interaction terms reported in the table, and the interaction terms between cohort and survey year indicators with the corresponding propagation controls (see Appendix B.2). Standard errors clustered by county are reported in parentheses (* $p < 0.1$, ** $p<0.05$, *** $p < 0.01$). The dependent variable in all columns is the *Support for Civil Rights Index* (see Section 5.2).

1980 U.S. Census (Ruggles et al., 2024). We match observations with potential exposure to *OI* using the 1,148 Principal Statistical Units (PSUs), the most disaggregated geographic identifier available, which combines entire counties or portions of adjacent counties. We define a marriage as interracial when one spouse belongs to the white majority and the other to a different racial group. Because such marriages were extremely rare during our study period, we restrict the analysis to areas with significant racial diversity, defined as PSUs with at least a 5% non-white population in 1940. As in previous analyses, we exclude individuals residing in a different state from their state of birth. The final sample includes 1,166,997 married individuals in 478 PSUs, with an average incidence of interracial marriage of 10.1 per 1,000 marriages.

Panel C of Figure 4 displays the average incidence of interracial marriage by cohort. As with interracial friendships, interracial marriage becomes substantially more common over time but remains rare, reaching only 16.3 per 1,000 marriages among the youngest cohort. Panel D reports the cohort-specific estimates from equation (1). We find no statistically significant differential exposure gradients for any cohort. Column (2) of Table 4 summarizes the effect for the target cohorts and similarly shows no detectable effect. Taken together, the results suggest that *OI* affected interracial social integration primarily at the margin of friendship rather than more intensive relationships such as marriage. This difference may partly reflect the substantially greater social and institutional barriers surrounding interracial marriage; until 1967, interracial marriage remained legally prohibited in several states.

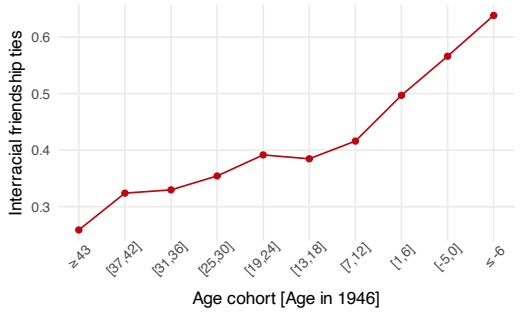
5.3 Political attitudes and behavior

We next ask whether the narrative shift in *OI* produced a narrowly circumscribed change in racial attitudes or a broader reorientation of political preferences and behavior. Table 5 reports cohort-study summary estimates for outcomes related to these dimensions using equation (2), following the same approach as Table 2.

We consider three dimensions. First, columns (1)–(2) examine whether exposure affected political support for actors associated with opposing positions on racial equality. On one side, we focus on support for political leaders who championed civil rights. On the other, we examine support for George Wallace, the segregationist governor of Alabama who became a national symbol of resistance to racial integration during the 1960s. Wallace gained national prominence through his opposition to school desegregation and subsequently carried this platform into national politics, including his 1968 third-party presidential campaign. We will discuss the electoral effects during these elections, along with mobilization in Section 5.4. Second, columns (3)–(4) study

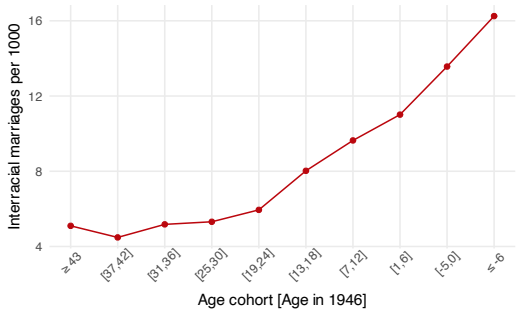
Figure 4: Interracial assimilation

A. Interracial friendships, by cohort



B. Cohort-study estimates on interracial friendships

C. Interracial marriages, by cohort



D. Cohort-study estimates on interracial marriages

Note. Panels A and B displays averages by age cohort in 1946. Panels B and D display the cohort-study exposure coefficients from equation (1) (see Section 4). In panel D, because estimates are based on a single census, controls include county-group, state-by-cohort, and state-by-year-of-first-marriage FEs, cohort-specific control interactions, and county-group clustering. Each coefficient measures how the relationship between potential exposure and the outcome within a cohort differs from that for individuals aged 19–24 in 1946 (indicated by the vertical line). Specifications include the controls and FEs described in Section 4. Error bars show 95% confidence intervals using standard errors clustered by county. *Interracial friendships* are defined using an indicator variable for whether the respondent reports having friends of different races. *Interracial marriages* are defined using an indicator variable for a marriage to someone of a different race, multiplied by 1,000. Appendix B.1 provides additional details on the variables.

Table 4: Cohort-study summary estimates on interracial assimilation

	Interracial friendships (1)	Interracial marriages (2)
A. Full sample		
Target × Exposure	0.029*	-0.094
	(0.016)	(0.480)
R^2	0.264	0.029
Observations	5,661	1,166,997
B. Restricted sample		
Target × Exposure	0.038**	0.098
	(0.019)	(0.534)
R^2	0.273	0.034
Observations	4,006	627,291

Note. Estimates are based on equation (2) using the full sample (panel A) and restricting the sample to target and older cohorts, aged 7–18 and 19+ in 1946, respectively (panel B). All specifications include the set of controls and FEs described in Section 4. In column (2), because estimates are based on a single census, controls include county-group, state-by-cohort, and state-by-year-of-first-marriage FEs, cohort-specific control interactions, and county-group clustering. *Target* is an indicator variable set to 1 for respondents aged 7–18 years in 1946, while *exposure* measures the share of the county population covered by the radio signal of stations that broadcast *OI* in 1946. Standard errors, clustered by county in column (1) and by household in column (2), are reported in parentheses (* $p < 0.1$, ** $p<0.05$, *** $p < 0.01$). The dependent variables, reported in the column headers, are defined as follows: (1) *interracial friendships* is a binary variable equal to 1 if the respondent reports having friends of different races and 0 if all friends are of the same race; (2) *interracial marriages* is a binary variable equal to 1 if one spouse belongs to the white majority and the other to a different racial group and 0 otherwise, multiplied by 1,000 to rescale the indicator. Appendix B.1 provides additional details about the variables.

whether exposure made individuals more politically engaged by examining their support for protests and participation in political campaigns. Third, columns (5)–(7) ask whether the effects extended to broader attitudes toward political authority and state institutions, focusing on trust in the federal government and views of the police and the military. Appendix B.3 reports the corresponding cohort-specific estimates from equation (1).

Among target cohorts, we find significant effects across all three dimensions. Relative to non-target cohorts and older cohorts, respectively, a one-standard-deviation increase in exposure increases favorable views of civil rights leaders by 0.08–0.10 standard deviations and reduces favorable views of George Wallace by 0.06–0.07 standard deviations. It also raises approval of protests by 0.07–0.12 standard deviations and participation in political campaigning by 0.06–0.07 standard deviations. Finally, exposure has no detectable effect on trust in the federal government, but it reduces favorable views of the police by 0.07–0.08 standard deviations and of the military by 0.05–0.07 standard deviations, pointing to a generalized decline in institutional trust associated with authority and coercion.

In panel B, restricting the comparison group to older cohorts yields similar or somewhat larger estimates for most outcomes, suggesting some spillovers to younger cohorts. As with support for civil rights, exposure gradients are highly stable among older

cohorts: they are jointly indistinguishable from zero for all outcomes, with no systematic linear gradient as cohorts approach the target ages (Appendix B.3).[10] This stability weighs against pre-existing cohort patterns along the dimensions studied in Table 5.

Together, these results show that the long-run effects of *OI* extend beyond attitudes toward racial equality to a broader set of political preferences and behaviors associated with the Civil Rights Movement, including support for its leaders, methods of political action, and views toward institutions implicated in the conflicts surrounding it. In Appendix B.5, we examine whether these broader changes translated into behavior by focusing on participation in the Vietnam War, actively opposed by prominent civil rights leaders, as a behavioral counterpart to the decline in pro-military sentiment. We proxy participation using military casualties obtained from the *Defense Casualty Analysis System (DCAS)*. This approach assumes that, conditional on serving, the likelihood of being wounded or killed was largely unrelated to exposure to *OI*. Participation in Vietnam was driven primarily by volunteers, as only 25% of those who served were draftees, compared with 66% during WWII (Davidson, 1988). Due to the age distribution of casualties during the war, we estimate equation (1) by comparing the target cohorts only with younger cohorts. Consistent with the attitudinal results, we find that greater exposure significantly reduces participation among the target cohorts.

5.4 Political mobilization at aggregate level

We next ask whether individual-level changes were sufficiently widespread to leave a detectable imprint on the political and civic life of exposed communities. We examine three manifestations of such effects: electoral outcomes, organized political mobilization around racial issues, and the salience of civil rights in local public discourse.

We implement an event-study DiD comparing counties over time according to their potential exposure to *OI* in 1946. Because the county-level analysis does not distinguish cohorts, estimates could reflect both the impact of *OI* on target cohorts and the broader effects of the stations that broadcast it. Several pieces of evidence point primarily to the former. First, direct listenership and information spillovers to adults appear limited (Section 2.1), and long-run individual effects are concentrated among target cohorts, with no differential effects among those who were already adults in 1946 (Section 5.2). Second, we find some spillovers to younger cohorts for broader political preferences (Section 5.3); these may contribute to county-level estimates but remain consistent with effects originating from *OI* rather than from the broadcasting stations

[10] The only exception is pro-police sentiment, for which the linear gradient is weakly significant at the 10% level, although the coefficients for older cohorts remain jointly indistinguishable from zero.

Table 5: Cohort-study summary estimates on broader political attitudes and behavior

	Political support		Political engagement		Attitudes towards institutions		
	Civil rights leaders	George Wallace	Approve of protest	Political campaigning	Trust in the federal government	Pro-police sentiment	Pro-military sentiment
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Full sample							
Target × Exposure	0.076*	-0.066**	0.073*	0.056*	-0.018	-0.074**	-0.053*
	(0.039)	(0.032)	(0.039)	(0.030)	(0.023)	(0.037)	(0.029)
R^2	0.258	0.247	0.237	0.132	0.196	0.170	0.198
Observations	5,304	7,883	5,487	7,840	9,644	6,404	7,333
B. Restricted sample							
Target × Exposure	0.102**	-0.062*	0.121***	0.073**	-0.017	-0.078**	-0.070**
	(0.048)	(0.035)	(0.043)	(0.034)	(0.026)	(0.033)	(0.032)
R^2	0.295	0.298	0.239	0.181	0.227	0.156	0.183
Observations	3,233	4,813	3,679	4,979	6,360	4,482	4,969

Note. Estimates are based on equation (2) using the full sample (panel A) and restricting the sample to target and older cohorts, aged 7–18 and 19+ in 1946, respectively (panel B). All specifications include the set of controls and FEs described in Section 4. *Target* is an indicator variable equal to 1 for respondents aged 7–18 in 1946, and *exposure* measures the share of the county population covered by the radio signal of stations that broadcast *OI* in 1946. Standard errors, clustered by county, are reported in parentheses (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). The dependent variables are defined as follows: (1) feelings toward Civil Rights leaders, ranging from 0 (unfavorable) to 100 (favorable); (2) feelings toward George Wallace, ranging from 0 (unfavorable) to 100 (favorable); (3) approval of protests and demonstrations, averaging available responses to two separately standardized items originally scored 1 (disapprove) to 3 (approve); (4) number of time the respondent participated in campaign activities, up to five activities; (5) trust in the federal government to do what is right, ranging from 1 (never) to 4 (always); (6) feelings toward the police, ranging from 0 (unfavorable) to 100 (favorable); and (7) feelings toward the military, ranging from 0 (unfavorable) to 100 (favorable). For ease of interpretation, all outcome measures are standardized around 0. Descriptive statistics and temporal coverage are presented in Appendix B.1.

more generally. Third, alternative radio content in 1946 produces no comparable effects on support for civil rights (Section 5.2), and our analysis of MBS programming confirms that *The Adventures of Superman* was its only production that year explicitly promoting racial tolerance. Finally, network affiliations changed substantially after 1946 (Appendix A.2), while the rapid diffusion of television during the 1950s reduced radio's centrality (Gentzkow, 2006).

For an outcome variable Y_{ct} for county c at time t, the event study approach estimates the following specification:

$$Y_{ct} = \sum_{\tau=\underline{T}}^{\bar{T}} \gamma_\tau S_\tau \cdot R_c + \mu_c + \mu_{t \times state} + (X_c \times S_t) + \epsilon_{ct} \tag{3}$$

where S_τ is an indicator variable for whether the observation refers to the period $\tau = t$ (the excluded period is the last period before 1946). The remaining terms are county FEs (μ_c), state by year FEs (μ_t), county-specific trends based on observed county characteristics ($X_c \times S_t$), and an error term clustered at the county level ($\epsilon_{c,t}$). X_c retains the county-level characteristics included in X_{ihct} in equation (1). All counties exposed to the intervention are "treated" simultaneously, which eliminates concerns about bias from treatment effect heterogeneity in staggered adoption designs. Figure 5 presents estimates using equation (3) for variables observed in more than two time periods. Table 6 presents summary estimates, pooling all post-1946 indicator variables in equation (3) into a single indicator variable.

We begin by examining electoral outcomes, focusing on the 1968 presidential election. Section 5.3 shows that exposure to *OI* during childhood later reduced the average support for George Wallace in the 1960s and 1970s. In 1968, Wallace ran for president as the American Independent Party candidate against Richard Nixon and Hubert Humphrey. His platform centered on opposition to federal civil rights enforcement and racial integration, winning 13.5% of the national popular vote and 46 electoral votes, carrying five Southern states. The 1968 election, therefore, provides a direct behavioral counterpart to the cohort-based estimates because the cohorts targeted by the program in 1946 were adults and eligible to vote; the votes of target cohorts constituted approximately 30% of the total votes in those elections (Appendix B.9).

We use the county-level vote share of Wallace and turnout as outcome variables. Wallace ran for president only in 1968, but for our DiD approach, we still need a pre-intervention measurement. We use the 1948 presidential election because 1948 and 1968 were the only presidential elections in our period in which a candidate ran on an

explicitly segregationist platform.[11] In 1948, Strom Thurmond ran as the candidate of the Dixiecrats against Harry Truman, Thomas Dewey, and Henry Wallace. His platform centered on preserving racial segregation and Jim Crow, and he received 2.4% of the national popular vote and 39 electoral votes, carrying four Southern states. Importantly, the 1948 election took place only two years after *OI*, when the cohorts targeted by the program were still too young to vote.[12] Therefore, the vote share for Thurmond and the turnout during the 1948 elections provide a useful comparison.

We gather information on county-level electoral returns and turnout in the presidential elections of 1948 and 1968 from Clubb et al. (1987). Due to regulatory barriers, Thurmond and Wallace ran only in a subset of states; therefore, we include in the analysis only the states in which both candidates appeared on the ballot. These states are Alabama, Arkansas, Florida, Georgia, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Texas, and Virginia.

Columns (1)–(2) in Table 6 present the results for the vote shares and turnout in these elections. From 1948 to 1968, an increase of one standard deviation in exposure reduces Wallace vote share by 2.4 percentage points. At the same time, we do not observe any effects on voter turnout. Thus, the electoral consequences of exposure appear to operate through the allocation of votes across candidates rather than through changes in aggregate political participation. These results complement the individual-level evidence in Section 5.3, showing that the effects on political preferences were sufficiently large to be reflected in county-level electoral outcomes. We discuss how the magnitude of these electoral effects relates to the cohort-study estimates in Section 5.5.

In columns (3)–(4), we examine whether these effects extended beyond electoral behavior to organized political mobilization around racial issues. We focus on the county-level presence of two organizations representing opposing sides of the struggle over civil rights: the KKK and the NAACP, the most prominent U.S. civil rights organization. The presence of local chapters captures a more sustained form of political engagement than voting, providing evidence on whether changes in racial attitudes translate into collective action. We obtain KKK chapter locations in 1932, 1942, 1965, 2000, and 2020, and NAACP branch locations in 1925, 1942, 1961, and 2020. Appendix B.5 provides details on data sources and descriptive statistics.

In the post-broadcast period, an increase of one standard deviation in exposure in 1946 leads to a significant reduction of 2.7 percentage points in the probability of having an active KKK chapter. Consistent with this result, we also observe a significant

[11] A broader classification of candidates across elections is difficult because explicit positions on segregation are often unavailable, particularly for losing candidates.

[12] Until 1971, the right to vote was generally granted to citizens aged 21 and older. In our sample, the sole exception was Georgia, which lowered its voting age to 18 in 1943.

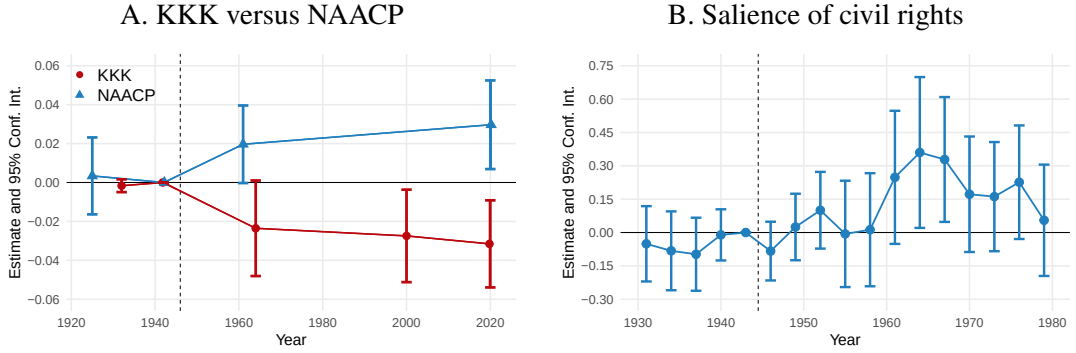
Table 6: Event study summary estimates on political mobilization

	Voting for segregation		**Mobilization**		
	Vote share	Turnout	KKK	NAACP	Salience of civil rights
	(1)	(2)	(3)	(4)	(5)
Post-1948 × Exposure	-0.024***	-0.003			
	(0.007)	(0.004)			
Post-1946 × Exposure			-0.027**	0.023***	0.182**
			(0.011)	(0.008)	(0.082)
Dependent variable mean	0.328	0.417	0.176	0.169	0.007
R^2	0.931	0.942	0.627	0.654	0.648
Number of counties	1,133	1,133	3,098	3,098	1,103
Observations	2,261	2,261	15,490	12,392	18,751

Note. Estimates are based on equation (3), pooling all post-intervention observations. *Post 1948* and *Post 1946* are indicators equal to 1 for observations after 1948 and 1946, respectively. Standard errors, clustered by county, are reported in parentheses (* p < 0.1, ** p<0.05, *** p < 0.01). The dependent variables are: (1) county-level vote share for Thurmond in 1948 and for Wallace in 1968; (2) voter turnout in these elections; (3) an indicator for at least one KKK chapter in the county at time t; (4) an indicator for at least one NAACP branch in the county at time t; and (5) an index defined by the first principal component of the salience of civil rights-related themes in local newspapers. The index is built aggregating salience across themes using RPCA, retaining counties observed in at least 50% of periods and winsorizing the series for each theme at the 1st and 99th percentiles. Section 5.4 details the construction of the newspaper measure, and Appendix B.7 reports estimates and robustness checks using alternative procedures.

increase of 2.3 percentage points in the probability that a county has an NAACP branch. Panel A of Figure 5 illustrates the temporal evolution of these effects. In line with the importance of target cohorts, the effects on both organizations appear in the 1960s and remain persistent thereafter. Before 1946, there were no significant differences.

Figure 5: Event study effects on political mobilization



Note. Estimates are based on equation (3). The 95% confidence intervals use standard errors clustered by county. In panel A, the dependent variables are *KKK* and *NAACP*, indicators for whether the county has at least one KKK chapter or NAACP branch at time t, respectively. In panel B, the dependent variable is the *salience of civil rights*, an index defined by the first principal component of the salience of civil rights-related themes in local newspapers. The index is built aggregating salience across themes using RCPA, retaining counties observed in at least 50% of periods and winsorizing the series for each theme at the 1st and 99th percentiles. Section 5.4 details the construction of the newspaper measure, and Appendix B.7 reports estimates and robustness checks using alternative procedures.

In column (5), we examine whether the effects extended to the local public discourse by increasing the salience of civil rights in newspapers. This outcome captures a

broader margin of social change than individual attitudes or political behavior: whether civil rights became a more prominent part of the information environment to which local communities were exposed. We collect data from more than 51 million pages published between 1930 and 1980 from `newspapers.com` and map newspapers to counties based on the location of their headquarters. For each county and 36-month period, we measure the salience of 25 civil-rights-related themes as the share of newspaper pages mentioning associated words. We aggregate these measures using regularized iterative principal component analysis (RPCA; Josse and Husson, 2012), which accommodates incomplete newspaper coverage across counties and periods. We use the first principal component, which explains 35.7% of the variation. Appendix B.7 describes the construction of the themes and shows robustness to alternative index-building procedures.

Panel B of Figure 5 plots event study estimates from equation (3). We find an effect on salience concentrated in the 1960s, at the height of the Civil Rights Movement and during a period in which target cohorts represented a substantial share of the adult population. These effects diminish and become statistically insignificant by the 1970s. Importantly, counties with varying degrees of exposure exhibit parallel trends before 1946. The summary estimate for the overall effect in the post-1946 period indicates that an increase of one standard deviation in exposure in 1946 leads to a significant increase in salience by 0.18 units.

5.5 Magnitude and interpretation

This section puts the magnitude of the effects documented above into perspective. In Section 5.2, we show that among the target cohorts, a one-standard-deviation increase in potential exposure to *OI* raises the *Support for Civil Rights Index* by 0.07 units relative to all non-target cohorts. Relative to the average dispersion in attitudes within the target cohorts, this corresponds to approximately 10%, indicating a moderate shift within the same generation (Appendix Figure B1). The effect is substantial relative to average differences across generations. Across approximately five decades of birth cohorts, the average index increases by about 0.28 units (Figure 3), so the estimated effect corresponds to approximately one quarter of the observed difference. These magnitudes are not driven by a single survey item, and the broader results reveal corresponding changes in political attitudes and behavior, suggesting a broader shift in racial and political attitudes rather than an unusually large response to a particular survey measure.

The long horizon is particularly important for interpreting these magnitudes. The estimated effects need not represent the mechanical persistence of a response formed in 1946. An initial change in social values can alter subsequent choices, including friend-

ships, social networks, political participation, and information exposure. These social interactions can reinforce or attenuate the initial effect over the life cycle. Consistent with this possibility, childhood exposure increased the probability of interracial friendship in adulthood while leaving educational attainment unchanged (Section 5.2). The long-run estimates capture the cumulative consequences of childhood exposure, including pathways through which initial changes in values shape subsequent experiences.

This interpretation is important when comparing the magnitudes obtained from attitudinal outcomes with our measure of revealed political behavior: Wallace's vote share in the 1968 presidential election. As shown in Section 5.4, a one-standard-deviation increase in potential exposure reduces support for segregationist candidates by 2.4 percentage points between the 1948 and 1968 elections. In Appendix B.9, we construct a long-run vote-shift rate that expresses this electoral effect relative to the counterfactual support for Wallace in the absence of any *OI* coverage. We estimate a rate of 6.4% for the overall electorate, increasing to 21.6% if the electoral response is concentrated entirely among the cohorts targeted by *OI* in 1946. These magnitudes are substantial, particularly given that they capture electoral behavior 22 years after the broadcasts.

Our measure is related to conventional persuasion rates but differs in its normalization and interpretation. Conventional persuasion rates adjust electoral effects for actual listenership, whereas our measure captures broader population-level effects that may extend beyond direct exposure through the diffusion of attitudes and values. Nevertheless, under the restrictive assumption that the long-run effects of *OI* operated exclusively through direct listenership, we obtain an illustrative persuasion rate of 23.5% (Appendix B.9). This magnitude is comparable to short-run persuasion rates estimated for influential radio broadcasts of the same period, including 36.8% for anti-Nazi radio in 1930s Germany (Adena et al., 2015) and 28% for Father Coughlin's broadcasts in 1930s U.S. (Wang, 2021). The comparison suggests that the persistent electoral effects of *OI* were meaningful, even more than two decades after the broadcasts.

Taken together, these comparisons show that the magnitudes across attitudinal and behavioral outcomes are mutually consistent with a persistent but not implausibly large effect of childhood exposure to *OI*.

6 Conclusion

This paper shows that fictional stories encountered during childhood can leave a lasting imprint on political preferences. We study an unusual historical experiment in which *The Adventures of Superman*, one of the most popular radio programs among children and adolescents in the U.S., abruptly introduced stories designed to confront prejudice

in American society. Decades later, we find that individuals exposed during childhood became more supportive of civil rights, more politically engaged, and less trusting of institutions. At the county level, greater exposure was followed by changes in political mobilization consistent with these individual-level patterns.

While our setting is historical, the findings underscore the potential for fictional stories encountered during formative years to shape political preferences in the long run. By engaging young audiences through characters and narratives rather than direct instruction, storytelling can expose them to social groups, conflicts, and moral values beyond their immediate experience. Our findings suggest that such exposure can leave persistent traces in social attitudes and political behavior decades later. These mechanisms are highly relevant for understanding how contemporary societies respond to current debates over discrimination and inclusion.

References

Aboud, F. E. (2012). Interventions to reduce prejudice and discrimination in children and adolescents. *Handbook of prejudice, stereotyping, and discrimination*, 310–327.

Adena, M., R. Enikolopov, M. Petrova, V. Santarosa, and E. Zhuravskaya (2015). Radio and the rise of the nazis in prewar germany. *The Quarterly Journal of Economics 130*(4), 1885–1939.

Adukia, A., A. Eble, E. Harrison, H. B. Runesha, and T. Szasz (2023). What we teach about race and gender: Representation in images and text of children's books. *The Quarterly Journal of Economics 138*(4), 2225–2285.

Alesina, A. and N. Fuchs-Schündeln (2007). Good-bye Lenin (or not?): The effect of communism on people's preferences. *American Economic Review 97*(4), 1507–1528.

Anderson, T. H. (2004). *The pursuit of fairness: A history of affirmative action*. Oxford University Press.

ANES (2021). ANES 2020 Time Series Study Full Release [dataset and documentation]. July 19, 2021 version.

Ang, D. (2023). The birth of a nation: Media and racial hate. *American Economic Review 113*(6), 1424–60.

Ang, D. and S. Chinoy (2024). Vanguard: Black veterans and civil rights after World War I. *Harvard University Working Paper*.

Bandura, A. and R. H. Walters (1977). *Social learning theory*, Volume 1. Prentice hall Englewood Cliffs, NJ.

Bartels, L. M. and S. Jackman (2014). A generational model of political learning. *Electoral Studies 33*, 7–18.

Bartley, N. and H. D. Graham (2019). *Southern politics and the Second Reconstruction*. JHU Press.

Bazzi, S., A. Ferrara, M. Fiszbein, T. Pearson, and P. A. Testa (2023). The other great migration: Southern whites and the new right. *The Quarterly Journal of Economics 138*(3), 1577–1647.

Benshoff, H. M. and S. Griffin (2021). *America on film: Representing race, class, gender, and sexuality at the movies*. John Wiley & Sons.

Bernini, A., G. Facchini, M. Tabellini, and C. Testa (2025). Black empowerment and white mobilization: The effects of the voting rights act. *Journal of Political Economy 133*(10), 3078–3131.

Bertrand, M. and E. Duflo (2017). Field experiments on discrimination. *Handbook of economic field experiments 1*, 309–393.

Billings, S. B., E. Chyn, and K. Haggag (2021). The long-run effects of school racial diversity on political identity. *American Economic Review: Insights 3*(3), 267–284.

Blouin, A. and S. W. Mukand (2019). Erasing ethnicity? propaganda, nation building, and identity in Rwanda. *Journal of Political Economy 127*(3), 1008–1062.

BMB (1949). State area reports: Study no. 2. Spring 1949, Broadcast Measurement Bureau, New York. Issued as separate volumes for each state.

Bonomi, G., N. Gennaioli, and G. Tabellini (2021). Identity, beliefs, and political conflict. *The Quarterly Journal of Economics 136*(4), 2371–2411.

Bowers, R. (2012). *Superman versus the Ku Klux Klan: the true story of how the iconic superhero battled the men of hate*. National Geographic Books.

Broadcasting (1946). *Issue of 13/05/1946*. The Weekly Newsmagazine of Radio, Broadcasting Publications Inc.

Broadcasting (1947). *1947 Yearbook Number*. The Weekly Newsmagazine of Radio, Broadcasting Publications Inc.

Brown, J. R., E. Cantoni, S. Chinoy, M. Koenen, and V. Pons (2023). The effect of childhood environment on political behavior: Evidence from young U.S. movers, 1992–2021. NBER Working Paper 31759, National Bureau of Economic Research. Revised August 2026.

Burris, S. E. and D. D. Brown (2014). When all children comprehend: increasing the external validity of narrative comprehension development research. *Frontiers in Psychology 5*, 168.

Bursztyn, L., G. Egorov, and S. Fiorin (2020). From extreme to mainstream: The erosion of social norms. *American Economic Review 110*(11), 3522–3548.

CAB (1946a). City reports – july 1945–february 1946. Collection of published reports, Cooperative Analysis of Broadcasting, Inc.

CAB (1946b). Program reports – february–july 1946. Collection of published reports, Cooperative Analysis of Broadcasting, Inc.

Calderon, A., V. Fouka, and M. Tabellini (2023). Racial diversity and racial policy preferences: the great migration and civil rights. *The Review of Economic Studies 90*(1), 165–200.

Campante, F., R. Durante, and A. Tesei (2022). Media and social capital. *Annual Review of Economics 14*, 69–91.

Cantoni, D., Y. Chen, D. Y. Yang, N. Yuchtman, and Y. J. Zhang (2017). Curriculum and ideology. *Journal of Political Economy 125*(2), 338–392.

Cantoni, D., A. Kao, D. Y. Yang, and N. Yuchtman (2024). Protests. *Annual Review of Economics 16*.

Chetty, R. and N. Hendren (2018). The impacts of neighborhoods on intergenerational mobility I: Childhood exposure effects. *The Quarterly Journal of Economics 133*(3), 1107–1162.

Chong, A. and E. La Ferrara (2009). Television and divorce: Evidence from brazilian novelas. *Journal of the European Economic Association 7*(2-3), 458–468.

Chyn, E. and K. Haggag (2023). Moved to vote: The long-run effects of neighborhoods on political participation. *The Review of Economics and Statistics 105*(6), 1596–1605.

Clubb, J. M., W. H. Flanigan, and N. H. Zingale (1987). *Electoral data for counties in the U.S.: Presidential and congressional races, 1840-1972*. Inter-university Consortium for Political and Social Research.

Dahl, R. E., N. B. Allen, L. Wilbrecht, and A. B. Suleiman (2018). Importance of investing in adolescence from a developmental science perspective. *Nature 554*(7693), 441–450.

Daniels, L. (1998). *Superman: The Complete History: The Life and Times of the Man of Steel*. Chronicle Books.

Davidson, P. (1988). *Vietnam At War: The History 1946-1975*. New York: Oxford University Press.

De Haven, T. (2010). *Our Hero: Superman on Earth*. Yale University Press.

DellaVigna, S., R. Enikolopov, V. Mironova, M. Petrova, and E. Zhuravskaya (2014). Cross-border media and nationalism: Evidence from serbian radio in croatia. *American Economic Journal: Applied Economics 6*(3), 103–132.

DellaVigna, S. and E. La Ferrara (2015). Economic and social impacts of the media. In *Handbook of Media Economics*, Volume 1, pp. 723–768. Elsevier.

Dinnerstein, L. (1995). *Antisemitism in America*. Oxford University Press.

Dippel, C. and S. Heblich (2021). Leadership in social movements: Evidence from the "forty-eighters" in the civil war. *American Economic Review 111*(2), 472–505.

Durante, R., P. Pinotti, and A. Tesei (2019). The political legacy of entertainment TV. *American Economic Review 109*(7), 2497–2530.

Enke, B. (2020). Moral values and voting. *Journal of Political Economy 128*(10), 3679–3729.

Esposito, E., T. Rotesi, A. Saia, and M. Thoenig (2023). Reconciliation narratives: The birth of a nation after the us civil war. *American Economic Review 113*(6), 1461–1504.

ETSI (2021). *Broadcast Sound Receivers; Part 2: AM broadcast sound service; Harmonised Standard for access to radio spectrum* (ETSI EN 303 345-2 V1.2.1 ed.).

Fouka, V. (2020). Backlash: The unintended effects of language prohibition in u.s. schools after World War I. *Review of Economic Studies 87*(1), 204–239.

Fryer Jr, R. G. (2007). Guess who's been coming to dinner? trends in interracial marriage over the 20th century. *Journal of Economic Perspectives 21*(2), 71–90.

Gagliarducci, S., M. G. Onorato, F. Sobbrio, and G. Tabellini (2020). War of the waves: Radio and resistance during World War II. *American Economic Journal: Applied Economics 12*(4), 1–38.

Gasser, L., Y. Dammert, and P. K. Murphy (2022). How do children socially learn from narrative fiction: Getting the lesson, simulating social worlds, or dialogic inquiry? *Educational Psychology Review 34*(3), 1445–1475.

Gentzkow, M. (2006). Television and voter turnout. *The Quarterly Journal of Economics 121*(3), 931–972.

Ghitza, Y., A. Gelman, and J. Auerbach (2023). The great society, Reagan's revolution, and generations of presidential voting. *American Journal of Political Science 67*(2), 520–537.

Giuliano, P. and A. Spilimbergo (2025). Aggregate shocks and the formation of preferences and beliefs. *Journal of Economic Literature 63*(2), 542–597.

Green, M. C. and T. C. Brock (2000). The role of transportation in the persuasiveness of public narratives. *Journal of Personality and Social Psychology 79*(5), 701–721.

Haines, M. R. (2010). Historical, demographic, economic, and social data: The United States, 1790–2002. Inter-university Consortium for Political and Social Research [distributor]. Version 3.

Hall, J. D. (2005). The long civil rights movement and the political uses of the past. *The Journal of American History 91*(4), 1233–1263.

Hooper Inc. (1949). *Collection of Hooperatings Comprehensive reports - Summer, 1947–1949*. New York: C.E. Hooper Inc. - Radio Audience Measurement.

ITU (1995). Prediction of sky-wave field strength at frequencies between about 150 and 1600 kHz. Recommendation ITU-R P.1147, International Telecommunication Union Radiocommunication Sector, Geneva.

ITU (2007). Ground-wave propagation curves for frequencies between 10 kHz and 30 MHz. Recommendation ITU-R P.368-9, International Telecommunication Union Radiocommunication Sector, Geneva.

ITU (2013). *Handbook on Ground-wave Propagation*. ITU.

Jensen, R. and E. Oster (2009). The power of TV: Cable television and women's status in india. *The Quarterly Journal of Economics 124*(3), 1057–1094.

Josse, J. and F. Husson (2012). Handling missing values in exploratory multivariate data analysis methods. *Journal de la société française de statistique 153*(2), 79–99.

Kaplan, E., J. L. Spenkuch, and C. Tuttle (2025). A different world: Enduring effects of school desegregation on ideology and attitudes. NBER Working Paper 33365, National Bureau of Economic Research. Revised August 2025.

Kearney, M. S. and P. B. Levine (2019). Early childhood education by television: Lessons from sesame street. *American Economic Journal: Applied Economics 11*(1), 318–350.

Kellogg, P. J. (1979). Civil rights consciousness in the 1940s. *The Historian 42*(1), 18–41.

Klein Goldewijk, K., A. Beusen, J. Doelman, and E. Stehfest (2017). Anthropogenic land use estimates for the holocene–hyde 3.2. *Earth System Science Data 9*(2), 927–953.

Kling, J. R., J. B. Liebman, and L. F. Katz (2007). Experimental analysis of neighborhood effects. *Econometrica 75*(1), 83–119.

Kuziemko, I. and E. Washington (2018). Why did the democrats lose the south? bringing new data to an old debate. *American Economic Review 108*(10), 2830–2867.

Levitt, S. D. and S. J. Dubner (2014). *Freakonomics: A Rogue Economist Explores the Hidden Side of Everything*. New York, NY: William Morrow.

Lowe, M. (2025). Has intergroup contact delivered? *Annual Review of Economics 17*(1), 321–344.

Michalopoulos, S. and C. Rauh (2024). Movies. NBER Working Paper 32220, National Bureau of Economic Research.

Michalopoulos, S. and M. M. Xue (2021). Folklore. *The Quarterly Journal of Economics 136*(4), 1993–2046.

National Authority (1945). *Standard Rate & Data Service - Radio Section*.

Olken, B. A. (2009). Do television and radio destroy social capital? evidence from indonesian villages. *American Economic Journal: Applied Economics 1*(4), 1–33.

Paluck, E. L. (2009). Reducing intergroup prejudice and conflict using the media: a field experiment in Rwanda. *Journal of personality and social psychology 96*(3), 574.

Raabe, T. and A. Beelmann (2011). Development of ethnic, racial, and national prejudice in childhood and adolescence: A multinational meta-analysis of age differences. *Child Development 82*(6), 1715–1737.

Radio Daily (1946). *The 1946 Radio Annual*. New York, NY.

Rao, G. (2019). Familiarity does not breed contempt: Generosity, discrimination, and diversity in delhi schools. *American Economic Review 109*(3), 774–809.

Reed, G. A. L. R. and K. F. Sander (1987). *Transmission and Propagation of Electromagnetic Waves*. Cambridge University Press.

Ruggles, S., S. Flood, M. Sobek, D. Backman, A. Chen, G. Cooper, S. Richards, R. Rodgers, and M. Schouweiler (2024). IPUMS USA: Version 15.0 [dataset]. Minneapolis, MN: IPUMS. https://doi.org/10.18128/D010.V15.0.

Schiappa, E., P. B. Gregg, and D. E. Hewes (2005). The parasocial contact hypothesis. *Communication monographs 72*(1), 92–115.

Schickler, E. (2016). *Racial realignment: The transformation of American liberalism, 1932-1965*. Princeton University Press.

Sears, D. O. and N. A. Valentino (1997). Politics matters: Political events as catalysts for preadult socialization. *American Political Science Review 91*(1), 45–65.

Siddique, A., M. Vlassopoulos, and Y. Zenou (2026). Leveraging virtual contact and social networks to foster interethnic harmony. *The Quarterly Journal of Economics 141*(2), 1449–1519.

Strömberg, D. (2015). Media and politics. *Annual Review of Economics 7*(1), 173–205.

US Bureau of the Census (1975). *Historical statistics of the U.S., colonial times to 1970*. Number 78. US Department of Commerce, Bureau of the Census.

Voigtländer, N. and H.-J. Voth (2015). Nazi indoctrination and anti-Semitic beliefs in Germany. *Proceedings of the National Academy of Sciences 112*(26), 7931–7936.

Wall, W. L. (2009). *Inventing the "American way": the politics of consensus from the New Deal to the civil rights movement*. Oxford University Press.

Wang, T. (2021). Media, pulpit, and populist persuasion: Evidence from father coughlin. *American Economic Review 111*(9), 3064–3092.

Whiteside, T. (1947). Up, up and awa-a-y. *New Republic 116*, 15–17.

Yanagizawa-Drott, D. (2014). Propaganda and conflict: Evidence from the Rwandan genocide. *The Quarterly Journal of Economics 129*(4), 1947–1994.

Zeng, J. and C. Duquennois (2026). Casting roles, casting votes: Lessons from sesame street on media representation and voting. *Working Paper*.

ONLINE APPENDIX

Supplementary material to *The Political Legacy of Fiction*

Alex Armand, Paul Atwell, Joseph Flavian Gomes, Giuseppe Musillo, Yannik Schenk

A Historical data sources and supporting evidence **0**

A.1 Historical sources and validation of radio coverage estimates 0

A.2 Radio antennas, propagation, and the *OI* stations 8

A.3 Content of *Operation Intolerance* 13

B Additional analysis **16**

B.1 Descriptive analysis of ANES data 16

B.2 Conditional balance of potential exposure R_c 17

B.3 Stability of the exposure gradient outside the target cohorts 19

B.4 Robustness to influential states and survey waves 24

B.5 Political and war mobilization 24

B.6 Robustness checks for the Support for Civil Rights index 26

B.7 Anticipation, spillover and long-run salience effects 28

B.8 Attitudes towards other groups and educational attainment 31

B.9 Long-run vote-shift rate 32

A Historical data sources and supporting evidence

A.1 Historical sources and validation of radio coverage estimates

Figure A1 shows examples of the digitized reports discussed in this section, which were produced by different agencies. First, **Hooper Inc.** published reports on the share of respondents who listen to a program at the time it airs, measured through telephone surveys in 33 cities geographically distributed throughout the nation. Listenership shares are aggregate statistics for the entire sample. Panel A in Figure A2 summarizes the listenership shares of programs produced by major radio networks in 1945–47. Panel B reports the number of listeners per listening set, distinguishing between adults and children/adolescents, and highlighting the slot of *The Adventures of Superman*. Panel C shows the share of weekly (Monday–Friday) hours of broadcasting by program category, while panel D provides an example of a radio schedule. Adventure fiction (tar-

geted at children) represented a significant share of the MBS productions. In the time slot with the highest share of children listening (5–6pm during the week), adventure fiction represented 100% of programming. Shows other than *The Adventures of Superman* in this time slot had elements of action and heroism, but they were not known to address issues of equality (Dunning, 1998).

Reports also allow for gathering information about the radio programs produced by MBS. The following table presents all programs for the period November–December 1946 with a listenership share larger than 6%. The audience for productions other than adventure fiction was primarily adults. There is no evidence that variety openly discussed equality, but we have evidence against news and commentaries discussing racial issues in a progressive way. We highlight six shows carrying the names of their commentators. Taylor, Lewis, and Close were conservative commentators, while Hale, Foster, and Heatter were less ideologically charged (Hilmes, 1997; Dunning, 1998).

Program	Genre and short description	Time (DoW)	Format	Share
The Shadow	DRAMA. Series feat. mind-clouding vigilante.	17:00 (Su)	30	35.5
True Detective Mysteries	CRIME SHOW. Series on real crimes.	16:30 (Su)	30	27.2
Adventures of the Falcon	DRAMA. Series feat. a detective.	20:30 (T)	30	25.1
Quick as a Flash	QUIZ. Game quiz show.	17:30 (Su)	30	24.5
Fulton Lewis	NEWS. Feat. Fulton Lewis.	19:00 (M–F)	15	23.0
Nick Carter, Master Detective	DRAMA. Feat. a private detective.	18:30 (Su)	30	22.6
Double or Nothing	QUIZ. Game quiz show.	19:00 (Su)	15	21.8
House of Mystery	DRAMA. Series feat. a scientist debunking phantoms.	16:00 (Su)	30	20.0
Those Websters	DRAMA. Sit-com feat. an American family.	18:00 (Su)	30	18.8
Queen for a Day	QUIZ. Contestants told personal stories for prizes.	14:30 (M–F)	30	18.6
Cedric Foster	NEWS. Feat. Cedric Foster.	14:00 (M–F)	15	17.9
Gabriel Heatter	NEWS. Feat. Gabriel Heatter.	21:00 (M–F)	15	16.1
John J. Anthony	VARIETY. Talk show about marital relations.	13:45 (M–F)	15	16.0
The Adventures of Superman	ADVENTURE FICTION. Series feat. Superman.	17:15 (M–F)	15	15.4
Tom Mix	ADVENTURE FICTION. Series feat. a cowboy.	17:45 (M–F)	15	15.3
Checkerboard Jamboree	VARIETY. Variety show.	13:00 (Sa)	60	15.2
Twenty Questions	QUIZ. Game quiz show.	20:00 (Sa)	30	14.5
Juvenile Jury	QUIZ. Quiz show with children.	13:30 (Su)	30	14.0
Arthur Hale	NEWS. Feat. Arthur Hale.	19:30 (TThSa)	45	13.9
Warden Lawes' Crime Cases	CRIME SHOW. True-crime show.	13:00 (Su)	15	13.8
Crimes of Carelessness	DRAMA. Portrayal of famous American fires.	15:30 (Su)	30	13.5
Captain Midnight	ADVENTURE FICTION. Series feat. a WWI aviator.	17:30 (M–F)	15	12.6
Casebook of Gregory Hood	DRAMA. Detective series feat. an antiques dealer.	20:30 (M)	30	12.3
Exploring the Unknown	DRAMA. Show exploring the natural world.	21:00 (Su)	30	11.3
Coke Club	VARIETY. Show feat. artist Morton Downey.	12:15 (M–F)	15	10.8
Treasure Hour of Song	MUSIC. Music program.	21:30 (Th)	30	10.3
Real Stories from Real Life	DRAMA. Soap opera based on true stories.	21:15 (M–F)	15	9.4
Buck Rogers in the 25th Century	ADVENTURE FICTION. Show feat. space adventures.	16:45 (M–F)	15	9.3
Henry J. Taylor	NEWS. Feat. H.J. Taylor.	19:30 (MF)	30	9.2
Victor H. Lindlar	VARIETY. Talk show feat. V.H. Lindlar.	11:45 (M–F)	15	9.0
Spotlight Bands	MUSIC. Music program.	21:30 (MWF)	30	8.7
Pilgrim Hour	RELIGION. Religious program.	12:00 (Su)	30	7.8
Judy 'n' Jill 'n' Johnny	MUSIC. Show feat. Johnny Desmond.	12:00 (Sa)	30	7.5

(continued on next page)

Program	Genre and short description	Time (DoW)	Format	Share
It's Up to Youth	VARIETY. Talk show about youth issues.	20:30 (W)	30	7.5
Inside of Sports	NEWS. Show covering sport news.	19:45 (M–F)	15	7.2
A brighter tomorrow	NEWS. Program on American success stories.	22:00 (Su)	30	7.2
Lutheran Hour	RELIGION. Religious program.	12:30 (Su)	30	7.0
Burl Ives	MUSIC. Folk music show.	20:00 (F)	15	6.9
Upton Close	NEWS. Feat. Upton Close.	22:15 (T)	15	6.4

Note. The radio programs are sorted based on the average listenership share. In column *DoW*, "M" indicates Monday, "T" Tuesday, "W" Wednesday, "Th" Thursday, "F" Friday, "Sa" Saturday, and "Su" Sunday. "M–F" indicates that the show is daily during weekdays. The column *format* reports the duration of each show in minutes. Data refer to the period November–December 1946 and are obtained from Hooper Inc. (1949). The column *share* is the average listenership share, which averages all available ratings (or *Hooperating*) reported for the period. When ratings are not available, we use the ratings for the period March–April 1947. The genre *News* includes commentaries. The genre *drama* includes mystery drama.

Second, the **Cooperative Analysis of Broadcasting (CAB)** measured listenership through random telephone calls, asking respondents whether they were listening to the radio when called and, if so, which program and station. Given limited telephone access in the 1940s, the survey covered radio owners with telephone access living in urban areas with populations of at least 50,000 and was representative of this population. Unlike Hooper, CAB reported geographically disaggregated data. We digitized two publications: the *City Report* and *Program Report* (CAB, 1946a,b). The bi-monthly City Report provides city-level station listenership shares by time slot. We obtained reports from July 1945 to February 1946, covering up to 82 cities, and linearly interpolated missing observations when the same city is observed in adjacent periods. The semi-monthly Program Report provides program-level listenership averaged across sampled cities. We obtained reports from February 1945 to July 1946. Before December 1945, data are available only for programs broadcast consistently over time. Thereafter, unassigned slots report listenership independently of the program broadcast ("Time Period Rating"). The resulting data, therefore, form a balanced panel only for slots broadcasting the same program throughout the period. Figure A3 shows listenership over time for programs broadcast between 5–6pm, including *The Adventures of Superman*.

Finally, we use data from the **Broadcast Measurement Bureau (BMB)**, a cooperative founded in 1945 by advertising and broadcasting associations to standardize audience measurements. BMB (1949) provides nationwide county-level measures of radio audiences using a retrospective ballot method in which respondents record their past listening behavior on printed forms. We digitize the complete survey from physical copies of the published reports preserved in the Wisconsin Historical Society archives, creating the first comprehensive county-by-station dataset of mid-century radio listening for the entire U.S. The reports provide total weekly audiences by station and locality (counties and, separately, larger cities), together with the daytime and nighttime frequency of listening (1–2, 3–5, and 6–7 times a week). Stations are reported only when at least 10%

of households in a locality listened at least once per week, so listenership rates are truncated at 10%. Using these data, we compile weekly listenership shares (i.e., share of households with radio sets reporting that they tuned in to a station at least once during an average week) for the 220 stations that aired *OI* in 1946 across 5,083 counties and larger cities. Figure A4 reports descriptive statistics for weekly listening frequencies and the resulting daily listenership measures.

Exploiting the wide geographical coverage of this information, we use it to validate R_c (Section 3.1) against realized listening rates. Some discrepancies may persist because the data refer to 1949, three years after the launch of *OI*, and changes in transmission infrastructure may have altered signal reach relative to 1946. Table A2 provides validation tests for our predicted coverage. Columns (1)–(2) of Table A2 regress actual on predicted reception. *Actual reception* is an indicator variable for whether a station–locality pair appears in the ratings, corresponding to a weekly audience of at least 10%. *Predicted reception* is an indicator variable for whether the predicted field strength at the locality centroid exceeds the signal strength threshold used in Section 3.1. Cities are assigned coverage based on their counties. Where the model predicts reliable reception, about 94% of station–locality pairs register a weekly audience of at least 10%, compared with fewer than 0.5% in areas predicted to be uncovered, potentially reflecting measurement noise from infrastructure changes between 1946 and 1949. Columns (3)–(4) replicate the listenership analysis of Section 5.1 using the BMB county-level data, allowing us to assess whether the relationship presented in the main text extends to the full set of counties. For each locality, we construct two measures of daily listenership to *OI* stations: the listenership share of the station with the largest share and the sum of shares across all *OI* stations (capped at 100%). Results confirm that greater exposure to the *OI* network predicts higher daily listenership.

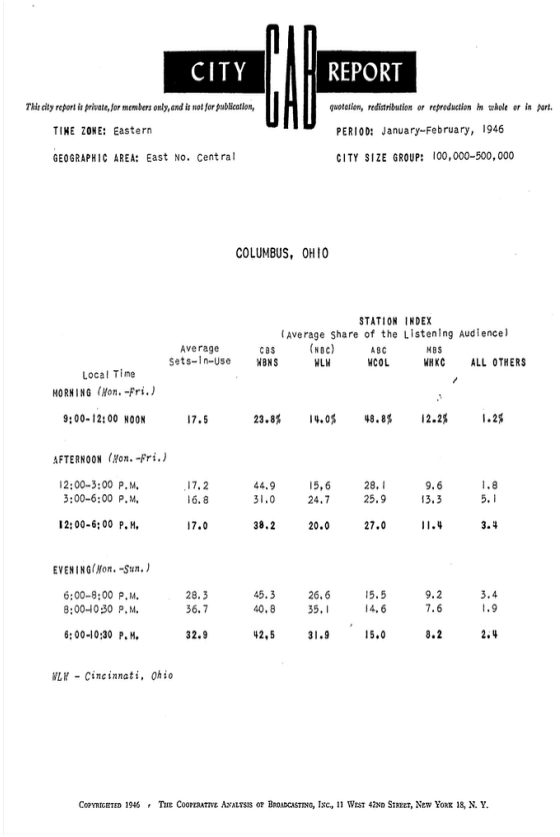
Table A2: Validation of signal predictions for the *Operation Intolerance* radio network

	Actual reception		**Daily listenership share of *OI* stations**	
			Station with largest share	Sum of all stations
	(1)	(2)	(3)	(4)
Predicted reception	0.941***	0.924***		
	(0.002)	(0.009)		
Constant	0.005***			
	(0.000)			
Exposure			0.022***	0.038***
			(0.003)	(0.004)
Observations	682,657	682,657	5,083	5,083
R^2	0.178	0.204	0.177	0.198
Locality + station FEs	–	Yes	–	–
Level of observation	Station-locality	Station-locality	Locality	Locality

Note. Columns (1)–(2) report OLS estimates. The dependent variable is *actual reception* is an indicator variable equal to 1 if a station–county pair appears in the ratings (i.e., has a local audience of at least 10%), and 0 otherwise. *Predicted reception* is an indicator variable equal to 1 if the predicted field strength at the locality centroid is greater than or equal to the same signal strength threshold used in Section 3.1, and 0 otherwise. Columns (3)–(4) estimate equation (4). The dependent variable is the daily listenership share of the radio stations that broadcast *OI* in 1946, using two alternative procedures to aggregate listenership within a locality (see Appendix A.1 for detailed definitions). Weekly listening frequencies from the data source are converted into daily station-level shares using midpoint weighting. *Exposure* denotes the share of the population in the county covered by the radio signal of stations that were part of *OI* in 1946. Standard errors (in parentheses) are clustered at the locality level in columns (1)–(2) and at the state level in columns (3)–(4). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure A1: Examples from historical reports of radio listenership

A. City Report



CITY CAB REPORT

This city report is private, for members only, and is not for publication, quotation, redistribution or reproduction in whole or in part.

TIME ZONE: Eastern PERIOD: January–February, 1946

GEOGRAPHIC AREA: East No. Central CITY SIZE GROUP: 100,000–500,000

COLUMBUS, OHIO

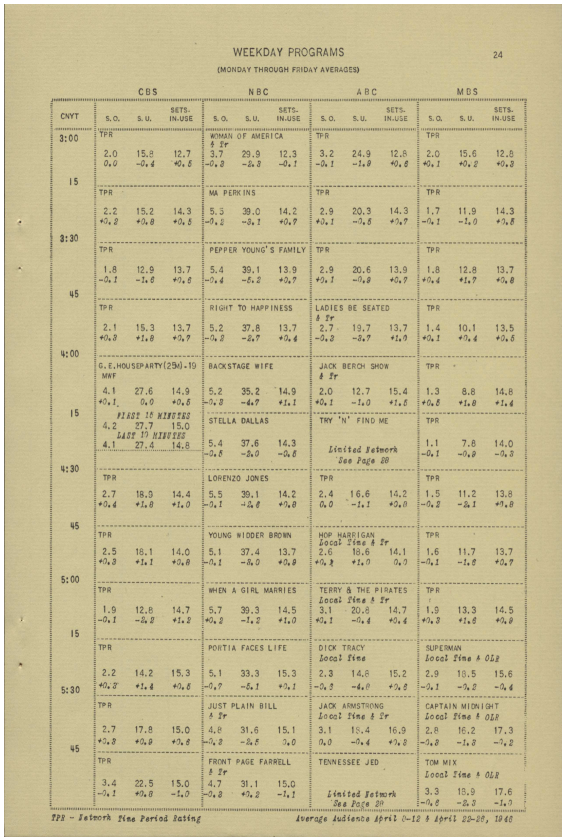
STATION INDEX
(Average Share of the Listening Audience)

Local Time	Average Sets-In-Use	CBS WBNS	(NBC) WLW	ABC WCOL	MBS WHKC	ALL OTHERS
MORNING (Mon.-Fri.)						
9:00-12:00 NOON	17.5	23.8%	14.0%	48.8%	12.2%	1.2%
AFTERNOON (Mon.-Fri.)						
12:00-3:00 P.M.	17.2	44.9	15.6	28.1	9.6	1.8
3:00-6:00 P.M.	16.8	31.0	24.7	25.9	13.3	5.1
12:00-6:00 P.M.	17.0	38.2	20.0	27.0	11.4	3.4
EVENING (Mon.-Sun.)						
6:00-8:00 P.M.	28.3	45.3	26.6	15.5	9.2	3.4
8:00-10:30 P.M.	36.7	40.8	35.1	14.6	7.6	1.9
6:00-10:30 P.M.	32.9	42.5	31.9	15.0	8.2	2.4

WLW - Cincinnati, Ohio

COPYRIGHTED 1946 , THE COOPERATIVE ANALYSIS OF BROADCASTING, INC., 11 WEST 42ND STREET, NEW YORK 18, N. Y.

B. Program Report



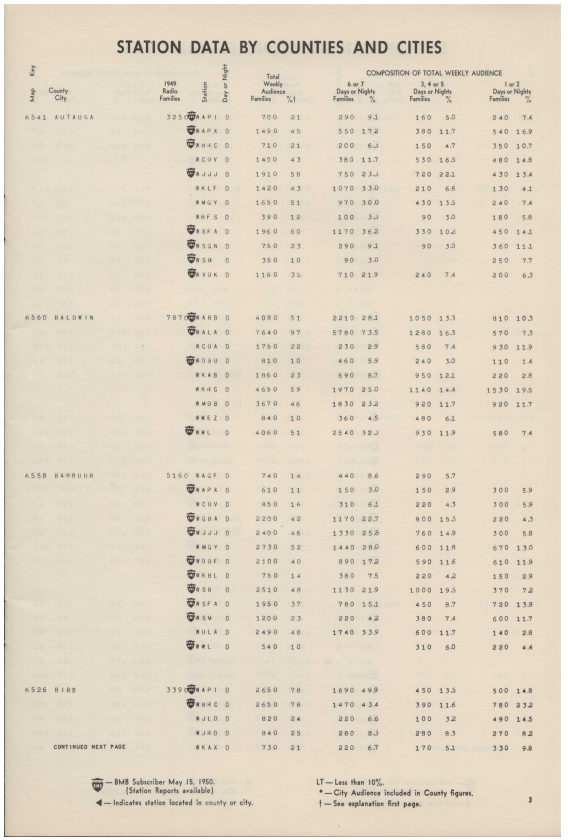
WEEKDAY PROGRAMS 24

(MONDAY THROUGH FRIDAY AVERAGES)

CNYT	CBS S.O.	CBS S.U.	CBS SETS-IN-USE	NBC S.O.	NBC S.U.	NBC SETS-IN-USE	ABC S.O.	ABC S.U.	ABC SETS-IN-USE	MBS S.O.	MBS S.U.	MBS SETS-IN-USE
3:00	TPR			WOMAN OF AMERICA & Tr			TPR			TPR		
	2.0	15.8	12.7	3.7	29.9	12.3	3.2	24.9	12.6	2.0	15.6	12.8
	0.0	-0.4	+0.5	-0.2	-2.3	-0.1	-0.1	-1.9	+0.2	+0.1	+0.2	+0.3
15	TPR			MA PERKINS			TPR			TPR		
	2.2	15.2	14.3	5.3	39.0	14.2	2.9	20.3	14.3	1.7	11.9	14.3
	+0.2	+0.8	+0.5	-0.2	-3.1	+0.7	+0.1	-0.5	+0.7	-0.1	-1.0	+0.8
3:30	TPR			PEPPER YOUNG'S FAMILY			TPR			TPR		
	1.8	12.9	13.7	5.4	39.1	13.9	2.9	20.6	13.9	1.8	12.8	13.7
	-0.1	-1.6	+0.6	-0.4	-5.2	+0.7	+0.1	-0.9	+0.7	+0.4	+1.7	+0.8
45	TPR			RIGHT TO HAPPINESS			LADIES BE SEATED & Tr			TPR		
	2.1	15.3	13.7	5.2	37.8	13.7	2.7	19.7	13.7	1.4	10.1	13.5
	+0.3	+1.8	+0.7	-0.2	-2.7	+0.4	-0.3	-0.7	+1.0	+0.1	+0.4	+0.5
4:00	G.E.HOUSEPARTY(25m)-19 MWF			BACKSTAGE WIFE			JACK BERCH SHOW & Tr			TPR		
	4.1	27.6	14.9	5.2	35.2	14.9	2.0	12.7	15.4	1.3	8.8	14.8
	+0.1	0.0	+0.6	-0.2	-4.7	+1.1	+0.1	-1.0	+1.5	+0.5	+1.8	+1.4
15	PLAZE 15 MINUTES			STELLA DALLAS			TRY 'N' FIND ME			TPR		
	4.2	27.7	15.0									
	LAST 15 MINUTES											
	4.1	27.4	14.8	5.4	37.6	14.3	Limited Network See Page 20			1.1	7.8	14.0
				-0.6	-5.0	-0.8				-0.1	-0.9	-0.8
4:30	TPR			LORENZO JONES			TPR			TPR		
	2.7	18.0	14.4	5.5	39.1	14.2	2.4	16.6	14.2	1.5	11.2	13.8
	+0.4	+1.8	+1.0	-0.1	-4.6	+0.8	0.0	-1.1	+0.8	-0.2	-2.1	+1.8
45	TPR			YOUNG WIDDER BROWN			HOP HARRIGAN Local Time & Tr			TPR		
	2.5	18.1	14.0	5.1	37.4	13.7	2.6	18.6	14.1	1.6	11.7	13.7
	+0.3	+1.1	+0.8	-0.1	-3.0	+0.9	+0.2	+1.0	0.0	-0.1	-1.6	+0.7
5:00	TPR			WHEN A GIRL MARRIES			TERRY & THE PIRATES Local Time & Tr			TPR		
	1.9	12.8	14.7	5.7	39.3	14.5	3.1	20.8	14.7	1.9	13.3	14.5
	-0.1	-0.2	+1.2	+0.2	-1.2	+1.0	+0.1	-0.4	+0.4	+0.3	+1.6	+0.9
15	TPR			PORTIA FACES LIFE			DICK TRACY Local Time			SUPERMAN Local Time & OLR		
	2.2	14.2	15.3	5.1	33.3	15.3	2.3	14.8	15.2	2.9	18.5	15.6
	+0.3	+1.4	+0.6	-0.7	-5.1	+0.1	-0.5	-4.0	+0.5	-0.1	-0.2	-0.4
5:30	TPR			JUST PLAIN BILL & Tr			JACK ARMSTRONG Local Time & Tr			CAPTAIN MIDNIGHT Local Time & OLR		
	2.7	17.8	15.0	4.8	31.6	15.1	3.1	18.4	16.9	2.8	16.2	17.3
	+0.3	+0.8	+0.8	-0.3	-2.6	0.0	0.0	-0.4	+0.8	-0.3	-1.8	-2.2
45	TPR			FRONT PAGE FARRELL & Tr			TENNESSEE JED			TOM MIX Local Time & OLR		
	3.4	22.5	15.0	4.7	31.1	15.0				3.3	18.9	17.6
	-0.1	+0.6	-1.0	-0.8	+0.2	-1.1	Limited Network See Page 20			-0.6	-2.3	-1.0

TPR - Network Time Period Rating Average Audience April 5-12 & April 22-28, 1946

C. State Area Report



STATION DATA BY COUNTIES AND CITIES

Map Key	County City	1949 Radio Families	Station	Day or Night	Total Weekly Audience Families	%	COMPOSITION OF TOTAL WEEKLY AUDIENCE: 6 or 7 Days or Nights Families	%	3, 4 or 5 Days or Nights Families	%	1 or 2 Days or Nights Families	%
6541	AUTAUGA	325	WAPI	D	700	21	290	9.1	160	5.0	240	7.4
			WAPX	D	1490	45	550	17.2	380	11.7	540	16.9
			WHHC	D	710	21	200	6.3	150	4.7	350	10.7
			WCOV	D	1400	43	380	11.7	530	16.5	480	14.8
			WJJJ	D	1910	58	750	23.3	720	22.1	430	13.4
			WKLF	D	1420	43	1070	33.0	210	6.6	130	4.1
			WMGY	D	1650	51	970	30.0	430	13.5	240	7.4
			WRFS	D	390	12	100	3.1	90	3.0	180	5.8
			WSFA	D	1960	60	1170	36.2	330	10.2	450	14.1
			WSGN	D	750	23	290	9.1	90	3.0	360	11.1
			WSM	D	350	10	90	3.0			250	7.7
			WYOK	D	1160	35	710	21.9	240	7.4	200	6.3
6560	BALDWIN	787	WABB	D	4080	51	2210	28.1	1050	13.3	810	10.3
			WALA	D	7640	97	5780	73.5	1280	16.3	570	7.3
			WCOA	D	1750	22	230	2.9	580	7.4	930	11.9
			WGBU	D	810	10	460	5.9	240	3.0	110	1.4
			WKAB	D	1860	23	690	8.7	950	12.1	220	2.8
			WKKG	D	4650	59	1970	25.0	1140	14.4	1530	19.5
			WMOB	D	3670	46	1830	23.2	920	11.7	920	11.7
			WMEZ	D	840	10	360	4.5	480	6.1		
			WWL	D	4060	51	2540	32.3	930	11.9	580	7.4
6558	BARBOUR	5160	WAGF	D	740	14	440	8.6	290	5.7		
			WAPX	D	610	11	150	3.0	150	2.9	300	5.9
			WCOV	D	850	16	310	6.1	220	4.3	300	5.9
			WGBA	D	2200	42	1170	22.7	800	15.5	220	4.3
			WJJJ	D	2400	46	1330	25.8	760	14.9	300	5.8
			WMGY	D	2730	52	1440	28.0	600	11.8	670	13.0
			WOOF	D	2100	40	890	17.2	590	11.6	610	11.9
			WRBL	D	750	14	380	7.5	220	4.2	150	2.9
			WSB	D	2510	48	1130	21.9	1000	19.5	370	7.2
			WSFA	D	1950	37	780	15.1	450	8.7	720	13.9
			WSM	D	1200	23	220	4.2	380	7.4	600	11.7
			WULA	D	2490	48	1740	33.9	600	11.7	140	2.8
			WWL	D	540	10			310	6.0	220	4.4
6526	BIBB	339	WAPI	D	2650	78	1690	49.9	450	13.5	500	14.8
			WHRC	D	2650	78	1470	43.4	390	11.6	780	23.2
			WJLD	D	820	24	220	6.6	100	3.2	490	14.5
			WJRD	D	840	25	280	8.3	280	8.3	270	8.2
CONTINUED NEXT PAGE			WKAX	D	730	21	220	6.7	170	5.1	330	9.8

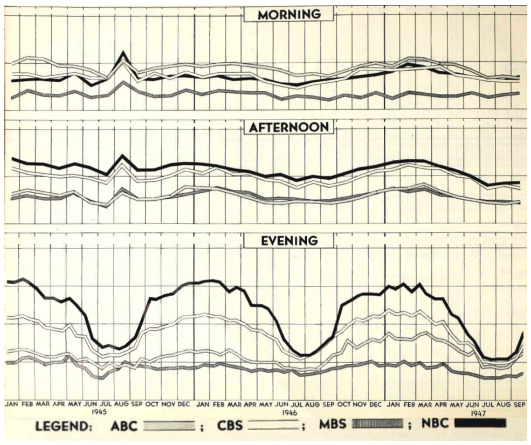
— BMB Subscriber May 15, 1950. (Station Reports available)
◀ — Indicates station located in county or city.

LT — Less than 10%.
* — City Audience included in County figures.
† — See explanation first page.

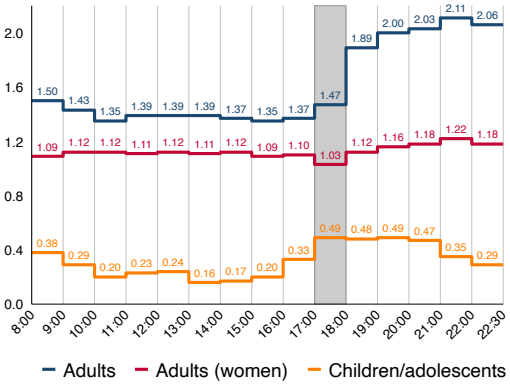
Note. Panel A and B provide two examples of information available from CAB (1946a,b). The sample covered by these reports includes the following cities, grouped by Census area: East North Central (Akron; Chicago; Cincinnati; Cleveland; Columbus; Dayton; Detroit; Evansville, Indiana; Grand Rapids; Indianapolis; Milwaukee; Racine; Rockford; South Bend; Springfield, Illinois; Springfield, Ohio; Toledo), East South Central (Birmingham; Jackson; Louisville; Memphis), Middle Atlantic (Altoona; Buffalo; Erie; Harrisburg; Johnstown; New York; Philadelphia; Pittsburgh; Reading; Rochester; Schenectady; Scranton; Syracuse; Trenton; Utica; Wilkes-Barre), Mountain (Denver; Salt Lake City), New England (Boston; Bridgeport; Hartford; Manchester; Portland; Providence; Springfield, Massachusetts; Waterbury; Worcester), Pacific (Fresno; Los Angeles; Portland; Sacramento; San Diego; San Francisco; Seattle; Spokane), South Atlantic (Atlanta; Baltimore; Columbia; Columbus; Jacksonville; Richmond; Washington, D.C.; Wilmington; Winston-Salem), West North Central (Des Moines; Duluth; Kansas City; Minneapolis and St. Paul; Omaha; Springfield, Missouri; St. Louis; Wichita), and West South Central (Dallas; Houston; Little Rock; New Orleans; Oklahoma City; San Antonio; Shreveport; Tulsa). Panel C provides an example of audience surveys from BMB (1949). The data include all stations with at least 10% of local households reporting they listen at least once per week, for all counties and, separately, cities with populations of 25,000 or more in metropolitan counties and 10,000 or more in non-metropolitan counties.

Figure A2: Listenership shares and MBS productions

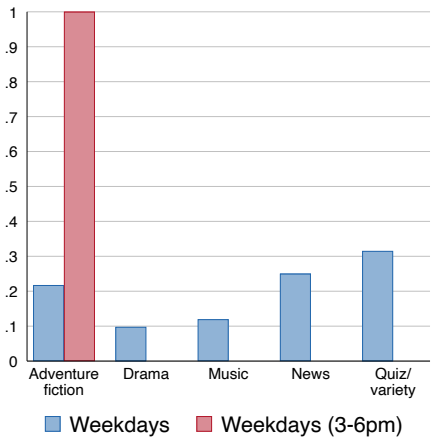
A. Listenership share by network (1945–47)



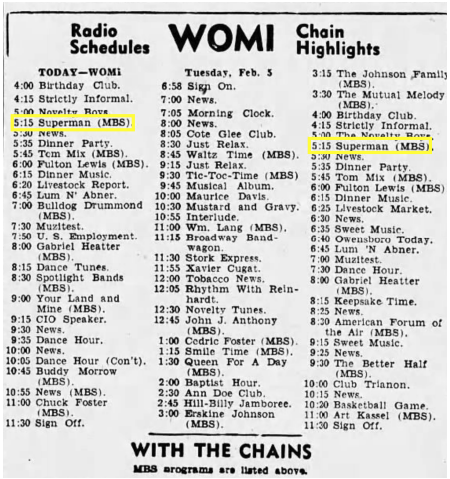
B. Number of listeners per listening set



C. Share of MBS productions (1946)



D. Example of schedule of MBS-affiliated



Radio Schedules — WOMI — Chain Highlights

TODAY—WOMI

4:00 Birthday Club.
4:15 Strictly Informal.
5:00 Novelty Boys.
5:15 Superman (MBS).
5:30 News.
5:35 Dinner Party.
5:45 Tom Mix (MBS).
6:00 Fulton Lewis (MBS).
6:15 Dinner Music.
6:20 Livestock Report.
6:45 Lum N' Abner.
7:00 Bulldog Drummond (MBS).
7:30 Muzitest.
7:50 U. S. Employment.
8:00 Gabriel Heatter (MBS).
8:15 Dance Tunes.
8:30 Spotlight Bands (MBS).
9:00 Your Land and Mine (MBS).
9:15 CIO Speaker.
9:30 News.
9:35 Dance Hour.
10:00 News.
10:05 Dance Hour (Con't).
10:45 Buddy Morrow (MBS).
10:55 News (MBS).
11:00 Chuck Foster (MBS).
11:30 Sign Off.

Tuesday, Feb. 5

6:58 Sign On.
7:00 News.
7:05 Morning Clock.
8:00 News.
8:05 Cote Glee Club.
8:30 Just Relax.
8:45 Waltz Time (MBS).
9:15 Just Relax.
9:30 Tic-Toc-Time (MBS)
9:45 Musical Album.
10:00 Maurice Davis.
10:30 Mustard and Gravy.
10:55 Interlude.
11:00 Wm. Lang (MBS).
11:15 Broadway Bandwagon.
11:30 Stork Express.
11:55 Xavier Cugat.
12:00 Tobacco News.
12:05 Rhythm With Reinhardt.
12:30 Novelty Tunes.
12:45 John J. Anthony (MBS).
1:00 Cedric Foster (MBS).
1:15 Smile Time (MBS).
1:30 Queen For A Day (MBS).
2:00 Baptist Hour.
2:30 Ann Doe Club.
2:45 Hill-Billy Jamboree.
3:00 Erskine Johnson (MBS).

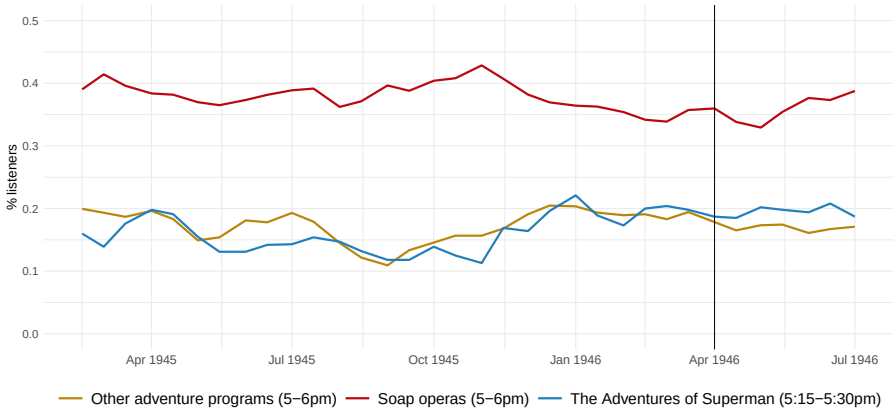
3:15 The Johnson Family (MBS).
3:30 The Mutual Melody (MBS).
4:00 Birthday Club.
4:15 Strictly Informal.
5:00 The Novelty Boys.
5:15 Superman (MBS).
5:30 News.
5:35 Dinner Party.
5:45 Tom Mix (MBS).
6:00 Fulton Lewis (MBS)
6:15 Dinner Music.
6:25 Livestock Market.
6:30 News.
6:35 Sweet Music.
6:40 Owensboro Today.
6:45 Lum 'N Abner.
7:00 Muzitest.
7:30 Dance Hour.
8:00 Gabriel Heatter (MBS).
8:15 Keepsake Time.
8:25 News.
8:30 American Forum of the Air (MBS).
9:15 Sweet Music.
9:25 News.
9:30 The Better Half (MBS).
10:00 Club Trianon.
10:15 News.
10:20 Basketball Game.
11:00 Art Kassel (MBS).
11:20 Sign Off.

WITH THE CHAINS

MBS programs are listed above.

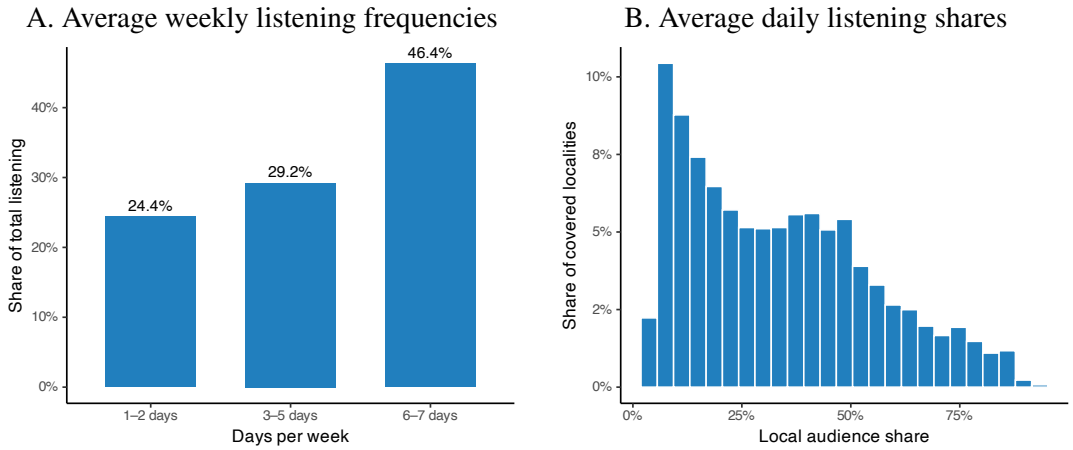
Note. Panel A reports average listenership shares by network and time of day (January 1945–September 1947), from the Summer 1947 Comprehensive Report. Panels B–C use November–December 1946 data (Hooper Inc., 1949): panel B shows listeners per radio set by age group, and panel C the share of weekday broadcasting hours by genre, overall and in the afternoon (*News* includes commentaries; *Drama* includes mystery drama). Panel D reproduces the WOMI schedule from the April 2, 1946 *Messenger-Inquirer* (Owensboro, Kentucky), distinguishing local programming from MBS productions (in parentheses).

Figure A3: Listenership share of *The Adventures of Superman* before and after *OI*



Note. The figure shows the time series of the average listenership share for different programs (group of programs) from CAB (1946b). *Other adventure programs* includes Captain Midnight (ABC/MBS), Dick Tracy (ABC/CBS), Jack Armstrong (ABC), Tennessee Jed (ABC), Terry and The Pirates (ABC), and Tom Mix (MBS). *Soap operas* includes Front Page Farrell (NBC), Just Plain Bill (NBC), Portia Faces Life (NBC), and When a Girl Marries (NBC). The vertical line indicates the launch of *OI*.

Figure A4: Distribution of listenership of the *OI* network

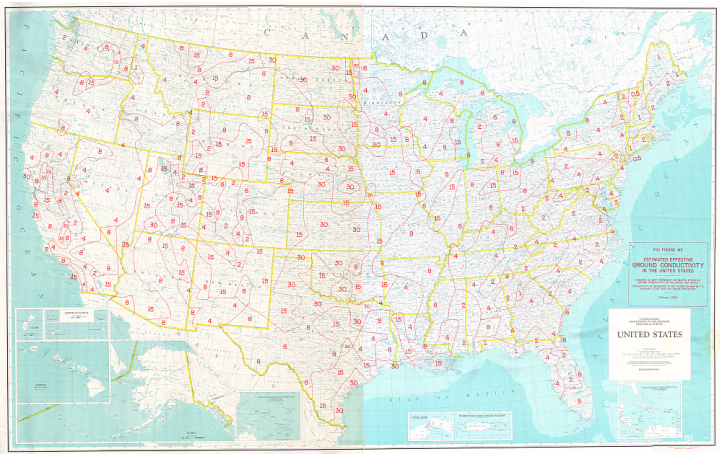


Note. Panel A shows the 1949 weekly listening frequency for stations airing *OI*, reporting the share of listeners tuning in 1–2, 3–5, or 6–7 days per week across all station–location pairs. Listeners may be counted multiple times if they listened to multiple stations. Panel B shows the distribution across places of the maximum daily audience share among *OI* stations, measured as a percentage of radio households; places with shares $\leq 10\%$ are excluded. Data source: BMB (1949).

A.2 Radio antennas, propagation, and the *OI* stations

Engineering models to accurately predict the signal of AM transmissions became available only in 1958 (Bremmer, 1958). Figure A5 shows the first ground conductivity map for the U.S., published by FCC (1954) and used as an input parameter in the radio-propagation model described in Section 3.1. Ground conductivity varies significantly, ranging from 0.5 to 30 mS per meter. Figure A6 provides examples of information used to identify the radio stations that broadcast *OI*. We validated these data using radio schedules from historical newspaper archives from `newspapers.com`. Search strings combined specific radio station identifiers obtained from previous steps with keywords such as "Superman" or "The Adventures of Superman". Searches were limited to the year 1946 and conducted state-by-state to ensure comprehensive geographic coverage.

Figure A5: Ground conductivity



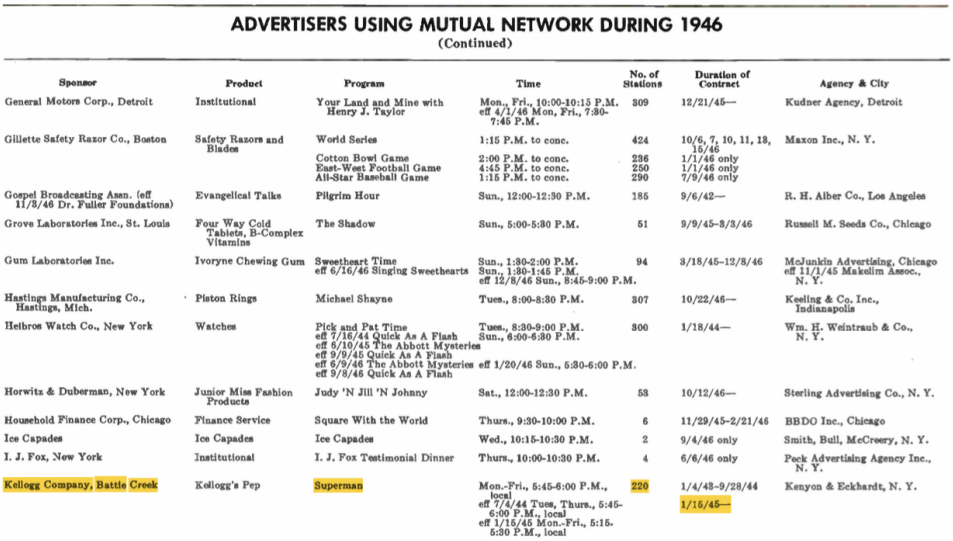
Note. Ground conductivity map published by FCC (1954).

We gathered information on 998 radio stations. We geo-referenced their transmitters using the precise address reported in Radio Daily (1946). Panel A in Figure A7 presents the geographical distribution of all radio antennas identified. Panels B–D present the geographical distribution of three propagation controls. First, a binary indicator variable for whether a county has at least one radio antenna that aired *OI* within its boundaries (panel B). Second, the natural logarithm of the distance (in km) between each county centroid and the nearest *OI* antenna (panel C). Third, the theoretical radio coverage under the assumption of perfect ground conductivity, computed assuming uniform ideal propagation conditions and assigning a conductivity value of 5 S/m—commonly used to model minimal-loss AM signal propagation (panel D).

Figure A8 reports descriptive statistics of the radio stations that broadcast *OI*, by year of establishment (horizontal axes). Establishment dates vary widely, reflecting the evolving nature of the network. Some stations date back to the 1920s, while others

Figure A6: Example of information digitized from historical publications

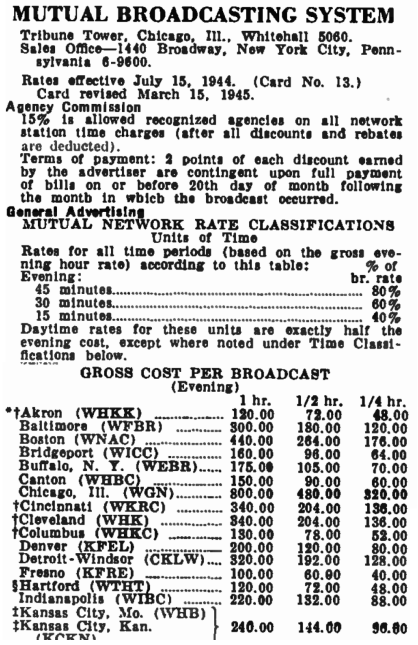
A. Broadcasting Yearbook



ADVERTISERS USING MUTUAL NETWORK DURING 1946
(Continued)

Sponsor	Product	Program	Time	No. of Stations	Duration of Contract	Agency & City
General Motors Corp., Detroit	Institutional	Your Land and Mine with Henry J. Taylor	Mon., Fri., 10:00-10:15 P.M. eff 4/1/46 Mon, Fri., 7:30-7:45 P.M.	309	12/21/45—	Kudner Agency, Detroit
Gillette Safety Razor Co., Boston	Safety Razors and Blades	World Series	1:15 P.M. to conc.	424	10/6, 7, 10, 11, 13, 15/46	Maxon Inc., N. Y.
		Cotton Bowl Game	2:00 P.M. to conc.	236	1/1/46 only	
		East-West Football Game	4:45 P.M. to conc.	250	1/1/46 only	
		All-Star Baseball Game	1:15 P.M. to conc.	290	7/9/46 only	
Gospel Broadcasting Assn. (eff 11/3/46 Dr. Fuller Foundations)	Evangelical Talks	Pilgrim Hour	Sun., 12:00-12:30 P.M.	185	9/6/42—	R. H. Alber Co., Los Angeles
Grove Laboratories Inc., St. Louis	Four Way Cold Tablets, B-Complex Vitamins	The Shadow	Sun., 5:00-5:30 P.M.	51	9/9/45-3/3/46	Russell M. Seeds Co., Chicago
Gum Laboratories Inc.	Ivoryne Chewing Gum	Sweetheart Time eff 6/16/46 Singing Sweethearts	Sun., 1:30-2:00 P.M. Sun., 1:30-1:45 P.M. eff 12/8/46 Sun., 8:45-9:00 P.M.	94	3/18/45-12/8/46	McJunkin Advertising, Chicago eff 11/1/45 Makelim Assoc., N. Y.
Hastings Manufacturing Co., Hastings, Mich.	Piston Rings	Michael Shayne	Tues., 8:00-8:30 P.M.	307	10/22/46—	Keeling & Co. Inc., Indianapolis
Helbros Watch Co., New York	Watches	Pick and Pat Time eff 7/16/44 Quick As A Flash eff 6/10/45 The Abbott Mysteries eff 9/9/45 Quick As A Flash eff 6/9/46 The Abbott Mysteries eff 9/8/46 Quick As A Flash	Tues., 8:30-9:00 P.M. Sun., 6:00-6:30 P.M. eff 1/20/46 Sun., 5:30-6:00 P.M.	300	1/18/44—	Wm. H. Weintraub & Co., N. Y.
Horwitz & Duberman, New York	Junior Miss Fashion Products	Judy 'N Jill 'N Johnny	Sat., 12:00-12:30 P.M.	53	10/12/46—	Sterling Advertising Co., N. Y.
Household Finance Corp., Chicago	Finance Service	Square With the World	Thurs., 9:30-10:00 P.M.	6	11/29/45-2/21/46	BBDO Inc., Chicago
Ice Capades	Ice Capades	Ice Capades	Wed., 10:15-10:30 P.M.	2	9/4/46 only	Smith, Bull, McCreery, N. Y.
I. J. Fox, New York	Institutional	I. J. Fox Testimonial Dinner	Thurs., 10:00-10:30 P.M.	4	6/6/46 only	Peck Advertising Agency Inc., N. Y.
Kellogg Company, Battle Creek	Kellogg's Pep	Superman	Mon.-Fri., 5:45-6:00 P.M., local eff 7/4/44 Tues, Thurs., 5:45-6:00 P.M., local eff 1/15/45 Mon.-Fri., 5:15-5:30 P.M., local	220	1/4/43-9/28/44 1/15/45—	Kenyon & Eckhardt, N. Y.

B. Standard Rate and Data Service



MUTUAL BROADCASTING SYSTEM

Tribune Tower, Chicago, Ill., Whitehall 5060.
Sales Office—1440 Broadway, New York City, Pennsylvania 6-9600.

Rates effective July 15, 1944. (Card No. 13.)
Card revised March 15, 1945.

Agency Commission
15% is allowed recognized agencies on all network station time charges (after all discounts and rebates are deducted).

Terms of payment: 2 points of each discount earned by the advertiser are contingent upon full payment of bills on or before 20th day of month following the month in which the broadcast occurred.

General Advertising

MUTUAL NETWORK RATE CLASSIFICATIONS
Units of Time

Rates for all time periods (based on the gross evening hour rate) according to this table:

Evening:	% of br. rate
45 minutes	80%
30 minutes	60%
15 minutes	40%

Daytime rates for these units are exactly half the evening cost, except where noted under Time Classifications below.

GROSS COST PER BROADCAST
(Evening)

	1 hr.	1/2 hr.	1/4 hr.
*†Akron (WHKK)	120.00	72.00	48.00
Baltimore (WFBR)	300.00	180.00	120.00
Boston (WNAC)	440.00	264.00	176.00
Bridgeport (WICC)	160.00	96.00	64.00
Buffalo, N. Y. (WEBR)	175.00	105.00	70.00
Canton (WHBC)	150.00	90.00	60.00
Chicago, Ill. (WGN)	800.00	480.00	320.00
†Cincinnati (WKRC)	340.00	204.00	136.00
†Cleveland (WHK)	340.00	204.00	136.00
†Columbus (WHKC)	130.00	78.00	52.00
Denver (KFEL)	200.00	120.00	80.00
Detroit-Windsor (CKLW)	320.00	192.00	128.00
Fresno (KFRE)	100.00	60.00	40.00
§Hartford (WTHT)	120.00	72.00	48.00
Indianapolis (WIBC)	220.00	132.00	88.00
‡Kansas City, Mo. (WHB) ‡Kansas City, Kan. (KCKN)	240.00	144.00	96.00

C. Radio Annual

MISSOURI

For Major Markets—Radio Homes 1946 Please Turn to Pages 247-255

K F V S
CAPE GIRARDEAU—EST. 1925

Frequency: 1400 Kc.....Power: 250 Watts
Owned By.....Oscar C. Hirsch
Operated By.....Oscar C. Hirsch
Address.....324 Broadway
Phone Number.....2104, 2105
Transmitter Location.....3½ miles west of Cape Girardeau on Highway No. 61
Time on the Air.....Unlimited
News Service.....UP
Transcription Service.....Standard Radio
Representative.....John E. Pearson

Personnel

President-General Manager....Oscar C. Hirsch
Station-Sales Manager.....Ralph L. Hirsch
Production-Program Director....Virginia Bohn
Chief Engineer.....R. L. Hirsch
Promotion-Publicity.....R. L. Hirsch

K F R U
COLUMBIA—EST. 1925
ABC

Frequency: 1400 Kc.....Power: 250 Watts
Owned By.....KFRU, Inc.
Address.....Ninth and Elm Sts.
Phone Number.....4141
Transmitter Location.....Hickman Hall, Stephens College
Time on the Air.....6:30 a.m. to 11 p.m.
Newspaper Affiliation....Star Times Publishing Company, St. Louis
News Service.....AP & UP
Transcription Service.....NBC Thesaurus
Representative.....John Blair & Co.

Personnel

President.....Elzey Roberts
General Manager.....Mahlon Aldridge
Program Director.....Harold Douglas
Musical Director.....Dorothy Jacques
Chief Engineer.....Robert Haigh
Record M. C.....Dorothy Jacques

K H M O
HANNIBAL—EST. 1941
MUTUAL

Frequency: 1340 Kc.....Power: 250 Watts
Owned-Operated By....Courier Post Publishing Co.
Address.....102½ S. Main St.
Phone Number.....3450-51
Transmitter Location.....Lindell Ave.
Time on the Air.....5:30 a.m. to 12 midnight: Sundays, 8 a.m. to 12 midnight
Newspaper Affiliation.....Hannibal Courier-Post
News Service.....AP
Transcription Service.....NBC & Cole
Representative.....John E. Pearson Co.

Personnel

General Manager-Sales Promotion Manager.....Wayne W. Cribb
Commercial Manager.....Chris Jensen
Program-Musical Director.....Dud Dawson
Production Manager.....Maxine Dierking
Publicity Director-Record M. C...Gene Hoenes
Chief Announcer.....Bob McCoy
Chief Engineer.....Ben Parrish

K W O S
JEFFERSON CITY—EST. 1936
MUTUAL BROADCASTING SYSTEM

Frequency: 1240 Kc.....Power: 250 Watts
Owned-Operated By.....Tribune Printing Co.
Business Address.....210 Monroe
Phone Number.....4000
Studio Address.....210 Monroe
Transmitter Location.....St. Mary's Blvd.
Time on the Air.....6 a.m. to midnight
Newspaper Affiliation...Jefferson City Capitol News & Post Tribune
News Service.....AP
Transcription Service.....Lang-Worth
Representative.....Sears & Ayer

Personnel

President.....R. C. Goshorn
General Manager.....R. L. Rose
Station Manager.....R. L. Rose
Commercial Manager.....R. L. Rose
Sales Promotion Manager-Publicity Director.....Natalie LePage
Program Director.....Ray Manning
Chief Engineer.....Harold White

W M B H
JOPLIN—EST. 1933
MUTUAL BROADCASTING SYSTEM

Frequency: 1450 Kc.....Power: 250 Watts
Owned-Operated By...Joplin Broadcasting Co.
Address....Friaco Bldg., Sixth and Main Sts.
Phone Number.....320
Transmitter Location.....1324 Roosevelt Ave.
Time on the Air.....6 a.m. to midnight
Newspaper Affiliation.....Joplin Globe Publishing Co.
News Service.....UP

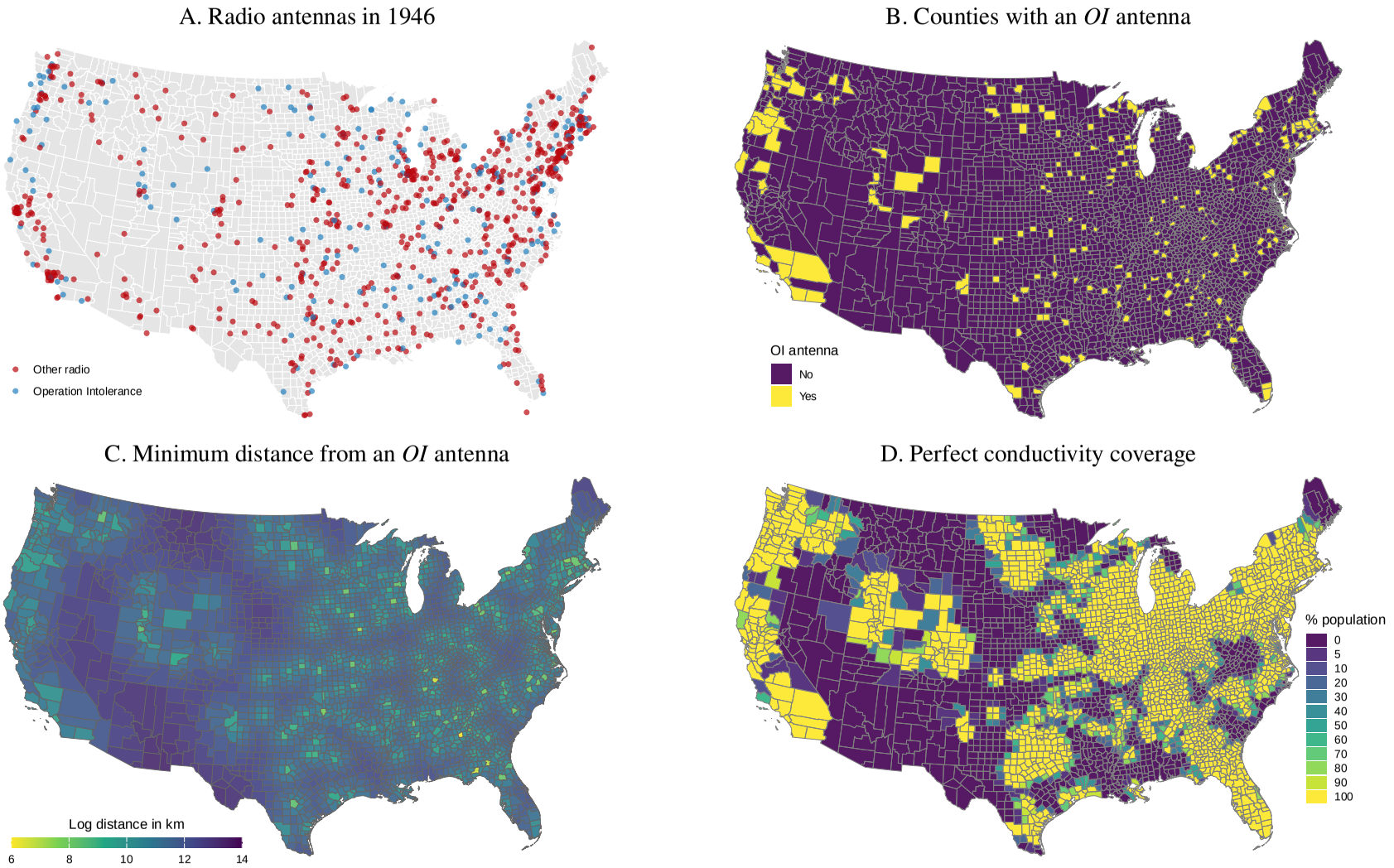
461

Note. Panel A is extracted from Broadcasting (1947), identifying Kellogg Company as sponsor of *The Adventures of Superman* and the broadcasting schedule in 1946. Panel B is extracted from National Authority (1945), allowing to identify the precise composition of broadcasting network of *The Adventures of Superman* in 1946. Panel C is extracted from Radio Daily (1946), showing some of the active antennas in Missouri, including technical specifications and the location. If precise location is not available we set it equal to the business address, or to the centroid of the respective county for the few remaining cases.

were established in the early 1940s. Regarding antenna location, 71% were in Northern states and 29% in Southern states (panel A), and about 47% were in counties containing a city (panel B). These figures suggest no systematic geographic targeting by the MBS network. In terms of network affiliation (panel C), 81% of the stations were exclusively affiliated with MBS in 1946, while the remainder also carried content from at least one

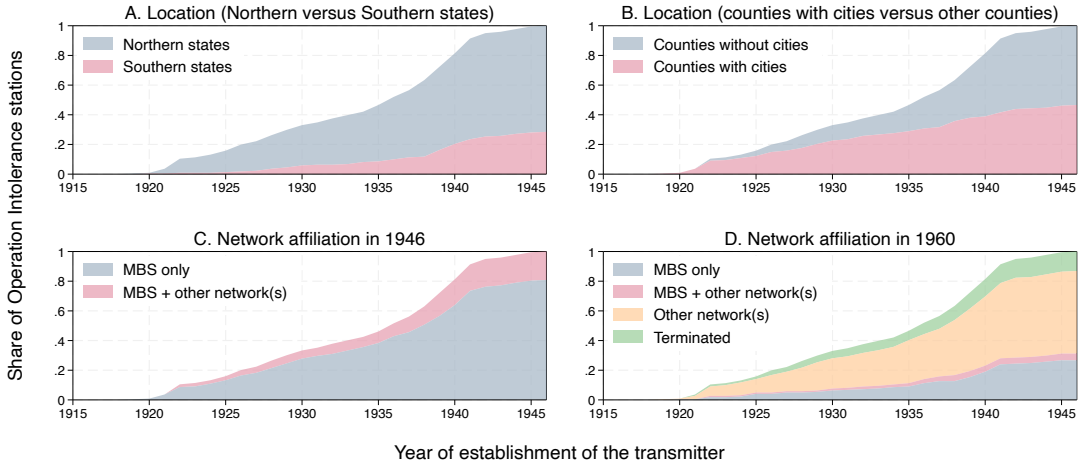
other national network. Panel D repeats this analysis for 1960, based on digitized data from Broadcasting (1960). Consistent with the fluid nature of the radio market during this period, only 27% of the stations were still solely affiliated with MBS by 1960; 5% were jointly affiliated with MBS and at least one other network; 56% had switched to another network; and 13% had ceased or relocated operations.

Figure A7: 1946 radio network and propagation controls for *OI*



Note. Panel A shows the distribution of antennas of the complete U.S. radio network as of 1946. Stations that broadcast *OI* are marked in blue, while those that did not are shown in red. For visual clarity, stations located in Canada are omitted. Panel B shows the distribution of counties that have an antenna airing the program placed within county boundaries. Panel C plots the logarithm of the distance (in kilometers) between each county centroid and the nearest *OI* broadcasting antenna. Panel D displays the geographical distribution of the population share covered at the county level, computed assuming ideal radio signal propagation conditions. This is calculated using the software *Field Strength Calculator One*, applying the algorithm used in Section 3.1. See Section 3 for details on the methodology used to construct these county-level measures.

Figure A8: Descriptive statistics of radio stations that broadcast *OI*



Note. The figure describes stations broadcasting *OI* by transmitter establishment year and location (Panels A and B) and network affiliation (Panels C and D). *Counties with cities* are defined according to the 1940 U.S. Census. *Southern states* include Alabama, Arkansas, Florida, Georgia, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, and Texas. *Other network(s)* includes ABC, NBC, and CBS. Section 3.1 describes the data sources.

A.3 Content of *Operation Intolerance*

Figure A9 provides examples of the coverage of *OI* in newspapers in 1946. The following summaries are exemplary story arcs from *OI*:

- **The Hate Mongers Organization (16/04/1946–20/05/1946).** This arc addresses the rise of racial and religious intolerance in postwar America. It opens with the firebombing of a Jewish-owned drugstore in Metropolis and follows newsboy Danny O'Neil, who witnesses the crime and is later attacked for speaking out. As Clark Kent investigates, he uncovers a secret hate group using teenage gangs to sabotage Unity House, a community center promoting interfaith unity. Through Jimmy Olsen's undercover work and Superman's intervention, the story exposes how hate groups manipulate youth and exploit silence in the face of bigotry. Paralleling real KKK activities, it stands as a powerful indictment of intolerance.
- **The Clan of the Fiery Cross (10/06/1946–01/07/1946).** This arc confronts racism and white supremacy. When Tommy Lee, a Chinese-American boy, replaces white teammate Chuck Riggs as pitcher for the Unity House baseball team, Chuck's resentment is fueled by his uncle, Matt Riggs, leader of the Clan of the Fiery Cross. The group spreads hate through violence and propaganda. Inspired by real Klan activity, the Lees face terror for being nonwhite Americans. Superman and the Daily Planet expose the Clan's lies and defend Tommy, while Chuck's eventual remorse underscores the possibility of moral growth and redemption.
- **George Latimer, Crooked Political Boss (03/09/1946–25/09/1946).** This arc links political corruption and racism in postwar America. Returning veterans protesting discriminatory hiring are met with violence and deceit. After Joe Martin is shot during a peaceful march, his Jewish friend Sam Robbins is framed by political boss George Latimer, who exploits the crisis to incite racial hatred and equate whiteness with patriotism. Superman exposes Latimer's manipulation and defends democratic values, echoing real civil-rights struggles.
- **Knights of the White Carnation (26/02/1947–17/03/1947).** In Metropolis, a hate group led by businessman Vincent Kirby targets four minority basketball players, fabricating corruption charges and spreading propaganda to have them expelled from the team. As violence escalates, Superman reveals the conspiracy. The story shows how racism thrives through scapegoating and fear and ends with the group's downfall, a clear condemnation of intolerance.

We conducted a content analysis based on transcripts of collected audio recordings of 1019 episodes broadcast on MBS between August 1942 and February 1949.[2] The

[2]The transcripts were collected from The Classic Archives Old Time Radio Channel and Nikola

Figure A9: Endorsements and success of *OI*

A. Endorsement of Harry S. Truman (U.S. President 1945–53)

Superman 'Reformer'

Will Combat Intolerance

Superman zooms off on a new adventure starting Tuesday, April 16 at 5:15 p. m. over KQV, with the avowed intention of combating bigotry and intolerance.

Superman is the first network children's radio series to take up the cudgels for tolerance, in support of a movement which has the blessing of every church and the endorsement of President Truman as well as a large group of organizations. Scripts for the new series were prepared with the co-operation of the National Conference of Christians and Jews.

"The plan for using this juvenile series to generate the principles of brotherhood has been in the works for some time," Robert Maxwell, producer of Superman, declared today. "We felt an obligation to perform a public service in addition to entertaining the youngsters. Superman reaches a large audience of children whose opinions and characters are being formed in a mold which will influence their entire lives. To influence them constructively is our purpose."

The new Superman theme has the hearty endorsement of the sponsor, whose president, W. H. Vanderploeg, declared: "For some time we have been planning a more direct approach to the problems

for some time," Robert Maxwell, producer of Superman, declared today. "We felt an obligation to perform a public service in addition to entertaining the youngsters. Superman reaches a large audience of children whose opinions and characters are being formed in a mold which will influence their entire lives. To influence them constructively is our purpose."

The new Superman theme has the hearty endorsement of the sponsor, whose president, W. H. Vanderploeg, declared: "For some time we have been planning a more direct approach to the problems now facing young boys and girls in their schools, their social clubs and their homes. We felt that given the proper entertainment values as a base, children would listen to a direct and exciting Superman series with a thoroughly American message of brotherhood. We hope, with the aid of our new Superman story approach, that the children of America will continue to be friendly with all other children, regardless of difference in race, creed or color."

The advertising agency for the company, in a statement heartily concurred with their client about the espousing of the new "Superman" story line. W. B. Lewis, vice president and radio director of the agency, said: "The story will be just as exciting as the usual Superman adventures. In fact, we think it will be even more exciting. The difference lies in the fact that Superman, if this first experiment is successful, will go to war against juvenile delinquency, racial intolerance, school absenteeism and the other problems of child behaviorism which bulk so large in the public consciousness."

B. Endorsement of Henry A. Wallace (U.S. Vice President 1941–45)

"Superman" Tolerance Campaign Wins Praise

The new tolerance theme recently inaugurated on the "Superman" radio series (WCLO, 5:15 p. m. Monday through Friday) has been the object of many accolades from organizations and educational groups throughout the country, including a special commendation from Secretary of Commerce Henry A. Wallace. In an effort to combat bigotry and racial prejudice, the adventures of the "man of tomorrow" have been turned from his more mythical enemies to real life foes of boys and girls the world over.

Citations and commendations have been received from the Calvin Newspaper Service, who state—"We applaud heartily this noble attempt to make better citizens of our children and to eradicate from their minds all thoughts of racial and religious intollerance," from the Associated Negro Press who commend the program and its sponsors for slanting the radio series in this vein, and from the Child Study Association of America who applaud the current Superman program.

Secretary of Commerce Wallace states—"I am happy to learn that you are using Superman for the purpose of teaching children that democracy includes the idea of tolerance and equal opportunity for all races, creeds and colors. It is much easier to plant the truth in young minds before anti-social teachings have taken firm root."

* * *

The Real Story!

The secret is out. Mrs. Fred Van Deventer, better known to "Twenty Questions" listeners as Florence Rinard, didn't go into radio for fame and fortune, but because she wanted to see her commentator husband once in a while. 'I got tired of being a golf widow by day

Note. Panel A is extracted from the edition of 14/04/1946 of The Pittsburgh Press (published in Pittsburgh, Pennsylvania). Panel B is extracted from the edition of 11/06/1946 of the Janesville Daily Gazette (published in Janesville, Wisconsin).

sample covers 64% of the 1592 episodes aired in the same period, while the remaining share is missing. When recordings of story arcs are only partially available, we use the available content for the classification. We analyze transcripts using a bag-of-words approach with 50 keywords related to intolerance and tolerance for racial, ethnic, and religious differences, returned by an AI prompt. Figure A10 provides the keywords and a graphical depiction of the frequency of these themes in the episodes.

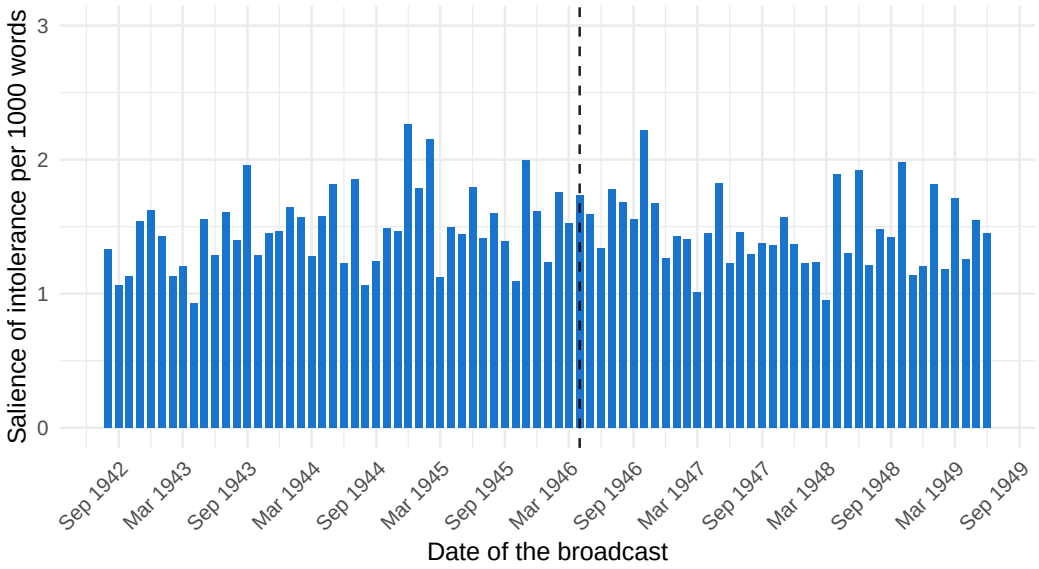
We apply the same methodology to *The Lone Ranger*, an ABC radio series featuring a masked former Texas Ranger who fought outlaws in the Old West. The show was a nationally broadcast and highly popular serialized adventure program with a large youth audience during the 1940s, making it a useful comparison. Figure A11 replicates Figure 1 in the main text, aggregating transcripts of 976 separate episodes into monthly indicators based on their broadcasting dates. We find no evidence of a paradigm shift, with the average salience of tolerance related keywords remaining consistently low and *decreasing* marginally by 0.47% (from 1.475 to 1.468 per 1000 words) after April 1946.

Tesla Wireless Radio, containing recordings of all available episodes.

Figure A10: Frequency of keywords related to tolerance and intolerance

Note. Word cloud illustrating the frequency of terms related to tolerance and intolerance. Frequencies are computed as count of keyword occurrences over the total number of words, removing stop words and applying basic stemming. Larger and more central words reflect higher frequencies. To build the word cloud, we use a dictionary of 50 keywords. The keywords are: acceptance, animosity, antisemitism, appreciation, bias, bigotry, chauvinism, closed-mindedness, community, compassion, cosmopolitanism, discrimination, disunity, diversity, division, dogmatism, empathy, empowerment, equality, equity, exclusion, fairness, gender equality, harmony, hate, homophobia, inclusion, individualism, inequality, injustice, integration, intolerance, justice, love, misogyny, open-mindedness, pluralism, prejudice, racism, respect, sectarianism, segregation, sexism, stereotyping, tolerance, tribalism, understanding, unity, and xenophobia. Keywords were obtained from ChatGPT-3.5 with the following request: "Give us a list of 50 words that are closely related to intolerance, bigotry, and prejudice, and their respective antonyms."

Figure A11: Content analysis of narrative in *The Lone Ranger*



Note. The figure illustrates monthly averages of the share of keywords related to tolerance or intolerance per 1,000 words in *The Lone Ranger* episode transcripts. The dashed line marks the launch of *OI*. The sample covers 31 August 1942–24 June 1949, matching the analysis window in Figure 1. We exclude *Chuck Wagon Champ*, *The Frontier Day Race*, and *A Racer of Turtles* because racing references generate false positives for the keyword "race."

B Additional analysis

B.1 Descriptive analysis of ANES data

The following table provides definitions of the variables used, including descriptive statistics. Panel A of Figure B1 shows the availability of each question in different survey rounds. We consider the survey waves in which we observe at least half of the questions used to build the *Support for Civil Rights Index* in each wave. Panel B shows the within-cohort distributions of the index. Figure B2 shows descriptive statistics of the variables used in the main text by cohort and by year of surveying. For comparability, we standardize all variables.

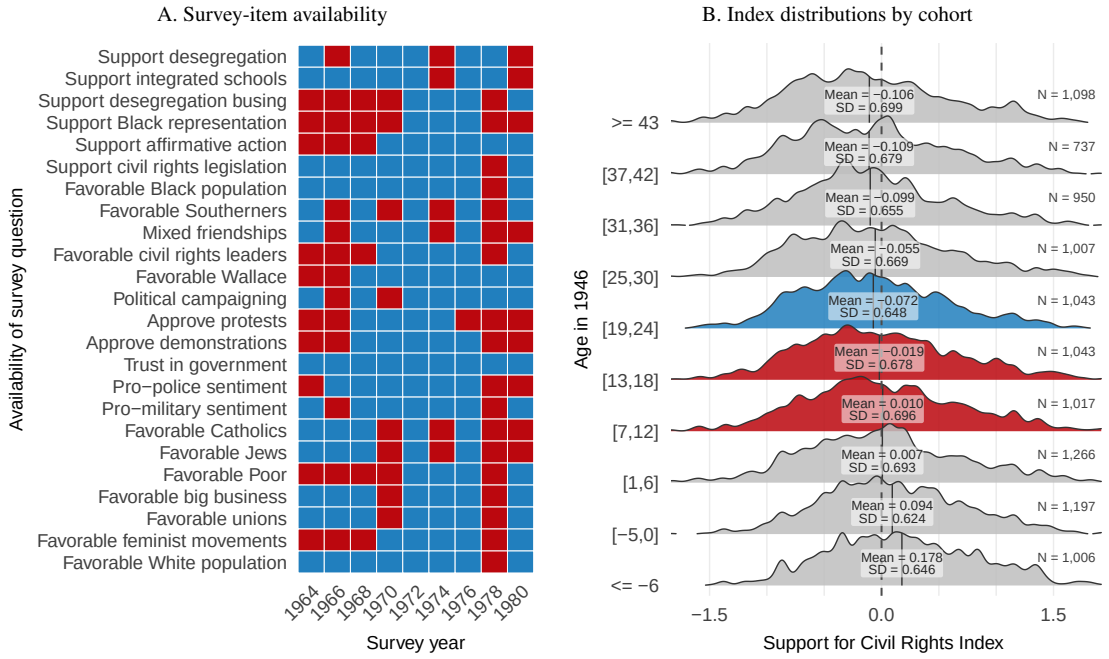
Variable	Description	μ	SD	N	ID
Approve of protests	Degree of approval of participation in protests and demonstrations. We average available, separately standardized approval of "protest meetings or marches that are permitted by the local authorities", and approval of "attempts at stopping the government from going about its activities with sit-ins, mass meetings, demonstrations". Both items range from 1 (disapprove) to 3 (approve).	1.64	0.56	5487	0601; 0603
Feelings towards (the) ...					
Black population	Feeling Thermometer (1–100) towards the Black population.	62.77	21.09	8145	0206
Civil Rights leaders	Feeling Thermometer (1–100) towards Civil Rights leaders.	44.74	26.56	5304	0216
George Wallace	Feeling Thermometer (1–100) towards George Wallace.	44.4	28.53	7883	0439
Pro-military sentiment	Feeling Thermometer (1–100) towards the military.	71.07	21.64	7333	0213
Pro-police sentiment	Feeling Thermometer (1–100) towards the police.	76.26	18.43	6404	0214
Southerners	Feeling Thermometer (1–100) towards Southerners.	63.24	20.12	5207	0208
Interracial friendships	Indicator variable equal to 1 if the respondent reports having friends of different races, and 0 if all friends share the same race.	0.41	0.49	5661	0866
Support for ...					
Affirmative action	Beliefs over whether the government should make efforts to improve the position of the Black population, ranging from 1 (Blacks should help themselves) to 7 (government should help Blacks).	3.61	1.89	6377	0830
Black representation	Beliefs over the influence of Blacks in American life and politics, ranging from 1 (too much influence) to 3 (too little influence).	1.98	0.79	3470	9274
Desegregation	Preference regarding racial desegregation, ranging from 1 (strict segregation) to 3 (desegregation).	2.23	0.67	6836	0815
Desegregation busing	Beliefs over whether the importance of racial integration of schools justifies busing children to schools out of their own neighborhoods, ranging from 1 (Keep children in neighborhood schools) to 7 (Bus to achieve integration).	1.81	1.60	4734	0817
Integrated schools	Indicator variable equal to 1 if the respondent believes that the government should ensure racially integrated schools.	0.45	0.50	5872	0816
Civil Rights legislation	Beliefs over whether civil rights leaders are pushing too fast or going too slowly, ranging from 1 (too fast) to 3 (too slowly).	1.55	0.64	7874	0814
Political campaigning	Activities undertaken (0–5): influencing voters, attending rallies, campaign work, displaying campaign material, and donating.	0.63	1.01	7840	0723
Trust in the fed. government	Degree of trust in the federal government doing the right thing, ranging from 1 (never) to 4 (always).	2.52	0.63	9644	0604
Appendix variables					

(continued on next page)

Variable	Description	μ	SD	N	ID
College attendance	Indicator equal to 1 if the respondent attended college, including some college, junior or community college, and completed college or advanced degrees.	0.25	0.43	8934	0140a
Educational attainment	Educational attainment, ranging from 1 (eight grades or less) to 7 (advanced degree).	3.17	1.72	8934	0140a
Feelings towards (the) …					
Big businesses	Feeling Thermometer (1–100) towards big businesses.	53.56	21.52	6976	0209
Catholics	Feeling Thermometer (1–100) towards people of Catholic religion.	65.22	21.47	5133	0204
Feminist movements	Feeling Thermometer (1–100) towards the feminist movement.	47.46	26.47	5286	0225
Jews	Feeling Thermometer (1–100) towards people of Jewish religion.	62.13	19.87	5075	0205
Poor	Feeling Thermometer (1–100) towards poor people.	73.49	17.32	4516	0223
Unions	Feeling Thermometer (1–100) towards labor unions.	54.54	23.33	7021	0210
White population	Feeling Thermometer (1–100) towards the white population.	78.81	17.72	8197	0207

Note. *ID* is the ANES code of the question, starting with *VCF*. Feelings are measured using the *Feeling Thermometer*, which measures respondents' affect toward a person or group on a 1–100 scale, with 1 indicates very cold/unfavorable feelings, 100 very warm/favorable feelings, and 50 neither warm nor cold. Thes variables are top-coded at 97.

Figure B1: Survey item availability and index distributions



Note. Panel A shows survey item availability across ANES waves; blue indicates availability. Panel B shows the distribution of the *Support for Civil Rights Index* by six-year age cohort in 1946. The index averages available standardized components, with higher values indicating greater support for civil rights. Red identifies the target cohorts (ages 7–18) and blue the reference cohort (ages 19–24). Solid marks indicate cohort means; the dashed line marks zero. Labels report cohort means, standard deviations in index units, and sample sizes. The youngest and oldest categories pool ages ≤ -6 and ≥ 43, respectively.

B.2 Conditional balance of potential exposure R_c

We examine the cross-sectional relationship between exposure and a broad set of pre-broadcast county characteristics from the 1940 Census, together with measures of pre-existing racism and political preferences. We conduct a standard conditional-balance

exercise used in the media literature. The objective is to assess whether, after accounting for factors associated with transmitter placement and the predictable geographic reach of the signal, the remaining variation in reception is systematically related to characteristics determined before *OI*. The assumption is that once these determinants of reception are held fixed, residual differences in exposure should not systematically predict characteristics determined before the broadcast. While we do not require conditional balance in our approach, the absence of such relationships provides evidence consistent with our identifying assumptions (see Section 4).

Columns (1)–(2) in Table B2 report summary statistics for pre-broadcast characteristics. Columns (3)–(5) report unconditional correlations withR_c. Columns (6)–(8) report the estimates from the following cross-sectional specification:

$$L_{cs} = \beta R_c + X'_{cs}\delta + \mu_s + \epsilon_{cs}, \tag{4}$$

where L_{cs} denotes a pre-broadcast characteristic of county c in state s, R_c is the exposure measure defined in equation (1), μ_s denotes state FEs, and ϵ_{cs} is an error term assumed to be clustered at the state level. X_{cs} is a set of controls that we build following the literature exploiting geographic variation in FM radio and TV reception and adapting it to AM broadcasting. We include the following *propagation controls*: the predicted radio coverage under perfect ground conductivity, which holds fixed the location and technical characteristics of transmitters while abstracting from spatial variation in ground conductivity, and therefore captures the geographic reach implied primarily by transmitter placement rather than actual propagation conditions (see Appendix A.2 for further details; this measure plays a role analogous to *free-field coverage* in studies of FM radio and television,as in Olken, 2009). an indicator for the presence of a broadcasting antenna in the county; and the minimum distance between an antenna and the county centroid. Propagation controls are also included in $X_{ihc,t}$ in equation (1). We additionally control for elevation, terrain ruggedness, and ground conductivity at the county centroid, which directly affects signal propagation.

Under conditional balance, $\beta = 0$. Table B2 shows that, unconditionally, exposure is strongly associated with several county characteristics. Conditional on controls, these relationships become substantially weaker for all pre-broadcast characteristics apart from one, which we discuss in Section 4.

Table B2: Conditional balance of the exposure to *OI*

	Pre-broadcast characteristics		Cross-sectional correlation with exposure: Unconditional			Conditional		
	mean	[st.dev.]	β	(s.e.)	R^2	β	(s.e.)	R^2
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Census characteristics								
Area ('000s squ. miles)	0.978	1.356	-0.269**	(0.129)	0.00	-0.056	(0.111)	0.54
Population ('000s per squ. mile)	0.195	1.984	1.161**	(0.559)	0.03	1.320	(0.897)	0.09
Population per dwelling	3.631	0.651	-0.243***	(0.065)	0.01	-0.074	(0.061)	0.35
% male population	0.513	0.020	-0.006***	(0.002)	0.01	-0.003	(0.002)	0.44
Average age (kids)	4.384	0.097	-0.005	(0.012)	0.00	-0.009	(0.008)	0.30
Average age (adults)	35.701	2.518	1.662***	(0.273)	0.04	0.329	(0.326)	0.66
% age $\geq$ 65 y.o.	0.071	0.023	0.008***	(0.003)	0.01	-0.003	(0.004)	0.57
% white population	0.848	0.168	0.022	(0.020)	0.00	-0.016	(0.012)	0.56
% white population (incl. non-native)	0.885	0.180	0.060***	(0.021)	0.01	0.007	(0.010)	0.62
% Black population	0.107	0.178	-0.054**	(0.021)	0.01	-0.009	(0.010)	0.66
% other race	0.008	0.043	-0.005	(0.004)	0.00	0.002	(0.003)	0.20
% illiterate	0.039	0.044	-0.018***	(0.004)	0.01	-0.002	(0.004)	0.48
% enrolled in school (5-17 y.o.)	0.791	0.104	0.066***	(0.014)	0.04	0.040	(0.025)	0.58
% owning their dwelling	0.498	0.118	-0.017	(0.015)	0.00	-0.019	(0.014)	0.51
% city population	0.061	0.194	0.225***	(0.025)	0.12	0.154***	(0.034)	0.19
% urban land	0.340	0.275	0.002	(0.036)	0.00	0.038	(0.035)	0.55
% unemployed	0.102	0.052	-0.013*	(0.007)	0.01	-0.009	(0.009)	0.30
# farms ('000s per squ. mile)	1.968	1.347	0.379**	(0.171)	0.01	-0.020	(0.170)	0.30
Average farm value (log)	15.650	1.755	0.663***	(0.209)	0.01	0.010	(0.286)	0.26
Violence and racism								
Lynchings	0.039	0.265	-0.014	(0.012)	0.00	-0.023	(0.016)	0.12
Lynchings (Black)	0.036	0.251	-0.012	(0.011)	0.00	-0.016	(0.015)	0.12
Green Book establishments	0.087	0.736	0.046	(0.031)	0.00	-0.050	(0.051)	0.02
Electoral outcomes								
Turnout	0.568	0.253	0.094***	(0.033)	0.01	0.014	(0.018)	0.80
% Democratic party	0.589	0.280	-0.085**	(0.038)	0.01	0.022	(0.020)	0.83
% Republican party	0.370	0.252	0.041	(0.034)	0.00	-0.019	(0.020)	0.79

Note. Census characteristics and radio network access are from the 1940 U.S. Census (Haines, 2010). Violence and racism measures include lynchings during 1931–1940 (Bailey et al., 2011) and Green Book establishments per county resident (Cook et al., 2023). Electoral outcomes refer to the 1940 Congressional elections (Clubb et al., 1987). Columns (1)–(2) report means and standard deviations; columns (3)–(5) report OLS coefficients, standard errors, and R^2 from regressions on *OI* exposure (R_c) without controls; and columns (6)–(8) report the corresponding estimates using equation (4). Standard errors in columns (3)–(8) are clustered at the state level, with one observation per county.

B.3 Stability of the exposure gradient outside the target cohorts

Figure B3 presents cohort-study estimates for each component of the *Support for Civil Rights Index* and the outcomes related to broader political attitudes and behavior. As discussed in Section 4, our identifying assumption is strengthened if the relationship between potential exposure to *OI* and later outcomes is stable across older cohorts. Table B3 reports the results of two stability tests for all outcomes in Tables 2, 4, and 5. First, we test whether exposure gradients among older cohorts jointly differ from those of the omitted cohort in equation (1). Using the full sample specification in equation (1), we jointly test $H_0^J : \gamma_\tau = 0$ for all $\tau \in \{25\text{–}30,\ 31\text{–}36,\ 37\text{–}42,\ 43+\}$, where γ_τ is the exposure gradient for cohort τ relative to the omitted cohort, aged 19–24 in 1946. Second, we test whether older cohorts exhibit a linear trend in the outcome variable. To do so, we restrict the sample to individuals aged 19 or older in 1946 and replace the cohort-specific interactions $\sum_\tau \gamma_\tau (D_\tau \times R_c)$ in equation (1) with $s_h \times R_c$, where $s_h = 0$

is for the omitted cohort aged 19 to 24 and is increased by one for each six-year cohort step toward younger cohorts. We, therefore, estimate the following specification:

$$Y_{ihct} = \lambda(s_h \cdot R_c) + \mu_c + \mu_h + \mu_{a_{it}} + \mu_{t\times state} + (X_{ihct} \cdot D_h) + (X_{ihct} \cdot S_t) + \epsilon_{ihct}, \quad (5)$$

where λ measures the change in the exposure gradient per cohort step toward the target cohorts and tests $H_0^L : \lambda = 0$. The first test allows arbitrary departures across older cohorts, while the second test has greater power against systematic directional patterns.

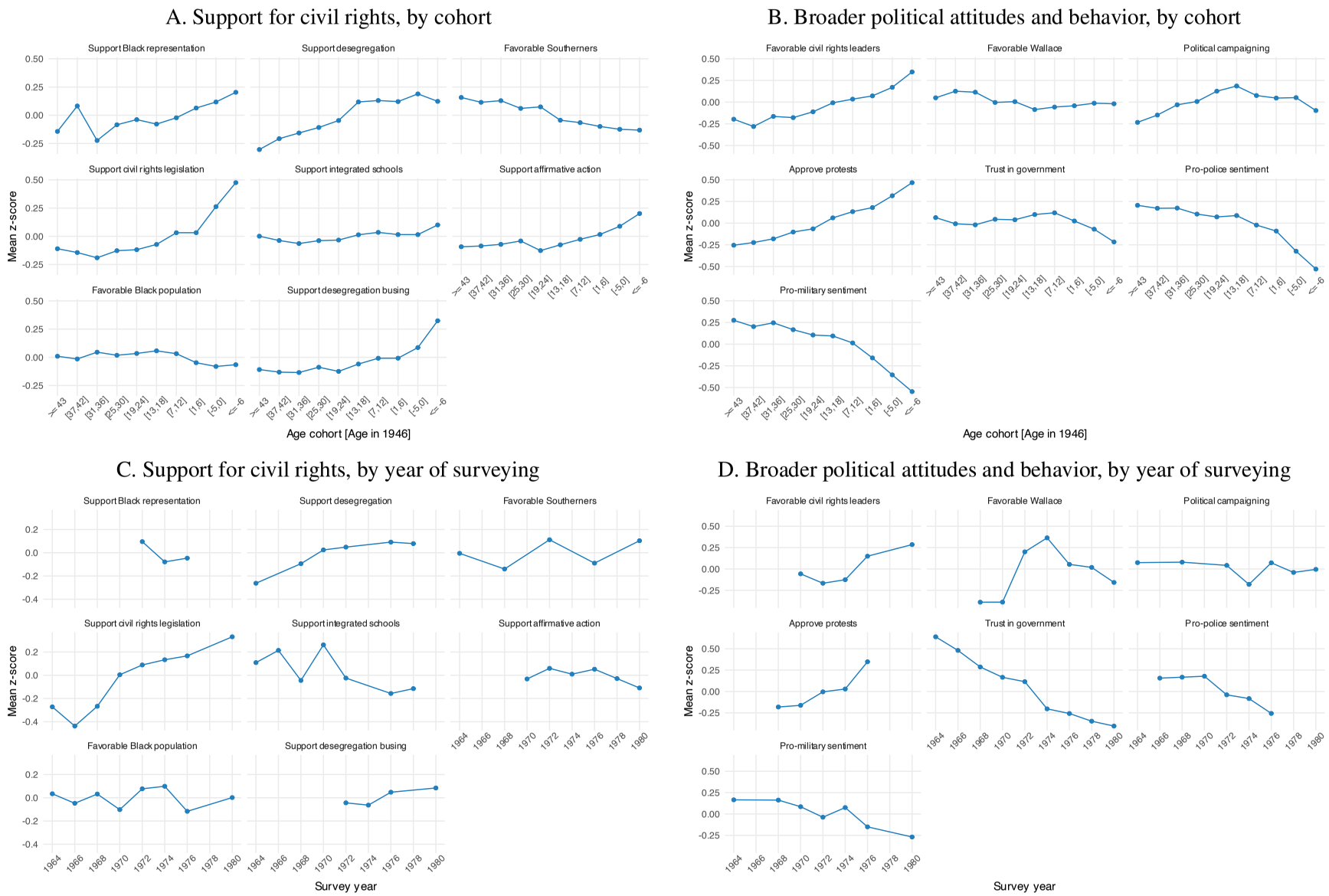
We complement these tests with an additional robustness exercise, assessing whether the results for the *Support for Civil Rights Index* are driven by any of its individual outcome components. Figure B4 reports the estimates for the effect on the target group by re-computing the index after excluding one component at a time and using an alternative index that excludes the two components for which either diagnostic rejects stability at the 10% level. Our main results are robust both to excluding any individual component and to simultaneously removing the components for which the older-cohort stability diagnostics may raise concerns.

Table B3: Tests for stability of exposure gradients across older cohorts

	Joint test	**Linear cohort gradient**	
Outcome	p-value	Slope λ (s.e.)	p-value
Approve of protest	0.853	0.005 (0.014)	0.730
Civil Rights leaders	0.911	0.004 (0.019)	0.838
Favorable towards Black population	0.796	0.000 (0.018)	0.991
Favorable towards Southerners	0.062	0.036 (0.015)	0.016
George Wallace	0.671	0.015 (0.012)	0.225
Interracial friendships	0.160	0.002 (0.008)	0.826
Interracial marriages	0.949	0.163 (0.221)	0.461
Political campaigning	0.834	0.014 (0.018)	0.419
Pro-military sentiment	0.542	0.008 (0.014)	0.579
Pro-police sentiment	0.647	0.026 (0.016)	0.096
Support for affirmative action	0.005	0.010 (0.021)	0.642
Support for Black representation	0.897	0.008 (0.028)	0.785
Support for Civil Rights Index	0.958	-0.005 (0.010)	0.594
Support for Civil rights legislation	0.726	-0.023 (0.016)	0.152
Support for desegregation	0.464	0.016 (0.018)	0.368
Support for desegregation busing	0.458	-0.000 (0.020)	0.996
Support for integrated schools	0.742	-0.008 (0.017)	0.615
Trust in the federal government	0.308	0.004 (0.012)	0.768

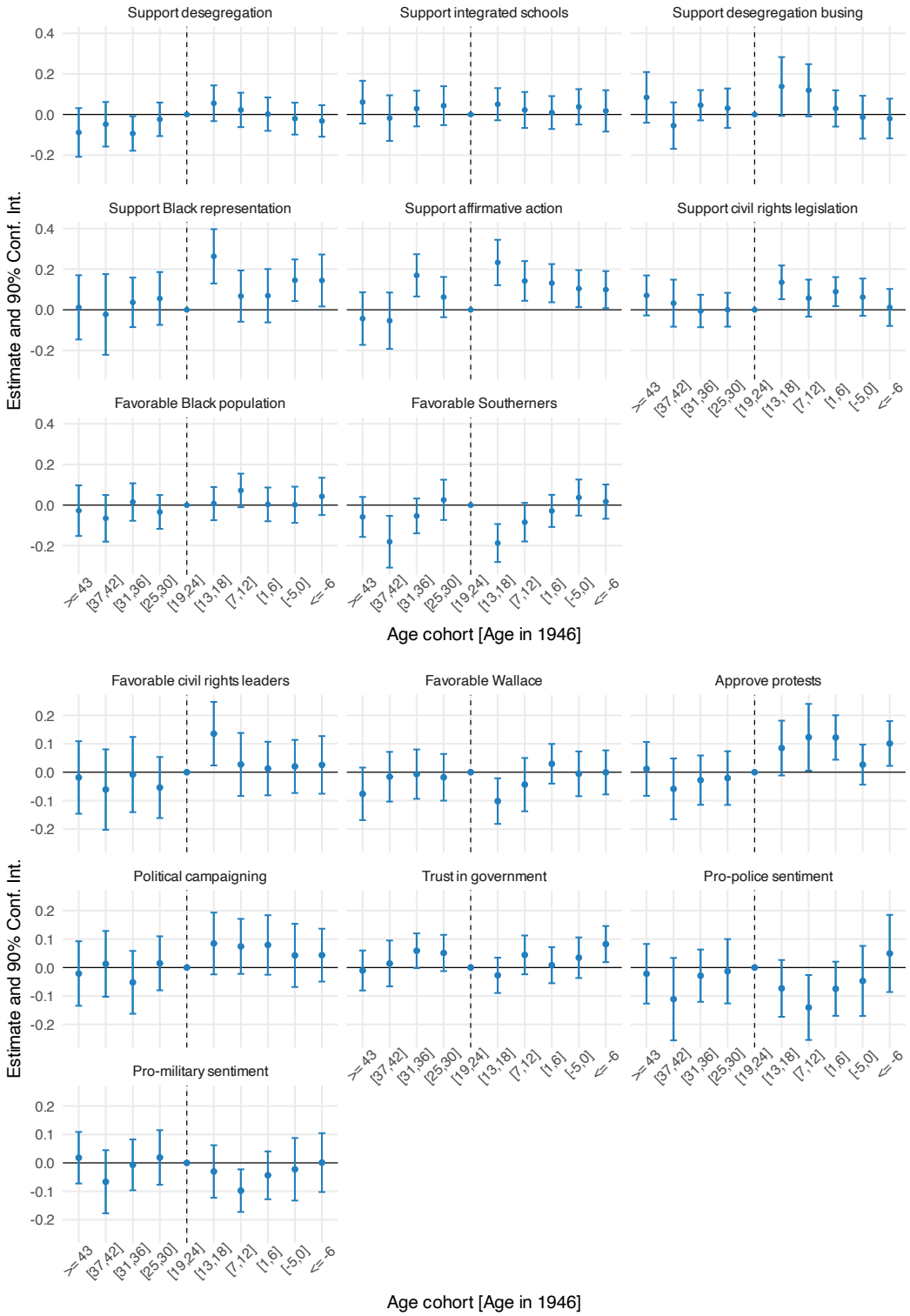
Note. The joint test reports the p-value from a Wald test that all exposure-by-cohort coefficients for cohorts older than the reference cohort (ages 19–24 in 1946) are jointly zero. The linear test estimates an exposure-specific linear gradient among the older placebo cohorts; the coefficient is expressed per six-year cohort step toward the target cohorts. Both tests use the cohort-study controls and fixed effects. The category aged 43 or older is pooled. The ANES estimates use county-clustered standard errors; the interracial-marriage estimates use household-clustered standard errors, as in the main-text table.

Figure B2: Descriptive statistics on attitudes, by cohort and survey year



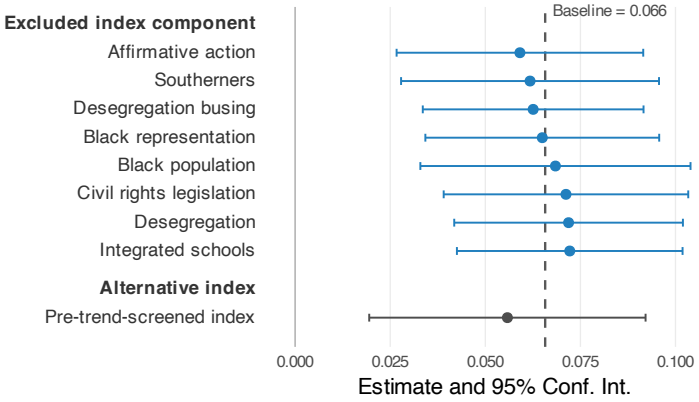
Note. Panels A and B show the mean attitudes by age cohort for each of the respective survey items. Cohorts are six-year age-in-1946 intervals, with ages ≤ -6 and ≥ 43 pooled in the endpoint categories. Panels C and D show mean attitudes by survey year. Outcomes are standardized and centered at 0. The definition of variables and data availability for each question is reported in Appendix B.1.

Figure B3: Cohort-study estimates by survey item



Note. The figure reports estimates from equation (1) separately for the components of the civil-rights index (Panel A) and the additional political-attitude outcomes (Panel B). The points represent the coefficients on the interactions between standardized Superman exposure and the age-cohort indicators. Respondents aged 19–24 in 1946 constitute the omitted reference cohort. The whiskers report 90% confidence intervals based on standard errors clustered at the county level. All outcomes are standardized z-scores. See Appendix B.1 for variable definitions.

Figure B4: Robustness of results to excluding individual components of the index

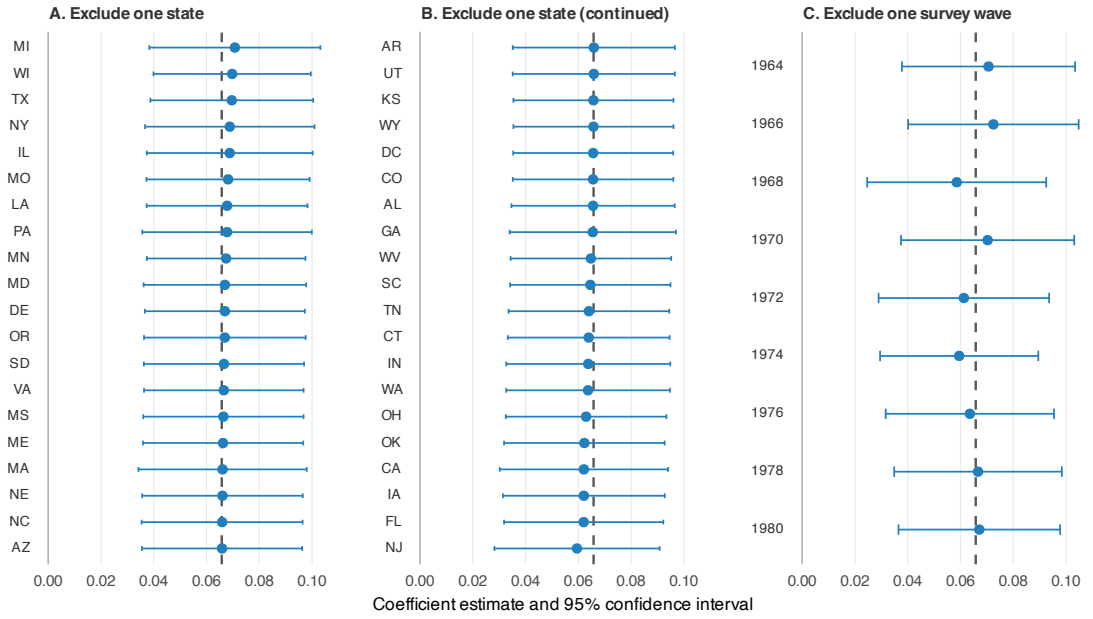


Note. The figure reports estimates of the interaction between the target-cohort indicator and exposure to *OI*. Target is an indicator equal to 1 for respondents aged 7–18 in 1946, while exposure is the standardized share of the county population covered by the broadcast signal of stations airing *OI* in 1946. We report estimates for the Support for Civil Rights Index, reconstructed after excluding each component in turn, and an alternative index that excludes attitudes toward affirmative action and attitudes toward Southerners. All specifications use the target-versus-comparison-cohort version of equation (1), include the controls and FEs described in Section 4, and retain all available cohorts. Whiskers report 95% confidence intervals based on standard errors clustered at the county level. The solid vertical line indicates zero, while the dashed vertical line indicates the baseline estimate reported in the main text.

B.4 Robustness to influential states and survey waves

To assess whether our findings are disproportionately driven by the geographic or temporal composition of the ANES sample, we conduct two leave-one-out exercises. First, we re-estimate the baseline specification after excluding each state in turn, testing sensitivity to influential state-specific circumstances. Second, we exclude each survey wave in turn, testing whether the result depends on a particular survey period or the outcomes available in that wave. Figure B5 shows that the estimates remain statistically significant and range from 0.058 to 0.072 index units across all exclusions, closely surrounding the full-sample estimate of 0.066. The result, therefore, cannot be attributed to any individual state or survey wave.

Figure B5: Leave-one-out robustness across states and survey waves



Note. The figure reports leave-one-out estimates of the interaction between the target-cohort indicator and exposure to *OI*. Panels A and B exclude each indicated state in turn, and Panel C excludes each indicated survey wave in turn. Target is an indicator equal to 1 for respondents aged 7–18 in 1946; exposure is the standardized share of the county population covered by the broadcast signal of stations airing *OI* in 1946. All specifications estimate equation (2), use all available cohorts in the remaining sample, and include the controls and FEs described in Section 4. Whiskers report 95% confidence intervals based on standard errors clustered at the county level. The solid vertical line indicates zero, and the dashed vertical line indicates the full-sample baseline estimate.

B.5 Political and war mobilization

We focus on two forms of mobilization. First, political mobilization in the form of pro- and anti-segregation organizations, measured by the presence of KKK and NAACP chapters. KKK chapter locations are obtained from Kneebone and Torres (2015) for 1932 and 1942, Committee on Un-American Activities (1967) for 1965, and SPLC

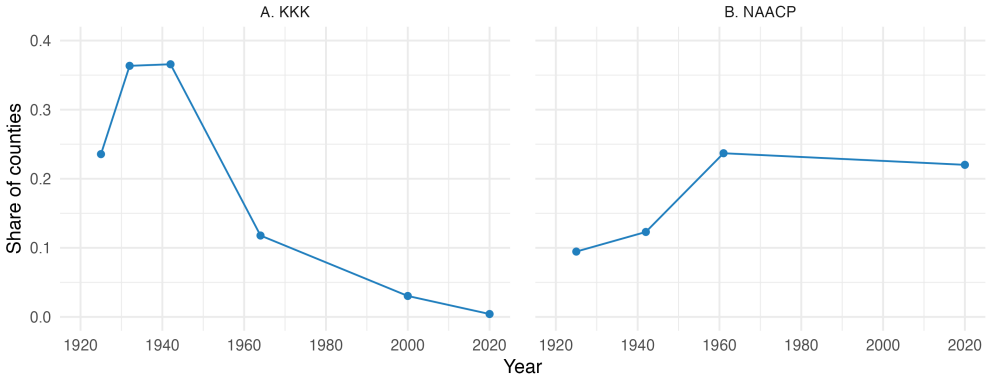
(2023) for 2000 and 2020. NAACP branch locations are obtained from Estrada and Hermida (2023) for 1925, 1942, and 1961, and from our web scraping for 2020. Figure B6 shows the share of counties with the presence of both groups. Second, we focus on the Vietnam War as a measure of war mobilization. Civil rights organizations, such as the *Nonviolent Coordinating Committee* (SNCC), explicitly advocated civil rights work as an alternative to the voluntary draft. DCAS records contain the service member's date of birth and the *home of record* address; that is, the place recorded as the home of the individual when commissioned, appointed, enlisted, inducted, or ordered on active duty. Due to the young age of most soldiers, we assume that this is the place where they grew up. DCAS does not provide information on whether the record refers to a draftee or a volunteer. Eligible men aged 17 or older had the option to volunteer. Alternatively, at age 18, men were required to report to their local draft board. Those classified as available for service could volunteer, allowing them to choose their service branch and serve for a shorter period. Draftees were typically assigned to the Army and served for up to three years. After removing casualties assigned to a hometown address that corresponds to a military base and those caused by self-inflicted harm, we obtain a total of 56,779 casualties (out of 57,241), of which 87% are white soldiers and 13% are African American soldiers. Casualties are concentrated in the period 1965–1973, with an average age at death of 23. Panel A in Figure B7 shows the geographical distribution of the average value (across all cohorts) of our main outcome (i.e., the ratio between the number of casualties in cohort h whose hometown was in county c, and the total number of casualties in cohort h, multiplied by 1,000). Panel B plots cohort- study estimates using equation (1) for participation in the Vietnam War. Table B4 provides the same estimate using equation (1) pooling the target cohorts, but restricting the sample to counties with a larger number of casualties.

Table B4: The effect on the participation in the Vietnam War, robustness checks

	Dependent variable: Vietnam War participation			
Counties selected:	$>$ 1 casualty	$>$ 5 casualties	$>$ 10 casualties	$>$ 25 casualties
	(1)	(2)	(3)	(4)
Expsoure $\times$ Target	-0.082***	-0.108***	-0.154***	-0.252***
	(0.022)	(0.029)	(0.041)	(0.073)
Dependent variable mean	1.62	1.62	1.62	1.62
R^2	0.756	0.763	0.775	0.802
Counties	2,639	1,651	972	410
Observations	55,419	34,671	20,412	8,610

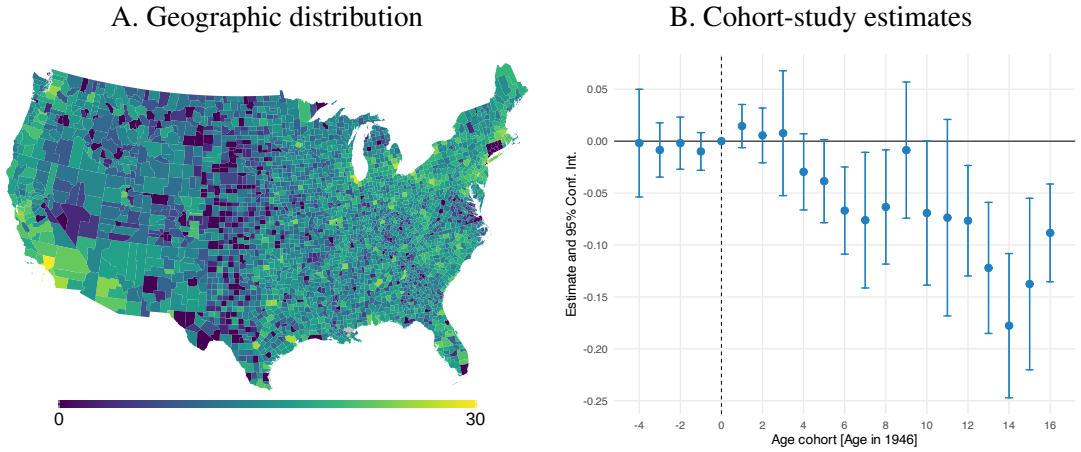
Note. The dependent variable is the county's percentage share of nationwide Vietnam War casualties within each cohort, multiplied by 1000. Specifications include county and state-by-cohort FEs, and cohort-specific controls. *Target* identifies cohorts aged 7–18 in 1946; *Exposure* is standardized county population coverage by stations broadcasting *OI*. County sample restrictions are indicated in the column headers. Standard errors are clustered by county and reported in parentheses (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Appendix B.5 describes the data.

Figure B6: Mobilization against and in favor of civil rights, over time



Note. The figure shows the share of counties where KKK (panel A) or NAACP (panel B) are present. Estrada and Hermida (2023) provide the list of active chapters in two periods: 1957–1963 (which we label as 1961), and in 1964. We do not use information for 1964 due to the large discrepancy between the number of geolocated chapters in this source and the aggregate number of chapters reported by NAACP for the same year. For the year 2020, we collected addresses of currently active branches by scraping all active NAACP websites, Google Maps locations, and the list of chapters registered in the U.S. Internal Revenue Services database. We geolocated each branch using the city reported in its address. Web scraping was performed in May 2023, but we assign the year 2020 because we do not observe the exact moment at which a branch became active.

Figure B7: Share of casualties (per 1,000 deaths) during the Vietnam War



Note. Panel A shows the geographical distribution of the share of casualties per county. The data source is the DCAS dataset (see Appendix B.5). In panel B, estimates based on equation (1). The dependent variable is the cohort-specific share of casualties assigned to a county based on the place of birth (multiplied by 1,000). We estimate equation (1) at the cohort-county level, thus comparing the distribution of the outcome variable within a cohort across counties with varying exposure of *OI* in 1946. The cohort -6 includes all cohorts whose age in 1946 was equal or smaller than -6. The cohorts 16 includes all cohorts whose age in 1946 was equal or larger than 16. The error bars indicate 95% confidence intervals. The vertical line indicates the cohort born when *OI* was launched.

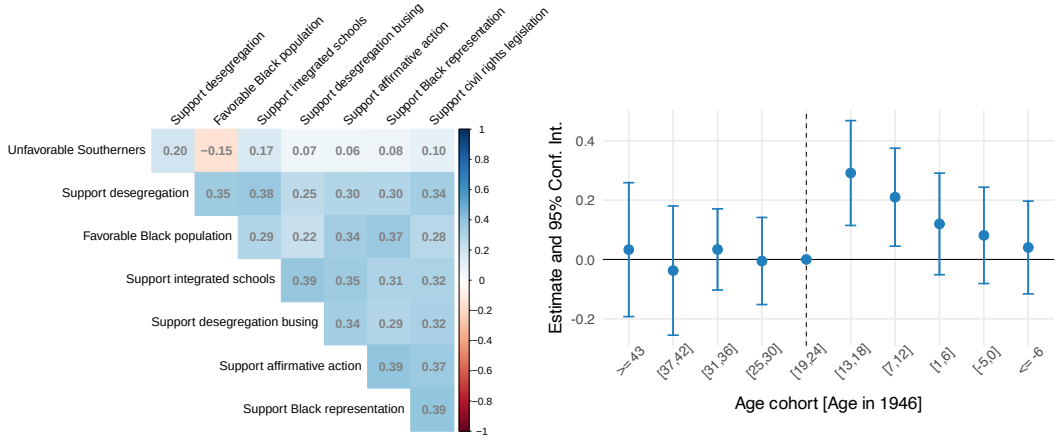
B.6 Robustness checks for the Support for Civil Rights index

Panel A of Figure B8 shows the partial correlation of each variable used to build the *Support for Civil Rights* index. To address the relatively high share of missing observations in our dataset, we complement the results presented in Figure 3 with an index built on a regularized iterative principal component analysis (RPCA; Josse and Husson, 2012). We use the first principal component, capturing 49.14% of the variation in our

sample. Panel B replicates panel B in Figure 3 using this index. Table B5 presents estimates using equation (2), while varying the set of controls and FEs compared to the specification presented in the main text. Table B6 shows the robustness of the results using survey weights, alternative coverage measures, and assuming spatial correlation in error terms with the Conley (1999) correction.

Figure B8: Support for Civil Rights Index, correlations and alternative index

A. Partial correlations B. Cohort-study estimates on RPCA index



Note. Panel A shows pairwise correlations across standardized survey items. Panel B presents cohort-study estimates for the RPCA-based *Support for Civil Rights index* (Appendix B.6), obtained from equation (1) by interacting age cohorts in 1946 with standardized county-level exposure to *OI*. Error bars represent 95% confidence intervals based on standard errors clustered at county level. The vertical dashed line identifies the omitted cohort (ages 19–24 in 1946). See Appendix B.1 for variable definitions.

Table B5: Robustness checks – alternative specifications for equation (2)

	Dependent variable: Support for Civil Rights Index							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Target × Exposure	0.067***	0.060***	0.059***	0.070***	0.066***	0.066***	0.063***	0.050***
	(0.013)	(0.015)	(0.015)	(0.016)	(0.015)	(0.015)	(0.016)	(0.017)
R^2	0.139	0.145	0.186	0.188	0.204	0.208	0.208	0.404
Observations	10,364	10,364	10,364	10,364	10,364	10,364	10,364	8,848
Propagation controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × survey-year FEs			Yes	Yes	Yes	Yes	Yes	Yes
Urban trends				Yes	Yes	Yes	Yes	Yes
Individual controls					Yes	Yes	Yes	Yes
Cohort × survey-year FEs						Yes		
Geographic controls							Yes	Yes
Extended individual controls								Yes

Note. Estimates are based on equation (2) and use all available cohorts. All specifications include county FEs. Column (1) includes survey-year and six-year cohort FEs. Column (2) adds the propagation controls discussed in Appendix A.2, interacted with cohort and survey-year indicators. Column (3) adds state-by-survey-year FEs. Column (4) adds interactions between cohort and survey-year indicators and the county population share living in cities in 1940. Column (5), our baseline specification, adds the baseline individual controls and their interactions with cohort and survey-year indicators. Column (6) allows each six-year cohort to have its own survey-year FE in an otherwise comparable specification. Column (7) adds geographic controls, including elevation, terrain ruggedness, and ground conductivity. Column (8) adds the extended set of individual controls, including race, employment status, income, marital status, education, religion, and an indicator for having grown up in a city. *Target* equals one for respondents aged 7–18 in 1946. *Exposure* is the standardized share of the county population covered by the broadcast signal of stations airing *OI* in 1946. Standard errors are clustered by county and reported in parentheses (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Descriptive statistics and the temporal coverage of the index components are reported in Appendix B.1.

Table B6: Robustness checks – weights, alternative radio coverage, and inference

	Dependent variable: Support for Civil Rights Index							
	Alternative measures of coverage					Conley Hsiang		
	Survey weights	Optimistic $\tau = 60$	Pessimistic $\tau = 72$	Area coverage	Binary	50km	100km	400km
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Target × Exposure	0.063***	0.051**	0.051***	0.057***	0.158***	0.066***	0.066***	0.066***
	(0.015)	(0.021)	(0.013)	(0.014)	(0.041)	(0.013)	(0.014)	(0.010)
R^2	0.208	0.203	0.204	0.204	0.204	0.204	0.204	0.204
Observations	10,364	10,364	10,364	10,364	10,364	10,364	10,364	10,364

Note. Estimates are based on equation (1) aggregating target and control cohorts into two groups. We include all available cohorts. All specifications include the set of controls and FEs described in Section 4. *Target* is an indicator variable equal to 1 for respondents who were 7–18 years old in 1946. *Exposure* denotes the share of the population in the county covered by the radio signal of *OI* in 1946. Standard errors, clustered by county, are reported in parentheses (* p < 0.1, ** p<0.05, *** p < 0.01).

B.7 Anticipation, spillover and long-run salience effects

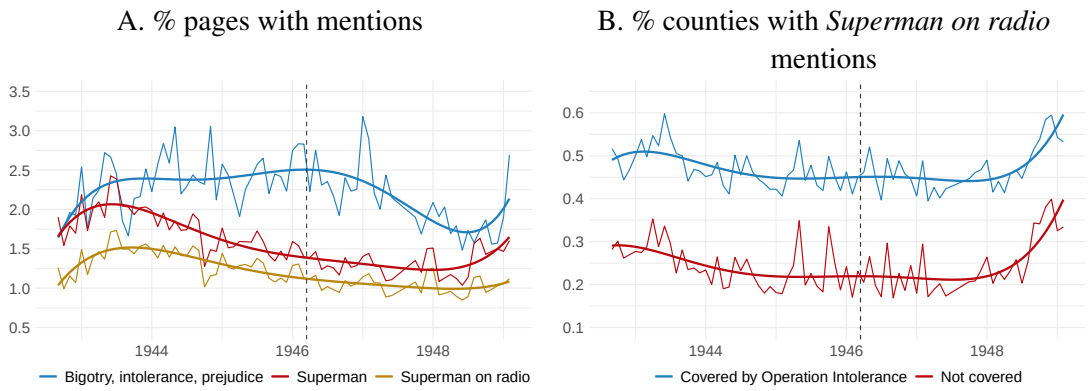
To analyze anticipation and spillover effects, we examine the share of newspaper pages that cover topics related to *OI* around its launch in April 1946. We focus on three topics. *Superman* refers to pages where the word *Superman* appears at least once. *Superman on radio* refers to pages where the words *Superman* and *radio* appear on the same page at least once, covering a wide range of references, including articles about Superman, radio schedules, and comic strips featuring Superman. *Bigotry, intolerance, and prejudice* refer to pages where any of the words—*bigotry*, *intolerance*, *prejudice*, or their derivatives—appear at least once.

Panel A in Figure B9 shows the share of pages that mentioned these themes from August 1942 to February 1949. Focusing only on *Superman on radio*, panel B in Figure B9 shows the share of counties in which local newspapers made at least one reference to this theme, depending on whether a county was covered by the broadcast in 1946. Overall, we do not observe different patterns following April 1946. We test whether *OI* brought about a short-term change in the salience of these themes in local newspapers using equation (3) applied to the data presented in Figure B9, transformed into quarterly data to minimize missing values. Table B7 presents the results for the share of pages with mentions in columns (1)–(3) and the probability of having at least one mention in columns (4)–(6).

To measure the long-run salience of civil rights, we focus again on local newspapers. Panels A and B in Figure B10 show descriptive statistics for the data gathered from the online archive `newspapers.com`. We use information from newspapers to build a measure of the salience of civil rights using the 25 themes.[3] The themes are

[3]The list is AI-generated by ChatGPT 3.5 using the following request: *I am an academic economist. I am analyzing a sequence of U.S. local newspapers published in the period 1930–1980. I am interested in finding articles promoting the narrative of the Civil Rights Movement and of racial tolerance, but I can only search for words or sequences of two words. Can you list the most important ones I should focus*

Figure B9: Newspaper coverage of Superman and topics related to *OI*



Note. Panel A displays the share of pages published in local newspapers that cover specific topics, with shares multiplied by 100. It covers August 1942 to February 1949. Panel B shows for each month the share of counties in which there is at least one reference to *Superman on radio* in the same page of a newspaper. Counties covered by *OI* are those in which a positive share of inhabitants was covered by *OI* in 1946. See Section 3.1 for the procedure to compute coverage. Thicker lines provide a smoothed version of each series. Section 5.4 provides further details about the source of newspaper data.

Table B7: The short-run effect of *OI* on local newspapers

	Share of pages with mentions of ...			**Any mention of ...**		
	Superman	Superman on radio	Bigotry, intolerance, and prejudice	Superman	Superman on radio	Bigotry, intolerance, and prejudice
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure × post	-0.033	-0.022	-0.047	0.007	-0.002	-0.002
	(0.051)	(0.043)	(0.031)	(0.007)	(0.008)	(0.005)
Dependent variable mean	0.999	0.695	2.08	0.589	0.470	0.901
R^2	0.557	0.587	0.639	0.516	0.546	0.338
Number of counties	906	906	906	906	906	906
Observations	24,462	24,462	24,462	24,462	24,462	24,462

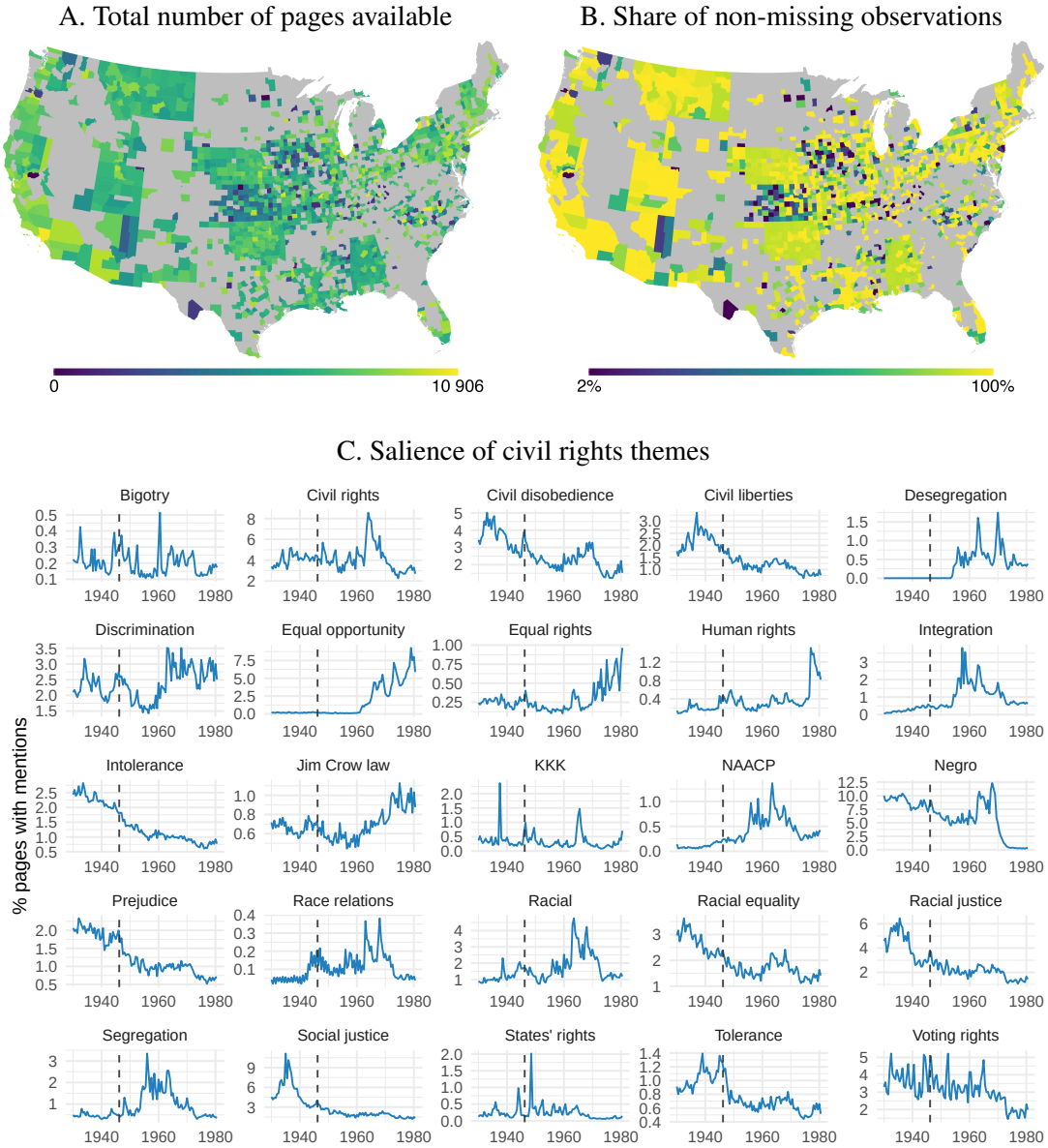
Note. Estimates based on equation (3) using quarterly data covering August 1942 to February 1949. Standard errors clustered at the county level are presented in parentheses (* p < 0.1, ** p<0.05, *** p < 0.01). *Post April 1946* is an indicator variable equal to 1 if the period of observation is posterior to the April 1946. *Exposure* is the share of the population in the county that was covered by the radio signal of *OI* in 1946. Dependent variables are reported in the column's header and defined as follows: (1)–(3) share of pages published in local newspapers that cover specific topics, with shares multiplied by 100; (4)–(6) indicator variable equal to 1 if there is at least one reference to the specific topic in the local newspapers.

presented in panel C, which summarizes the salience of each topic over time. Following Beach and Hanlon (2022), we measure salience using a bag-of-words approach. For each county c and theme k, we count the number of newspaper pages mentioning the words associated with that theme, $pages_{kc,t}$, and divide it by the total number of newspaper pages published in the county, $all_pages_{c,t}$. Our corpus contains more than 51 million pages published between 1930 and 1980. To reduce noise, we aggregate the monthly data into 36-month periods relative to the launch of *OI*, indexed by y. We define the salience of theme k in county c and period y as the ratio between $pages_{kc,t}$ and $all_pages_{c,t}$, and we compute this measure for 25 civil-rights-related themes observed between 1930 and 1980. Panel C of Figure B10 shows the share of total pages

on? Restrict the result to words that were in use throughout the period and not only after the 1950s.

published in each half-year that contain a specific theme included in our measure of the salience of civil rights in local newspapers.

Figure B10: Local newspapers and salience of civil rights themes, 1930–1980



Note. Panel A shows the geographical distribution of the total number of pages per county in the period January 1930 – December 1980. Panel B shows the share of non-missing observations, when we compute the total number of pages at yearly level using all available data within each county-year. The number of pages is aggregated within each county by year relative to the launch of *OI* in April 1946. Gray areas are counties without any data in the archive. Panel C shows the share of the total pages published in each six-month period that contain a specific theme. The share is aggregated for the whole U.S. and by semester. Each data point includes one year relative to the launch of *OI* in April 1946. For instance, the data point for year 1946 includes the period April 1946 - March 1947. The vertical line indicates the year 1946, when *OI* was launched. Searches include alternative terms and expressions within each theme. For example, the KKK theme includes "klan," "klux," "lynching," and "lynched," while racial equality combines race" or racial" with equal*" or inequal*."

In the main text, we aggregate each measure using the regularized iterative principal component analysis (RPCA), retaining counties observed in at least 50% of peri-

Table B8: The effect of *OI* on the salience of civil rights

	Salience of civil rights on local newspapers							
Method:	RPCA		RPCA		PCA		PCA	
Counties selected (obs. per county):	≥ 1		$\geq 75\%$		100%		100%	
Themes selected:	All		All		Top 75% of themes		Top 50% of themes	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post 1946 × Exposure	0.156**		0.168*		0.176*		0.193**	
	(0.076)		(0.089)		(0.092)		(0.085)	
Period 1946–55 × Exposure		0.049		0.035		0.005		0.051
		(0.067)		(0.066)		(0.077)		(0.072)
Post 1955 × Exposure		0.192**		0.212**		0.233**		0.240**
		(0.088)		(0.105)		(0.106)		(0.097)
Dependent variable mean	0.008	0.008	0.005	0.005	-0.008	-0.008	-0.008	-0.008
R^2	0.594	0.594	0.686	0.687	0.762	0.762	0.780	0.781
Number of counties	1,467	1,467	804	804	483	483	483	483
Observations	24,939	24,939	13,668	13,668	8,211	8,211	8,211	8,211

Note. Estimates based on equation (3). Columns (1), (3), (5) and (7) pool post-1946 observations; columns (2), (4), (6) and (8) estimate separate effects for the two post-intervention periods. Standard errors clustered at the county level are presented in parentheses (* $p < 0.1$, ** $p<0.05$, *** $p < 0.01$). *Post 1946* equals 1 for periods beginning in April 1946 or later. *Period 1946–55* equals 1 for periods covering April 1946–March 1955. *Post 1955* equals 1 for periods beginning in April 1955 or later. *Exposure* is the standardized share of the population in the county that was covered by the radio signal of *OI* in 1946. Dependent variables are reported in the column's header and defined as follows: (1)–(2) index built using RPCA including all themes and selecting counties with at least one non-missing observation; (3)–(4) index built using RPCA including all themes and selecting counties with at least 75% of non-missing observations; (5)–(6) index built using PCA excluding search series whose pre-1946 99th-percentile salience is at or below the 25th-percentile cutoff; (7)–(8) index built using PCA excluding search series whose pre-1946 99th-percentile salience is at or below the 50th-percentile cutoff. Appendix B.7 provides additional details about the variables.

ods, winsorizing each series at the 1st and 99th percentiles, and using the first principal component. Table B8 instead provides event study estimates using alternative indices. Columns (1), (3), (5) and (7) provide estimates for the post-1946 periods using equation (3) and pooling all post-1946 observations, while columns (2), (4), (6) and (8) split the estimate of the effect in two periods, the 1946-1955 period and the post-1955 period (with 1955 being a crucial year due to the Montgomery bus boycott).

B.8 Attitudes towards other groups and educational attainment

Table B9 provides cohort-study estimates on the effect of *OI* on favorable feelings toward other groups. Columns (1)–(2) focus on attitudes towards the main contemporary religious minority groups. Columns (3)–(4) provide estimates related to societal and economic inequality. Columns (5)–(6) present estimates for attitudes towards feminist movements and the white population. Columns (7)–(8) provide estimates on educational outcomes, measured in the sample of respondents aged 25 and older.

Table B9: Attitudes toward other groups and educational attainment

	Attitudes toward other groups						Education	
	Catholics	Jews	Poor	Big business	Feminist movements	Whites	Educational attainment	College attendance
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Target × Exposure	-0.016	0.003	0.006	-0.024	0.003	-0.035	0.003	0.013
	(0.035)	(0.032)	(0.042)	(0.029)	(0.041)	(0.029)	(0.027)	(0.013)
R^2	0.213	0.195	0.199	0.182	0.231	0.208	0.289	0.190
Observations	5,133	5,075	4,516	6,976	5,286	8,197	8,934	8,934

Note. Estimates use all available cohorts and the controls and fixed effects described in Section 4. Attitude outcomes and educational attainment are standardized z-scores; college attendance is a binary indicator. Both education outcomes are restricted to respondents aged 25 or older. *Target* is an indicator variable equal to 1 for respondents who were 7–18 years old in 1946. *Exposure* denotes the standardized share of the population in the county covered by the radio signal of *OI* in 1946. Standard errors are clustered by county and reported in parentheses (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Appendix B.1 provides variable definitions.

B.9 Long-run vote-shift rate

To quantify the long-run electoral effects of *OI*, we construct a long-run vote-shift rate, defined as the reduction in support for segregationist candidates associated with greater childhood exposure to the broadcast, relative to the counterfactual support for such candidates. This measure builds on the distinction between average treatment effects and persuasion among individuals whose behavior could change (see, e.g., Jun and Lee, 2026), but it differs from conventional persuasion rates in both interpretation and normalization. While persuasion studies often focus on short-run behavioral responses through changes in information, beliefs, or salience, our setting concerns the formation of values during childhood and their persistent influence on political choices decades later. Therefore, rather than adjusting for the actual listenership of a message, as in persuasion rates, we normalize the electoral effect by counterfactual support in the absence of any exposure to avoid reflecting effects that might operate beyond direct exposure to a message over very long horizons.

Let v_0 denote the counterfactual Wallace vote share in the 1968 presidential election, evaluated at the minimum observed level of geographic exposure. We define the long-run vote-shift rate (LVR) as

$$LVR = -\frac{\beta}{v_0},$$

where β is the DiD estimate of the effect of a one-standard-deviation increase in geographic radio coverage (R_c in the main text) on segregationist voting between 1948 and 1968. The rate expresses the estimated electoral shift as a proportion of counterfactual Wallace support.[4]

[4]Our measure uses the difference-in-differences estimate $\beta = \partial v_{1968}(s)/\partial s - \partial v_{1948}(s)/\partial s = -0.024$, where v_{1968} and v_{1948} denote Wallace and Thurmond vote shares, respectively. It therefore captures the differential exposure gradient in segregationist voting across elections, rather than the exposure gradient in Wallace support alone.

We estimate v_0 using the estimates presented in column (1) of Table 6, evaluating predicted voting outcomes at the minimum observed level of exposure in the states where both Thurmond in 1948 and Wallace in 1968 appeared on the ballot. Aggregating predicted votes yields $v_0 = 0.377$. Our estimates in Section 5.4 indicate that a one-standard-deviation increase in exposure reduces segregationist vote shares by 2.4 percentage points. The corresponding long-run vote-shift rate for the overall electorate is therefore $LVR^{All} = \frac{0.024}{0.377} = 0.064$. Thus, the estimated electoral shift corresponds to 6.4% of counterfactual Wallace support.

We supplement this statistic with two alternative exercises. First, we compute an alternative long-run vote-shift rate that attributes the aggregate electoral response entirely to the target generation. The 6.4% LVR assumes that the effect of *OI* spreads across all cohorts. While we cannot identify the contribution of each cohort to the vote share in 1968, our cohort-study findings indicate that the effects of *OI* were concentrated among these cohorts. Assuming that the electoral response is concentrated entirely among the target cohorts provides a benchmark for the maximum magnitude of their electoral response, provided that any effects on other cohorts operate in the same direction. We first compute the population aged 29–40 in 1968, corresponding to individuals aged 7–18 in 1946. This cohort represented approximately 13.7% of the total U.S. population of 201.2 million (Bureau of Census, 1968a). Applying this proportion to the population of the selected states (49.7 million; Bureau of Census, 1969a) yields approximately 6.8 million individuals aged 29–40. Assuming a turnout rate of 53.6%, we estimate that 3.6 million individuals in this cohort voted in the 1968 presidential election.[5] We compare this estimate with the approximately 12.4 million votes cast in the selected states, computed by applying the turnout rate in the selected states (46.5%; Bureau of Census, 1969b) to their voting-age population of 26.6 million (Bureau of Census, 1968b). Under these demographic and turnout assumptions, we estimate that the target cohorts accounted for approximately 29.5% of votes cast in the selected states in 1968. Letting β^T and β^O denote the electoral effects for target cohort voters and other voters, respectively, and assuming that the target cohort's share of voters does not vary with geographic exposure, the aggregate effect can be decomposed as $\beta = 29.5\%\beta^T + 70.5\%\beta^O$. Further assuming that the electoral response is concentrated entirely among the target cohorts, such that $\beta^O = 0$, we obtain:

$$\beta^T = \frac{\beta}{p} = \frac{-0.024}{0.295} = -0.081.$$

[5] Using Bureau of Census (1969b), we estimate turnout among those aged 29–40 from the reported age-group rates (18–24 and 25–44), then adjust it proportionally for the lower turnout in the selected states relative to the U.S. overall (46.5% versus 63.3%).

To normalize this effect, we assume that, absent *OI*, target-cohort voters would have supported Wallace at the same rate as the overall electorate, implying $v_0^T = 0.377$. The corresponding long-run vote-shift rate is $LVR^T = \frac{0.024}{0.295 \times 0.377} = 0.216$. Under these assumptions, a one-standard-deviation increase in childhood exposure to *OI* is associated with a reduction in segregationist voting equivalent to 21.6% of the target generation's counterfactual Wallace support.

Second, we complement the LVR with a conventional persuasion rate, under the restrictive assumption that the long-run electoral effects of *OI* operate exclusively through direct listenership. Following Enikolopov et al. (2011) and assuming no effect on turnout, the persuasion rate for a message discouraging support for a candidate is

$$f = -\frac{1}{v_0}\frac{dv/ds}{de/ds}, \tag{6}$$

where dv/ds is the effect on segregationist voting and de/ds is the corresponding change in listenership. Setting $dv/ds = \beta$, this measure differs from the LVR by its additional normalization for listenership.

Because individual listenership of *OI* is unobserved, we approximate it using census data on population by age and radio ownership (Haines, 2010), combined with predicted radio coverage. Specifically, we assume that members of the target cohorts in radio-owning households could listen to the broadcast when coverage reached at least 50% of the county population. We estimate the relationship between this proxy for potential listenership and standardized geographic exposure using equation (4), obtaining $de/ds = 0.271$. This calculation provides an illustrative benchmark rather than an identified listener-level persuasion rate, as it relies on assumptions about actual listening behavior and excludes potential indirect effects through social interactions. We obtain $\frac{de}{ds} = 0.271$. By replacing it in the formula for f, we obtain a rate of 23.5%. However, the estimate should be interpreted as an illustrative benchmark rather than a conventional persuasion rate, given the possibility of indirect effects through social interactions.

Appendix Bibliography

Bailey, A. K., S. E. Tolnay, E. Beck, and J. D. Laird (2011). Targeting lynch victims: social marginality or status transgressions? *American Sociological Review 76*(3), 412–436.

Beach, B. and W. W. Hanlon (2022). Historical newspaper data: A researcher's guide and toolkit. NBER Working Paper 30135, National Bureau of Economic Research.

BMB (1949). State area reports: Study no. 2. Spring 1949, Broadcast Measurement Bureau, New York. Issued as separate volumes for each state.

Bremmer, H. (1958). Propagation of electromagnetic waves. In *Elektrische Felder und Wellen/Electric Fields and Waves*, pp. 423–639. Springer.

Broadcasting (1947). *1947 Yearbook Number*. The Weekly Newsmagazine of Radio, Broadcasting Publications Inc.

Broadcasting (1960). *1960 Yearbook Number*. The Weekly Newsmagazine of Radio, Broadcasting Publications Inc.

Bureau of Census (1968a). Estimates of the population of the U.S., by age, race, and sex: July 1968. Summary report, U.S. Department of Commerce.

Bureau of Census (1968b). Estimates of the population of voting age for states: November 1968. Summary report, U.S. Department of Commerce.

Bureau of Census (1969a). Estimates of the population of states, by age: 1965–1968. Summary report, U.S. Department of Commerce.

Bureau of Census (1969b). Voting and registration in the election of november 1968. Summary report, U.S. Department of Commerce.

CAB (1946a). City reports – july 1945–february 1946. Collection of published reports, Cooperative Analysis of Broadcasting, Inc.

CAB (1946b). Program reports – february–july 1946. Collection of published reports, Cooperative Analysis of Broadcasting, Inc.

Clubb, J. M., W. H. Flanigan, and N. H. Zingale (1987). *Electoral data for counties in the U.S.: Presidential and congressional races, 1840-1972*. Inter-university Consortium for Political and Social Research.

Committee on Un-American Activities (1967). *Present Day Ku Klux Klan Movement – Report by the U.S. Congress Committee on Un-American Activities, House of Representatives, Ninetieth Congress, First Session*. Washington DC: U.S. Government Printing Office.

Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics 92*(1), 1–45.

Cook, L. D., M. E. Jones, T. D. Logan, and D. Rosé (2023). The evolution of access to public accommodations in the united states. *The Quarterly Journal of Economics 138*(1), 37–102.

Dunning, J. (1998). *On the air: The encyclopedia of old-time radio*. Oxford University Press.

Enikolopov, R., M. Petrova, and E. Zhuravskaya (2011). Media and political persuasion: Evidence from russia. *American Economic Review 101*(7), 3253–3285.

Estrada, J. and A. Hermida (2023). NAACP Branches Database - Mapping American Social Movements Project. Civil Rights and Labor History Consortium / University of Washington. Retrieved on 03/05/2023 from https://depts.washington.edu/moves/NAACP_database.shtml.

FCC (1954). Map of effective ground conductivity in the U.S. for AM broadcast stations. Federal communication commission, U.S. Department of the Interior Geological Survey.

Haines, M. R. (2010). Historical, demographic, economic, and social data: The United States, 1790–2002. Inter-university Consortium for Political and Social Research [distributor]. Version 3.

Hilmes, M. (1997). *Radio Voices: American Broadcasting, 1922-1952*. Minneapolis: University of Minnesota Press.

Hooper Inc. (1949). *Collection of Hooperatings Comprehensive reports - Summer, 1947–1949*. New York: C.E. Hooper Inc. - Radio Audience Measurement.

Josse, J. and F. Husson (2012). Handling missing values in exploratory multivariate data analysis methods. *Journal de la société française de statistique 153*(2), 79–99.

Jun, S. J. and S. Lee (2026). Learning the effect of persuasion via difference-in-differences. *https://arxiv.org/pdf/2410.14871*.

Kneebone, J. and S. Torres (2015). Data to support the "mapping the second ku klux klan, 1919-1940" project. History Data.

National Authority (1945). *Standard Rate & Data Service - Radio Section*.

Olken, B. A. (2009). Do television and radio destroy social capital? evidence from indonesian villages. *American Economic Journal: Applied Economics 1*(4), 1–33.

Radio Daily (1946). *The 1946 Radio Annual*. New York, NY.

SPLC (2023). SPLC hate map. Southern Poverty Law Center. Accessed on May 24th 2023.